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Final Report

**Wetland Geomorphic Linkages to Juvenile Salmonids and Macroinvertebrate
Communities in Headwater Streams of the Kenai Lowlands, Alaska**

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Study History: The Kenai Lowlands is a geographic region covering 810,000 acres of the western Kenai Peninsula, Alaska. In 2004, years of effort to classify and map wetland plant communities of the Kenai Lowlands culminated in the production of the Kenai Lowlands Wetland Management Tool (Gracz et al. 2004). Eight different geomorphic settings, supporting 70 different wetland plant community types were identified, covering 42% of the Kenai Lowlands area. The majority of these wetlands are in near pristine condition; however, human activity on the Peninsula is increasing rapidly, accompanied by increased impacts to wetlands. The goal of this project was to attribute the Kenai Lowlands Wetland Management Tool with data on landscape linkages to juvenile salmonids and macroinvertebrate communities in headwater streams. Providing this information in a spatially explicit format will allow planners, managers, developers, regulators and property owners to better understand the potential effects of development on stream biota. Field work for this 1 year project was conducted on the Kenai Peninsula, Alaska, USA from May 2006 through July 2006.

Project Abstract: We investigated relationships between wetland geomorphic settings, and headwater stream characteristics, with the goal of being able to attribute the Kenai Lowlands Wetland Management Tool, a GIS based wetland classification and map for the lower Kenai Peninsula. Thirty streams representing relic glacial drainageway, relic glacial lakebed, kettle and mixed-low-wetland-cover geomorphic settings were sampled across the five major drainages of the lower Kenai Peninsula. Stream reaches were sampled for abiotic and habitat characteristics, macroinvertebrate and fish assemblages, and characterization of riparian plant communities. Thirty-seven landscape metrics were calculated using GIS software for each stream segment sampled. Data was analyzed using a combination of Principal Component Analysis (PCA), non-metric multidimensional scaling (nMDS), multiple regression and regression tree analysis (RTA). Results of the PCA showed strong correlations between topography and wetland geomorphic setting, and between topography and in-stream abiotic and habitat characteristics. Ordination of the riparian vegetation data clearly showed that almost all sites were dominated by a single species, *Calamagrostis canadensis*. Regression analysis revealed a topographic index (a measure of the hydrologic condition that is based on upstream drainage area and slope) that had a strong correlations between stream water chemistry variables; hydrology and stream substrates. Streams with a low topographic index (higher gradient, low wetness) had higher velocities, dissolved oxygen, and nitrite-nitrate N, lower temperatures, more woody debris and more gravelly substrates. Higher topographic index streams (lower gradient, high wetness) had lower velocities, higher dissolved organic carbon, higher temperatures and more peat dominated substrates. Macroinvertebrate assemblages varied with the topographic index as did the fish assemblage. Sites with a low topographic index supported higher densities of small salmonids; and higher topographic index streams supported fewer larger fish.

Key Words: Kenai Peninsula, Alaska, relic glacial landscapes, watersheds, wetlands, headwater streams, riparian, juvenile salmon, macroinvertebrates

Project Data: *Description of data- Description of data-* data were collected in the field for fish, invertebrates and vegetation. Fish were identified in the field, with digital images taken as reference. Macroinvertebrate identifications were completed at Baylor University in Waco, Texas, where a voucher collection is housed. Plant analyses were conducted by the Smithsonian Environmental Research Center in Edgewater, Maryland. Water samples were analyzed by the Cook Inlet Keeper in Homer, Alaska. *Format* - All data were entered as Excel spreadsheets. *Custodian* – contact Coowe Walker, Kachemak Bay Research Reserve, 95 Sterling Highway, Suite 2, Homer, AK 99603.

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Executive Summary

The Kenai Lowlands is a geographic region covering 810,000 acres of the western Kenai Peninsula, Alaska. In 2004, years of effort to classify and map wetland plant communities of the Kenai Lowlands culminated in the production of the Kenai Lowlands Wetland Management Tool (Gracz et al. 2004). This effort resulted in the identification of 8 different geomorphic settings, supporting 70 wetland plant community types, covering 42% of the Kenai Lowlands area. The majority of these wetlands are in near pristine condition. However, human activity on the Peninsula is increasing rapidly, accompanied by increased impacts to wetlands. The goal of this project was to attribute the Kenai Lowlands Wetland Management Tool with data on wetland geomorphic linkages to juvenile salmonids and macroinvertebrate communities in headwater streams of the Kenai Lowlands, Alaska. Providing this information will substantially increase the usefulness of the Wetland Management Tool for regulatory and conservation planning needs, by providing managers, regulators and property owners with spatially explicit information on how wetlands and streams function in supporting stream biota.

There is growing evidence headwater streams and associated riparian areas are important for maintaining healthy fish populations and stream ecosystems (Bedford and Goodwin 2003; Lowe and Likens 2005; Richardson 2000). The lower Kenai Peninsula of Alaska supports many anadromous salmon runs, including king (*Oncorhynchus tshawytscha*), silver (*O. kisutch*), steelhead (*O. mykiss*), pink (*O. gorbuscha*) and Dolly Varden (*Salvelinus malma*). Headwater streams make up a large proportion of the stream networks these salmonids utilize, however, there is scant specific information on how fish use the habitats provided in headwater stream systems. Land use on the southern Kenai Peninsula has changed dramatically over the past 15 years. Increased forest practices in response to the spruce bark beetle epidemic, and increased development activities associated with a growing population create the potential for widespread anthropogenic disturbances. Upper reaches of stream systems may be subjected to the most severe impacts because large portions of these upper watersheds are in private ownership, and there are few prescribed stream protections.

With this project we open the window into the linkages between wetland geomorphic setting and instream abiotic characteristics and biotic communities in headwater streams. We used a stratified sampling design to investigate 30 headwater streams associated with the four dominant wetland geomorphic settings, and across the five major drainage basins, of the lower Kenai Peninsula. At each sample site, we recorded abiotic and biotic variables, following EMAP protocols, and characterized riparian vegetation communities. Using a Geographic Information System (GIS) we calculated 37 landscape metrics for each sampling site. A Principal Component Analysis (PCA) of these variables revealed strong relationships between topographic index (a measure of how wet a point should be based just on upstream drainage area and slope), watershed slope and wetland class. A second PCA of landscape metrics and 67 reach scale variables revealed strong instream habitat and abiotic characteristics that are unrelated to basin but strongly related to landscape features (slope, elevation, topographic index). Using multiple linear regression and Regression Tree analysis (RTA) to model the relationships between variables revealed strong correlations between hydrology (velocity, depth); water chemistry variables (dissolved oxygen, nutrients, temperature, pH); habitat characteristics (substrate, canopy cover, woody debris) and topographic index. Streams with low topographic index (high gradient streams) were shallower, had higher velocities, higher dissolved oxygen, higher nitrite-nitrate N, lower temperatures, lower pH, more woody debris and more gravelly substrates. Lower gradient streams (higher topographic index) were deeper, had lower velocities, higher

dissolved organic carbon, higher temperatures, higher pH and more peat dominated substrates. Macroinvertebrate assemblages changed corresponding to these trends.

Streams became slightly more acidic in wetter, low gradient watersheds, suggesting a greater influence of organic acids on stream water chemistry. Trends towards lower dissolved oxygen, higher temperatures, and higher dissolved organic carbon, also suggest more interaction between wetland organic matter and shallow groundwater in watersheds with lower topographic relief. Higher gradient streams had much higher percentages of gravel and cobble substrates, and much higher tree cover. This turns out to be a critical finding because of the requirement of fine gravel, erosional habitats for spawning salmonids and successful rearing of eggs. Tree cover was also strongly related to channel characteristics (undercut banks, mineral substrate, woody debris), habitat characteristics which favor growth of small salmonids. Lower gradient streams had much higher percentages of peat substrates. These findings show that streams in most low gradient watersheds began to resemble depositional, wetland habitats compared to the erosional, mineral substrate streams at higher gradients.

Differences in stream characteristics were embodied in fish assemblages also; with low topographic index streams supporting higher densities of small salmonids; and higher topographic index streams supporting fewer but larger fish. First-year cohos were essentially absent in peaty streams with soft bottoms and streams with sufficient biological oxygen demand to reduce DO below 100% saturation during daylight hours. Both of these responses make ecological and physiological sense because streams with mostly peat substrates did not support adequate mixing of water near the sediment interface nor fine, clean mineral substrates for spawning adults. Depressed dissolved oxygen during daylight probably corresponded to even lower levels during night, which would stress juveniles, particularly during the earliest stages of development. A strong negative exponential relationship indicated that streams with high densities of cohos tended to support small, first-year fish, whereas streams dominated by larger, second or third year fish had low overall densities. This relationship suggests the biomass per kilometer of stream reach was density dependent and that different streams supported different size classes of fish, yet similar levels of population biomass. No overall relationship was evident for biomass/km and landscape or environmental variables.

The Wetland Management Tool was attributed in 1 km headwater stream segments, using topographic index to predict instream abiotic characteristics, habitat features, macroinvertebrate and fish assemblages. Although wetland class was not the best predictor of stream functions, it was clearly tied to topographic index, and an important contributor to many of the instream habitat features, water chemistry and hydrology variables. These associations were explained in dialogue boxes associated with the attribution of the Wetland Management Tool. The addition of the results of this project to the Wetland Management Tool provides regulators, managers, developers and property owners with the ability to predict instream habitat conditions, macroinvertebrate and fish use in headwater stream areas of the Anchor River, Stariski Creek, Deep Creek, Ninilchik River. This information also provides the framework for future investigations of anadromous watersheds on the Kenai Lowlands, which will hopefully lead the way to forecast and mitigate ecological consequences of development activities in the headwater regions.

Introduction

Salmonid ecosystems throughout the world have experienced serious declines in productivity and diversity in recent history, due in large part to the impacts of human activities, such as watershed development, forestry practices and fishing (Naiman and Bilby 1998). Dramatic declines and extinctions within native Pacific Northwest salmon populations outside of Alaska caused by human pressures has prompted intense focus on conservation of wild salmonid populations during the last ten or so years. Although most Alaskan salmon populations are healthy so far, increasing landscape development in some areas, coupled with global climate change and natural variability raises the serious possibility of future population declines.

Human pressures on the Kenai Peninsula of Alaska (Figure 1) are mounting. Currently, bountiful salmon populations are sustained by the watersheds of the Kenai Lowlands, providing millions of dollars to local economies through sport and commercial fisheries, and related businesses, as well as important subsistence resources for Natives and other groups. However, impending development may adversely affect salmon populations in this area in the future.



Figure 1. The Kenai Peninsula, located in southcentral Alaska, has abundant salmon populations and is an area of active human population growth.

To maintain healthy salmonid ecosystems, we need to understand watershed ecological processes and spatial complexity (Fausch et al. 2002). Headwater streams typically occupy a large proportion of total channel length, however they are often overlooked in studies of river ecological processes (Lowe and Likens 2005). In Alaska, the combined contribution of headwater streams to fueling stream energy may be large (Richardson 2000; Bryant, et al. 2004). There is growing recognition that stream headwaters, and surrounding wetlands, exert a strong influence on downstream physical and chemical water properties by moderating water temperatures, nitrogen transformation and phosphorous retention, factors that can strongly affect stream productivity, and provide critical rearing and forage support habitat for young salmon (Bedford and Goodwin 2003, Steel et al.

2003, Wipfli and Gregorovich 2002). For coho (*Onchorynchus kisutch*) and chinook (*O. tshawytscha*) salmon, species that spend several years as juveniles in freshwater streams before out-migrating to the ocean, headwater stream habitats may be especially important.

The five major drainages of the Kenai lowlands occupy roughly 167,000 hectares (412,500 acres), embedded into a relict glacial landscape with a high percentage (42%) of wetlands in the landscape (Gracz et al. 2004) (Figure 2). Unlike many parts of Alaska, land ownership where these headwater streams are located is predominately private, and there are very few stream protection requirements to limit adverse impacts from development. These areas have seen a rapid rise in population growth, which is expected to continue (KBRR and NOAA/CSC 2001), leading to a high potential for considerable changes to these streams and their surrounding watersheds. The majority of these streams are not even officially recognized as bearing salmon, a designation which provides the only legal protection for salmon habitat in Alaskan streams. Only 13% of the headwater streams on the Kenai Lowlands have documented salmon presence, although it is likely that juvenile salmon, especially coho, are present in many of them. Indeed, the Alaska Anadromous Stream Catalogue, documents coho salmon populations in all five of the major streams throughout the lower and middle reaches, making it highly likely that coho are present as juveniles in the upper reaches.

US EPA Region 10 has devoted considerable time and financial resources to developing a Geographic Information System (GIS) based wetland map that can be used as the framework for management decisions in this area. This map, called the Kenai Lowlands Wetland Management Tool, identifies eight different geomorphic settings, supporting 70 different wetland plant community types. Since completion of the map, a primary goal of EPA Region 10 has been to understand the relationship between wetlands in the landscape and stream water chemistry, hydrology, and biota.

The primary goal of this study is to model the relationship between dominant wetland geomorphic settings in watersheds identified in the Kenai Lowlands that support stream headwaters (e.g., relict glacial lakebeds [LB], relict glacial drainageways [DW], discharge slopes [DS], and mixed wetland-upland complexes) to in-stream habitat, water chemistry, and ecological attributes (e.g., juvenile coho relative abundance). This information will provide people involved with land-use decisions with information on ecological processes at watershed and reach scales that can provide a solid foundation for management, planning and conservation efforts to sustain salmonid populations on the lower Kenai Peninsula.

Objectives

1. Identify stream water chemistry, hydrology, and habitat characteristics associated with macroinvertebrate and juvenile salmonids assemblages.
2. Attribute (provide data to associate with GIS map shape files) the Kenai Lowlands Wetland Management Tool with pertinent results to aid in planning, management and regulatory decision-making.

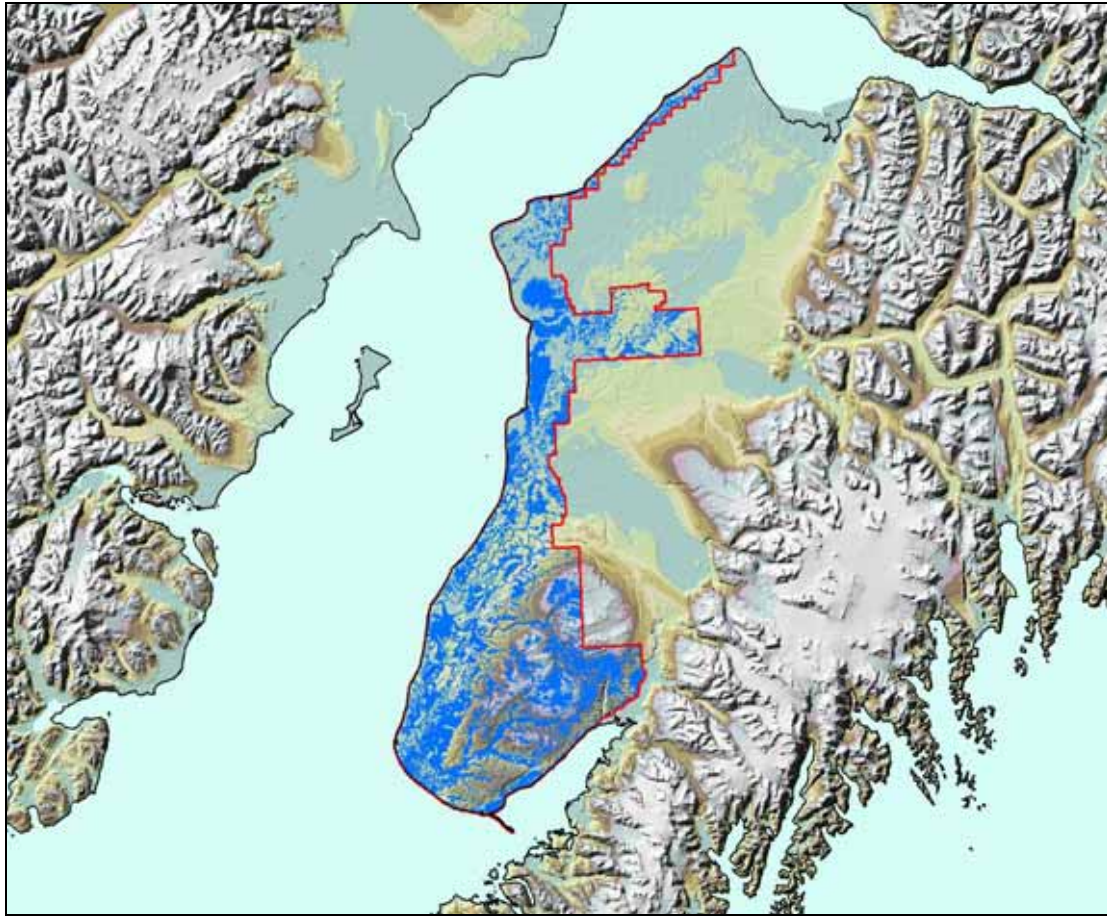


Figure 2. The Kenai Lowlands (shown bordered in red) includes all of the terrain west of the Kenai National Wildlife Refuge. Wetlands occupy forty-two percent of the Kenai Lowlands area.

Overview of Study Area

The southern part of the Kenai Peninsula, bounded by Cook Inlet to the west and the Caribou Hills to the east has gently rolling topography, with wide river valleys and extensive peat wetland systems. Elevations range from around 950 meters to sea level (Mauger 2005). Climate in the study area is transitional between maritime and continental, in general becoming more continental towards the north. Homer, located at the southern end of the Peninsula, has an average winter (January) temperature of -5.2°C and average summer (July) temperatures of 11.9°C . The average annual precipitation in Homer is 61.7cm, with the majority of rain occurring in fall (September through November). Snowmelt and ice breakup contribute to high stream flows in spring (April-May) (Mauger 2005).

The southern Peninsula supports many wildlife species including a wide variety of seabirds, waterfowl, shorebirds, raptors, and songbirds, moose, black and brown bears, fox, lynx, coyote, and a variety of small mammals. The streams in this area support numerous small fish and salmonid species including king (*Oncorhynchus tshawytscha*), silver (*O. kisutch*), steelhead (*O. mykiss*) pink (*O. gorbuscha*) and Dolly Varden (*Salvelinus malma*).

The economic base for communities on the southern Peninsula is largely composed of commercial fishing, sport fishing and tourism. Land ownership in the area includes Federal, Native Corporation, State, Borough and private holdings. Land use on the southern Kenai Peninsula has changed dramatically over the past 15 years, especially due to increased forest practices in response to the spruce bark beetle epidemic, and to increased development activities associated with a growing population. Population growth is expected to continue (KBRR and NOAA/CSC 2001).

Methods

Site Selection

The study area for this project includes the headwaters for the major anadromous stream drainage basins of the Kenai Lowlands: North Fork Anchor River, South Fork Anchor River, Stariski Creek, Deep Creek and Ninilchik River (Figure 5). First-order streams were identified from USGS DLG data (basically from 1:25,000 topographic maps) on these streams. A very preliminary filter for accessibility was done by choosing only streams within 3 km of a road. Points were placed every 250 meters along these streams, and drainage basins were calculated for each of these points, using the DLG data. These drainage basins were then hand-checked and edited where necessary by overlaying them on the best-available USGS topographic maps. For each stream, this resulted in a series of nested, overlapping drainage basins, with the furthest upstream basin the smallest, and each basin downstream including the area of all those upstream. The total area of each basin, and the area covered by each wetland type within each polygon was calculated, and this data was placed in an Excel spreadsheet. In summary, we ended up with 118 probably-accessible headwater streams, with 1,111 points and associated drainage basins on those streams, and a good spread across the major river systems (roughly proportionate to system size) , although the accessible 1st order streams in the Ninilchik and Deep Creek drainage basins tended to be lower in the watershed.

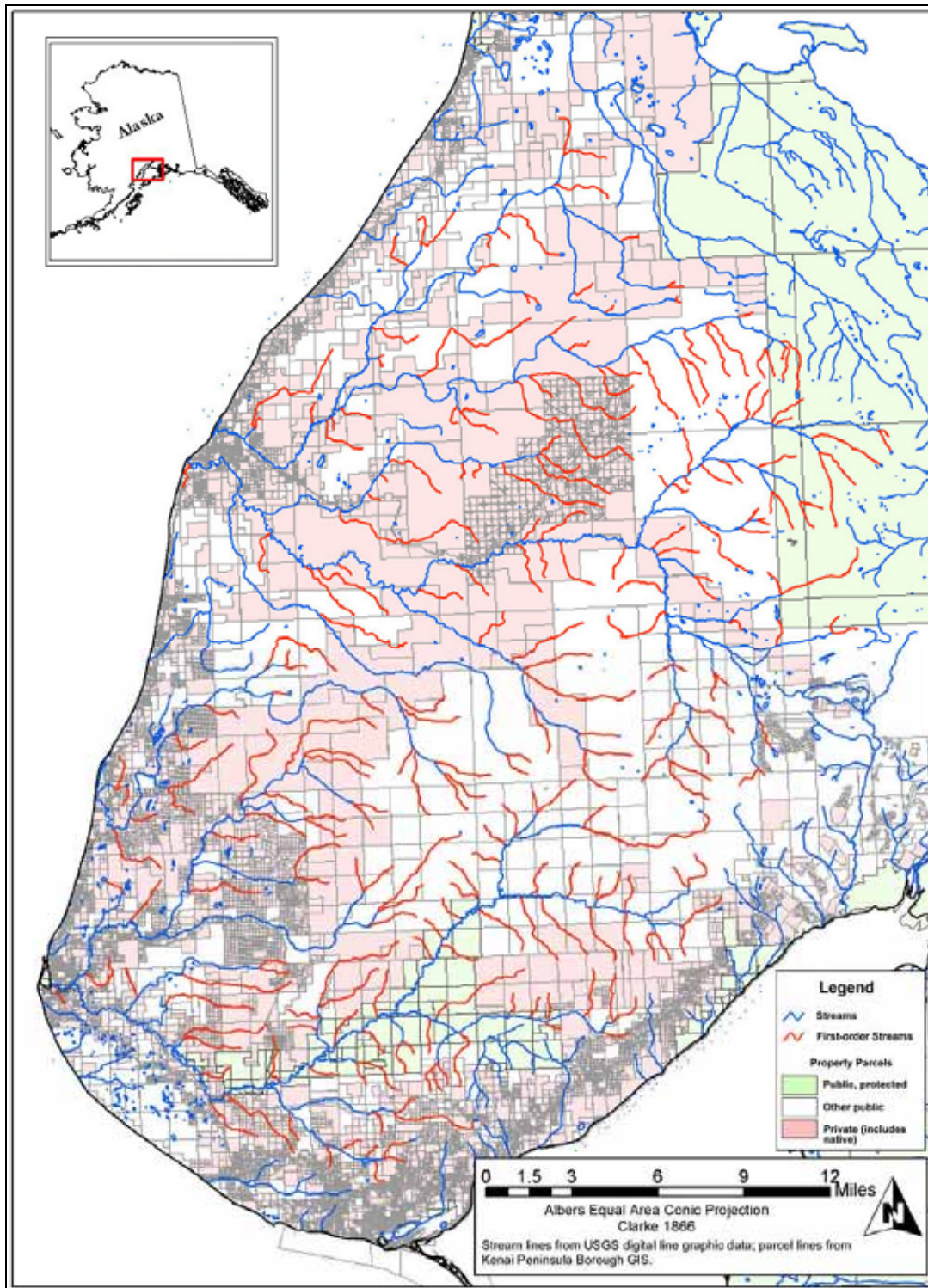


Figure 3. Project study area, showing drainages, first-order streams (red segments) and property ownership. Note the high percentage of private (pink), and unprotected public (white) lands.

Once potential drainage basins had been identified that could be reasonably be accessed, the watershed data was then analyzed to see if we could identify general groupings based on dominant wetland classes. We employed a series of analyses and graphical evaluations, including hierarchical agglomerative cluster analysis to identify groups, non metric multidimensional scaling (nMDS) ordination to look at the groups in an ordination space, and classification tree analysis to identify potential cutoff values of wetland percentages that relate to the groups (Figure 4). A few clear patterns emerged from these analyses:

- 1) Generally, there is considerable overlap of wetland classes within watersheds (something that we anticipated).
- 2) Despite the overlap in classes, there are four relatively distinct and widespread groups based on the percentages of wetland cover within watershed boundaries (with a fifth group of watersheds with no defining characteristics based on wetland type). These groups are Lakebed (LB), Drainageway (DW), Discharge Slope (DS), and Kettle (KT).
- 3) LB, DW, and DS wetlands often dominate the overall wetland cover in a watershed and appear to be the three most distinct groups, even though they often overlap to varying degrees in watersheds. LB and DS particularly tend to co-occur, but LB has usually 2x more cover in most of the watersheds classified as the LB group (25-45% LB compared to 7-23% DS, interquartile range). The DS group is distinguished by having 7-20% DS cover but little other wetland cover in the watershed. Thus, the difference between the LB and DS groups is that both have similar amounts of DS, but only the LB group has much LB cover. Finally, the general pattern of low LB and DS but high DW characterizes the DW group.
- 4) There is a large group of watersheds that have relatively low wetland cover and/or a heterogeneous mix of low-abundance classes such as Late Snow Fend (LS), Headwater Fen (HF), Depression (DP), and Riparian (RI) that probably will limit our ability to relate any particular wetland class to stream characteristics in these systems.
- 5) Watersheds with high percentages of kettle (KT) wetlands also emerge as a potential "class" but these also tend to co-occur with DS, DW, or LB wetlands. KT dominated watersheds also tend to be larger.
- 6) The Riparian (RI) class tends to be found across all watersheds in similar abundance (typically 4-8%) and thus will not likely be useful for site selection.

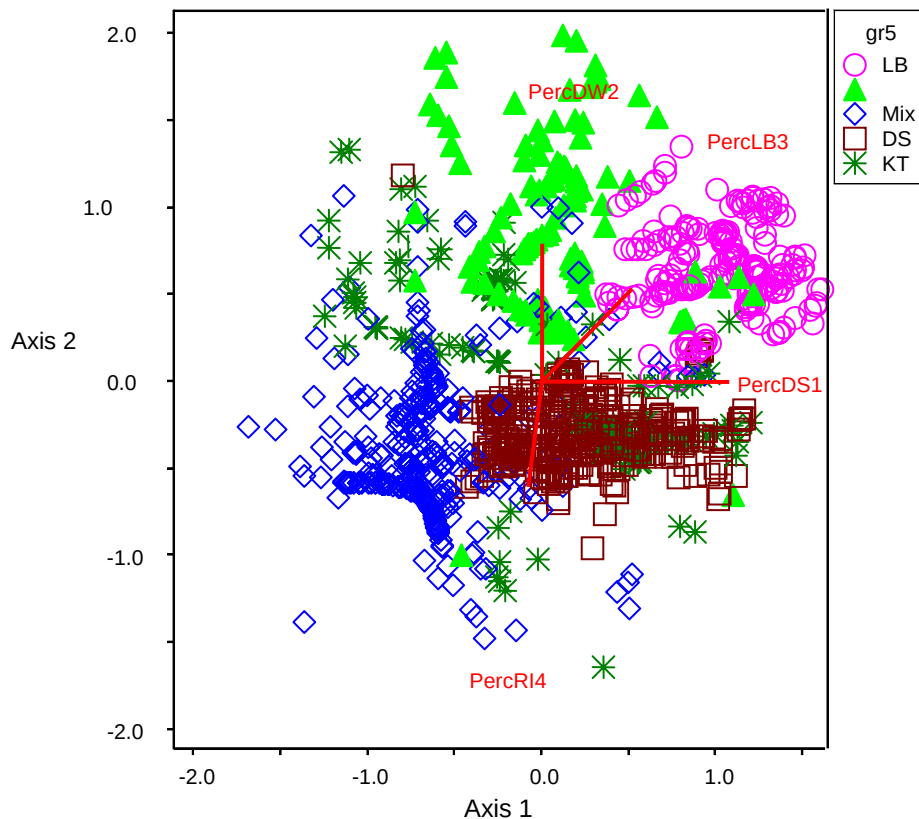


Figure 4. NMDS Ordination of the 5 watershed groups defined by their similarity in percent wetland cover. Points close together in the ordination are similar in wetland composition, where LB=lakebed, DW= drainageway, Mix= mixed wetland class/overall low wetland cover, DS= drainage slope, KT= kettle

We mapped the cluster groups across the five major drainage basins in order to visualize the spatial arrangement of the potential sites (Figure 5). From this map, it appeared that the northern two drainage basins (Deep Creek and Ninilchik) have more mixed-low wetland cover watersheds and more kettle dominated watersheds than the southern three drainage basins (Stariski Creek, North Fork Anchor and South Fork Anchor). We mapped the percentage of each wetland class in each drainage basin, and found that indeed, the northern two drainages have lower overall percentage of wetlands and greater percentage of kettle wetlands than the southern three drainages (Figure 6).

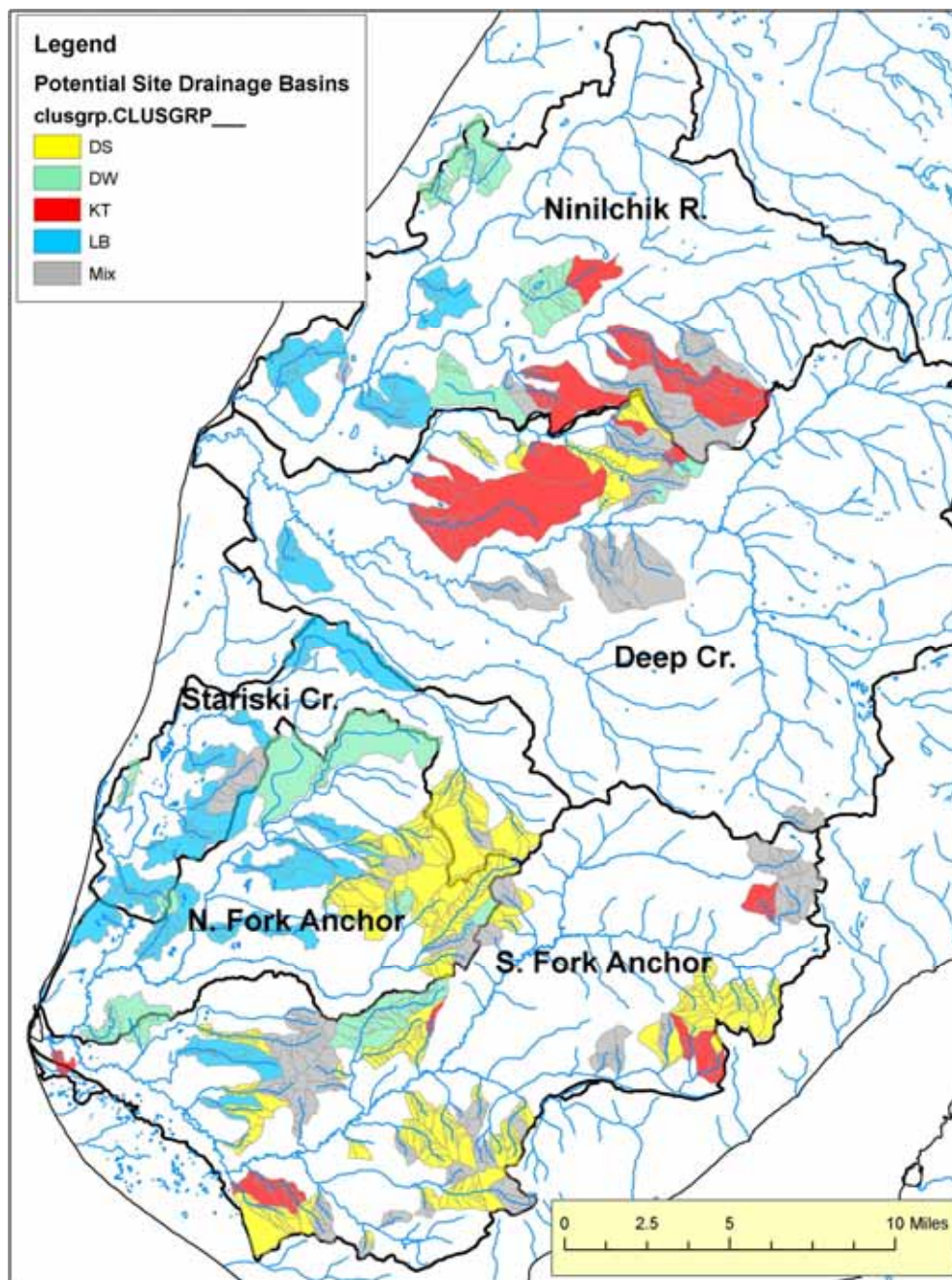


Figure 5. Map showing the spatial arrangement of the wetland groups that emerged from the cluster analysis

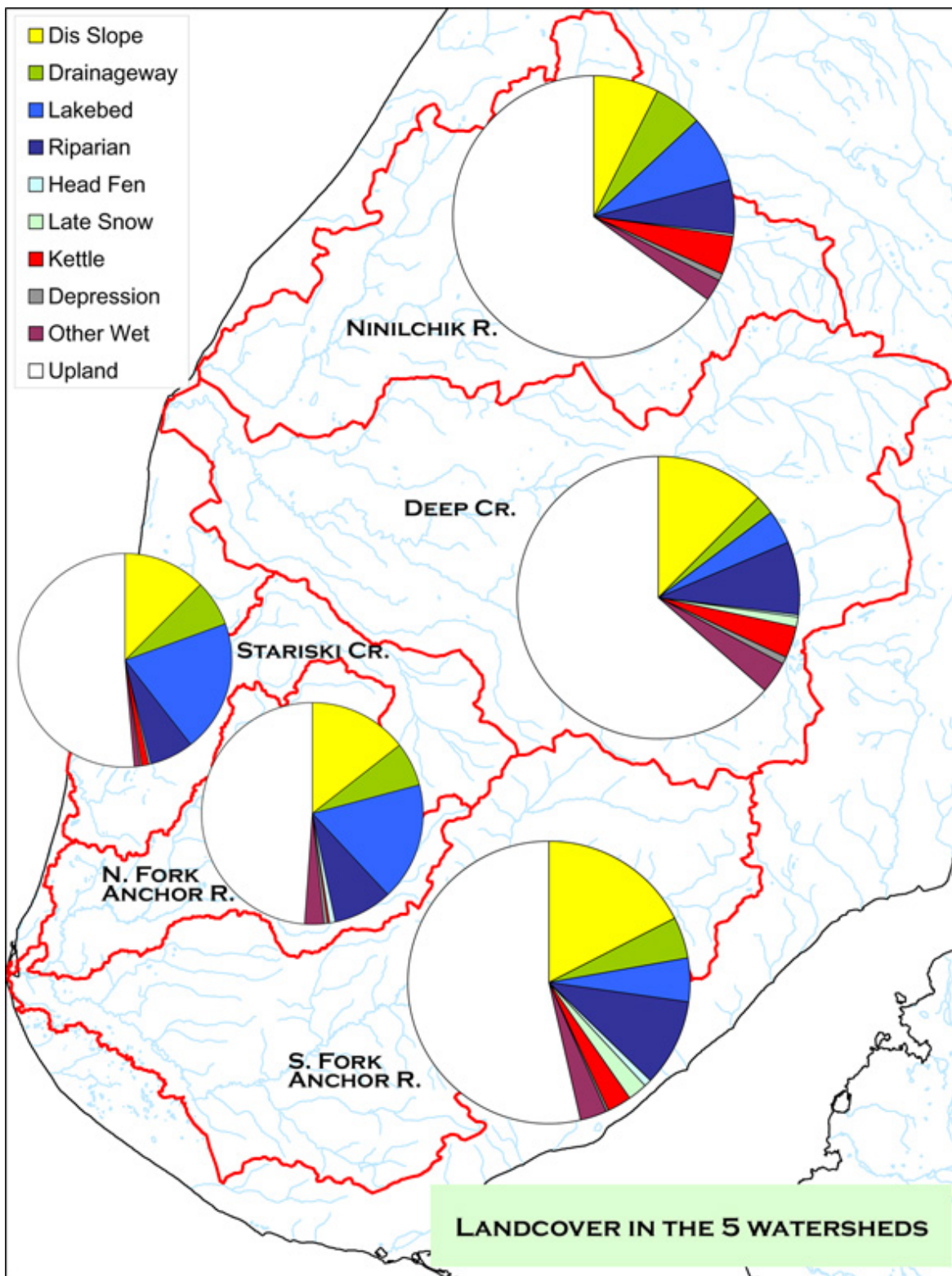


Figure 6. Percentage landcover type in each of the five major drainage basins of the study area.

Our goal was to develop models that can be used to attribute as many of the Wetland Management Tool wetland polygons as possible. Models can only be used to make predictions using the same independent variables that went into the model, and those measurements must be in the same range of those used to build the model. This means that geomorphic classes that aren't well represented in our study will be difficult to attribute unless they co-occur with other classes that are well represented and used in the models. For that matter, watersheds with classes that are represented may still be difficult to attribute if relationships to functions are weak in our models. Conversely, if there are strong patterns between one or more of the major geomorphic classes and stream functions such as fish abundance, we should be able to use that model to attribute most of the headwater stream segments with predicted/potential functional values using % cover of each class deemed important in the model.

In general, the LB wetlands always have DS wetlands associated with them. The difference between the LB and DS groups is that the DS group is only represented by DS wetlands. The DS-only watersheds also appear to be distributed mainly in the upper portions of the major drainages, which will also distinguish them from the LB/DS mixed watersheds. The ordination analysis revealed several kettle-group watersheds intermingling with DS and DW watersheds; these are probably DS or DW dominated watersheds that have relatively low kettle cover, but enough KT to distinguish them from the other DS or DW watersheds. This was also illustrated in the classification tree, as KT watersheds were some of the last to be sorted out.

We decided that our best option was to pick two watersheds from each major drainage (five drainages) that are each reasonably well represented by LB, DS, DW, and mixed-but-low-total-wetland-cover classes (4 classes), for a total of 40 watersheds. We categorized the size distribution of each type. We then picked the mean size and looked at those sites that are close to the mean size to determine how much watershed is upstream of each one. This allowed us to determine the total area (size of each area identified plus the size of the associated upstream watershed) of potential study sites. By going through this exercise, we determined how many potential sites occur in each of the five major drainages (where they are located and the size of the source areas that drain into them). With this information in hand, we then determined which sites could be reached with reasonable effort. We then gave the sites that fit the 'access' criteria numbers and randomly picked the ones that we would sample (Figure 7).

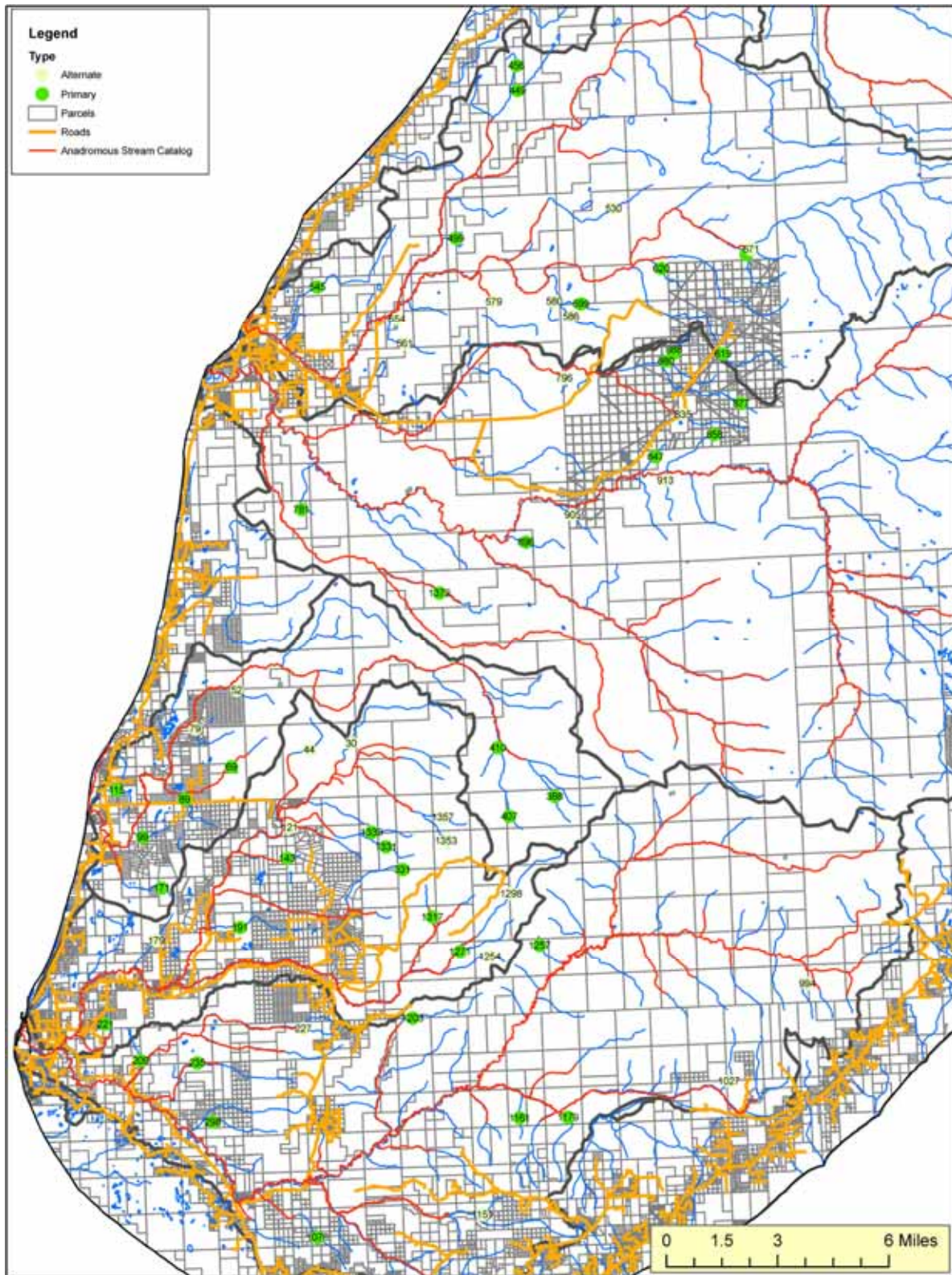


Figure 7. Map showing final primary (bright green) and alternate (light green) site selection for the forty sample sites. Road access, which was an important consideration for this expansive area, is shown in orange.

Station Set Up

Establishment of station boundaries and sampling procedures followed a slight modification of methods described by U.S. EPA's EMAP protocol (Lazorchak et al 1998). Each site encompassed a 250-m reach of stream (stream-channel distance). Stream reaches were marked with 11 equidistant cross-section locations (transects perpendicular to the stream channel). Cross-section locations were spaced 25 m apart. These locations served as a sampling framework for all physical, chemical and biological measurements.

Sampling was accomplished using a field crew of 4 individuals. Sampling was structured into 5 major components in the following order: (1) establishing site boundaries and marking transects (1-2 persons), (2) electrofishing/seining of juvenile salmon (minimum 2 persons), (3) riparian assessment (1-2 persons), (4) stream physical habitat, water chemistry, and discharge (1-2 persons), and (5) benthic macroinvertebrate sampling (1-2 persons). A pair of activities typically took place simultaneously to maximize sampling efficiency, but coordination between the two sampling teams was important to ensure safety and data quality.

Upon arriving at a site, 1-2 persons began laying out the site boundaries while the remaining 2 persons prepared equipment and data sheets and measured *in situ* water chemistry for electrofishing (see Fish Sampling). Once the first few transects in the downstream portions of the 250-m reach had been marked, the electrofishing team began sampling while the others continued laying out the remaining transects and recording general information about the site per Lazorchak et al (1998). Once the site layout was completed, these persons conducted the riparian assessment (see Riparian Assessment) while the electrofishing team completed its work in the stream. Physical habitat, water chemistry samples, discharge (see Physical Habitat, Water Chemistry, and Discharge) were collected or measured by the riparian team following completion of the riparian assessment and commenced either after the electrofishing is completed or at lower end of the reach, at least 100 m away from where the electrofishing was taking place upstream. Following electrofishing, this team collected macroinvertebrate samples (see Benthic Macroinvertebrate Sampling) while the other individuals completed the physical habitat sampling.

Fish Sampling

Prior to sampling, *in situ* specific conductance, dissolved oxygen, temperature, and pH were measured using a YSI 556 multiprobe, or YSI 660 Sonde. *In situ* water chemistry was measured at this time to assess conductivity of the stream, which influences selection of appropriate voltage settings on the backpack electrofisher unit prior to sampling. The YSI multiprobe was calibrated daily in accordance with QA/QC procedures.

Juvenile fish species composition and density estimations were established using a DC-pulsed backpack electrofishing unit (Smith-Root, Inc.) and supplemental seining when necessary. Fish sampling methods followed Lazorchak et al. (1998), with very few modifications. Lazorchak et al. (1998) recommends a minimum reach length of 150-m, or 40 times the average stream width for larger streams, for characterizing stream fish communities. Because all of the streams encountered in this study averaged < 4 m in width, we sampled 150 m of the 250-m stream reach by sampling three 50-m subreaches. Subreaches that corresponded to the following transect markers: 0–50 m, 100–150 m, and 200–250 m. Subreaches located between 50 to 100 m and 150 to 200 m (100 m total) were avoided during electrofishing and reserved for macroinvertebrate sampling to minimize disturbance to substrate to these small streams due to intensive wading during electrofishing (see Benthic Macroinvertebrate Sampling).

Fish were identified, enumerated, measured and released as described in Lazorchak et al. (1998). Fish species density and community composition was estimated at the scale of the entire reach.

Benthic Macroinvertebrate Sampling

Benthic macroinvertebrate sampling was conducted by the electrofishing team following fish sampling and began at the 50-m transect. Stream macroinvertebrate taxa composition, density, and biomass were sampled using a modification of methods described in Lazorchak et al. (1998). One surber sample (0.092 m² surface area; 350-µm mesh) was collected from each of the 50, 75, 100, 150, 175, and 200 m transects, and an additional sample was collected from a midpoint between each of the 50-75, 75-100, 150-175, and 175-200 m transects, totaling 10 surber samples for the entire reach. Each sample was sieved in the field and composited into one sample to represent the entire reach. All macroinvertebrate and organic matter samples were kept on ice until returning to the lab, where the composite samples were preserved in 10% buffered formalin stained with rose bengal in the laboratory (King and Richardson 2002).

Benthic Macroinvertebrate Laboratory Identification And Enumeration

In the laboratory, composite samples were rinsed through a 350 µm mesh sieve to remove residual formalin. Samples were subsampled to achieve 500 individuals using fixed-count methods described in King and Richardson (2002). Sample material was distributed evenly in a gridded pan. Numbered squares in each pan were selected using randomly selected numbers. Material from each randomly selected square was carefully removed and sorted under stereomicroscope at 10X magnification. This process was repeated until 500 individuals are removed from the sample, or the entire sample is sorted, whichever comes first. Number of squares was used to estimate the fraction of the total sample area subsampled, which was used to estimate relative densities of organisms among sites. Macroinvertebrates were identified to the lowest practical level of identification, usually genus. Samples were shipped to Baylor University via the US Postal Service.

Quality Assurance/Quality Control for sorting and taxonomy was conducted following Barbour et al. (1999). Briefly, 10% of the sorted samples were examined by a qualified member of the laboratory to ensure that the sorter missed no more than 10% of the organisms contained in the randomly subsampled grids. A voucher collection was maintained by the Aquatic Ecology lab at Baylor University. Identification was confirmed by qualified experts in the laboratory. Funding did not permit independent verification of identifications by outside experts.

Physical Habitat, Water Chemistry, and Discharge

Physical habitat measurements followed Lazorchak et al. (1998) and Barbour et al. (1999). Lazorchak et al. (1998) identified seven general physical habitat attributes important to the ecology of wadeable streams:

- Channel Dimensions
- Channel Gradient
- Channel Substrate Size and Type
- Habitat Complexity and Cover
- Riparian Vegetation Cover and Structure
- Anthropogenic Alterations
- Channel-Riparian Interaction

The habitat characterization protocol developed for EMAP (Lazorchak et al. 1998) differs from other rapid habitat assessment approaches (e.g., Barbour et al. 1999) by employing a systematic spatial sampling design that minimizes bias in the placement and positioning of measurements. The time commitment to gain repeatability and precision is greater than that required for more qualitative methods. In EMAP studies, two

people typically complete the specified physical habitat, water chemistry and discharge measurements in about three hours of field time.

Physical habitat measurements were made at two scales of resolution along the length of the reach; the results were later aggregated and expressed for the entire reach. The following features were measured or visually estimated at systematic cross-section locations along the reach:

- Maximum (thalweg) depth (cm)
- Wetted width (m)
- Channel cross-section dimensions
- Bank height
- Bank undercut distance
- Bank angle
- Velocity (m/s)
- Presence of soft sediment
- No. of large woody debris within bankfull channel
- No. of large woody debris above bankfull channel
- Riparian canopy density (densiometer, %)
- Straight-line distance (sinuosity; *likely during riparian assessment*)
- Substrate size classes
- Embeddedness (%)
- Areal cover of fish concealment features
- Aquatic macrophyte cover and type
- Filamentous algae and periphyton cover
- Human disturbances: type and proximity to channel

Lazorchak et al. (1998) describes in detail methods for each measurement. The logistics and work flow proceeded in the following manner. Two people proceeded upstream from the downstream end of the sampling reach making observations and measurements at the chosen increment spacing. One person was in the channel making width and depth measurements and determining whether soft/small sediment was present under his/her staff. Width, depth, and sediment measurements took place both at each primary cross-section location (11 locations) and a midpoint between each transect (10 additional locations; 21 total measurements). The other person recorded these measurements and tallied large woody debris. At the 11 primary cross-sectional locations, one person proceeded with the channel cross-section dimension, substrate, bank, and canopy cover measurements. The second person recorded those measurements. Both persons worked together to measure and record a velocity measurement in the thalweg (the deepest part of the stream channel) at a depth of 0.6 thalweg depth using a portable flowmeter and visually estimated instream fish cover, aquatic macrophytes and periphyton, human disturbance, substrate particle size class, and embeddedness.

Water-chemistry samples were collected at the upstream end of the reach (250 m marker) following completion of the physical habitat survey, and delivered to the Cook Inlet Keeper water quality laboratory in Homer for analysis of total and soluble reactive phosphorus, total nitrogen, nitrate-nitrite nitrogen, ammonium-nitrogen, dissolved organic carbon, and total alkalinity.

Discharge measurements were made after collecting the chemistry samples. Discharge was estimated at an optimal cross section location based on channel morphology (but not necessarily at a marked transect).

Discharge was estimated using the velocity-area method. Velocities were estimated using a Marsh-McBirney electromagnetic flowmeter following procedures in Lazorchak et al. (1998).

Riparian Assessment

We characterized the wetland and upland habitats within 100 m of the streams to evaluate the landscape for potential inputs to the stream from vegetation that would fall into it as well as discharges of surface and groundwater that would deliver water from adjacent ecosystems.

Each 250 m reach of stream was divided into five 50 m segments. On each side of the stream, we sampled each segment. Our objective in this sampling was to quantify the vegetation characteristics of the near-stream riparian area to quantify similarities and differences between the different categories of headwater stream systems.

1) Landscape (vegetation) characterization of the riparian zone

In each segment, we identified the distinct vegetation zones within 100 m on both sides of the stream. These measurements involved identifying and measuring (with meter tapes or range finders) the boundaries of distinct vegetation types between the stream and a distance of 100 m from the stream. The result was a map that could be used to quantify the percent of each segment that is occupied by each vegetation type (wetland as well as terrestrial habitats).

2) Standing live and dead trees, and shrubs

Trees are an important component of many riparian systems. They can influence microclimate and they are sources of organic matter and nutrients to streams. In addition to organic matter and nutrients, coarse woody debris that falls into the stream or is carried to the stream by animals (i.e., beaver) can provide important in-stream structural habitat. In each segment we determined the basal area of standing live and standing dead trees (defined as single stems ≥ 10 cm DBH) that can potentially contribute directly to the streams if they were to fall. For purposes of sampling, we assumed that only trees that were within 20 m of the stream had the potential to fall across or into the stream. This was done by measuring the DBH of each tree that potentially could fall into or across the stream. We identified each tree and determined the basal area (m^2/ha) for each species, all living trees, all standing dead trees, and all trees combined.

Shrubs (defined as single stems < 10 cm DBH) also have the potential to influence streams. At the mid-point of each segment, we established a 2m wide X 10 m long transect perpendicular to the stream and count the number of shrubs. The number of shrubs was used to determine density/ m^2 .

3) Surface and subsurface hydrologic connections

Lateral flows that enter the stream can influence in-stream processes by providing nutrients, dissolved organic matter, sediments, and particulate organic matter. Many of the vegetation types that are within the 100 m sampling area in each segment may have seasonal or perennial hydrologic connections with the stream.

We walked the length of each 50 m segment to determine the number of seasonal or perennial channels that are present. We walked the length of each channel located to determine, as much as possible, where a distinct flow feature begins. When that point was located, we determined how far it is from the stream (with a range finder or measuring tape) and identified the type of vegetation that was present at the point of origin. We also determined if each flow feature discharges directly to the stream or ends on the

floodplain adjacent to the stream. After we had measured existing surface connections, we walked the stream channel to count the number of subsurface discharges, if any, entering the stream.

4) Vegetation

In each segment, we collected cover data for herbaceous species and seedlings of woody species. In the middle of each 2 X 10 m transect used to sample shrubs, a single 1 X 1 m plot was established to determine the percent cover of herbaceous vegetation and seedlings of woody species. Cover estimates were made for each species. At a small subset of sites, we collected newly expanded leaves of the dominant herbaceous species for nutrient analysis. All samples were dried at 60 C, ground and analyzed for nutrients (nitrogen, phosphorus, Calcium, magnesium, potassium) at the analytical services laboratory at Pennsylvania State University.

Analysis

The primary goal of this study was to model relationships between wetland geomorphic classes in watersheds to instream habitat, water chemistry, and ecological attributes (e.g., juvenile coho relative abundance). Use of continuous class percentage cover rather than attempting to arbitrarily classify each watershed into discrete classes and compare means among classes recognizes that every watershed had more than one geomorphic class as well as different spatial arrangements of classes within its boundaries. We used two complementary statistical approaches for modeling relationships between wetland geomorphic class percentages and major drainage basins (predictors) and riparian and in-stream response variables: multiple linear regression and Regression Tree analysis (RTA). Summaries of data from individual streams served as observations in the analysis (n=30); i.e., the analysis was not pseudoreplicated in any way.

The use of multiple regression depended largely on the distribution of the data and whether the data suitably met the parametric assumptions of the analysis. Skewness and kurtosis of variables were assessed prior to analysis. Appropriate transformations (e.g., log10) were applied to predictors when necessary to help data meet assumptions. Correlations among variables were first evaluated to determine collinearity and spatial dependency problems. Regression models were constructed following the watershed analysis framework outlined in King et al. (2005).

RTA was used as an alternative modeling approach to multiple regression for variables that did not meet assumptions of multiple regression, or for other variables that are more successfully represented using a tree-based underlying response model. RTA explains variation of a single response variable using one or more predictor variables. Response variables were continuous (regression tree) in this study. RTA works by recursively partitioning data into two mutually exclusive groups by selecting a predictor variable that best explains variation in the response variable (i.e., greatest reduction in deviance). The process is repeated until the tree can no longer be grown based on a set of stopping rules and cross-validation of the model. The method works particularly well when critical levels of a predictor result in an acute, threshold change in the mean and variance of the response. The objective of the method is to explain as much variation (r^2) in the response variable as possible while minimizing the size of the tree. This is analogous to incorporating explanatory variables into multiple regression. However, RTA has been shown to have numerous advantages over linear regression approaches, including (1) making very few assumptions about the properties of the data, (2) the detection of change in both the mean and variance of response variables, and (3) superior ability to reveal higher-order interactions (De' Ath and Fabricius 2000; Olden and Jackson 2002, Urban 2002).

We required that terminal nodes (leaves) and branches of each RTA model had no fewer than 5 and 10 observations, respectively. We cross-validated each tree model to determine the most appropriate size of the

tree (i.e., number of explanatory variables included in the model). Cross-validation was conducted by randomly partitioning the data into 10 groups of equal or similar size and creating a cross-validation regression tree with only nine of the 10 groups. This cross-validation tree was subsequently used to predict response-variable data from each of the stations remaining in the tenth group. The process was repeated 10 times so that each of the 10 groups of sites was used as the cross-validation group once. We retained predictors that resulted in an average overall model r^2 within one standard error of the minimum model r^2 among all possible trees (1 SE rule; Breiman et al. 1984, De'Ath and Fabricius 2000).

Response variables in the multiple regression or RTA models included:

- total juvenile salmonid density
- density of Dolly Varden <10 cm
- density of Dolly Varden >10 cm
- density of coho <10 cm
- density of coho >10 cm
- macroinvertebrate density
- macroinvertebrate biomass
- macroinvertebrate community dissimilarity
- macroinvertebrate taxonomic richness
- number of Ephemeroptera, Plecoptera, and Trichoptera (EPT) taxa
- % macroinvertebrate functional feeding groups
- water chemistry variables
- large woody debris
- instream habitat variables
- riparian zone vegetation variables

For some sets of correlated ecological indicators, principal components analysis (PCA) or nonmetric multidimensional scaling (nMDS) was used to extract axis scores that could be used as a response in the multiple regression or RTA analysis. PCA was limited to sets of variables such as physical habitat that were likely to meet the underlying linear response model required by the technique (McCune and Grace 2002). PCA was used only to analyze abiotic variables (instream habitat, water chemistry) to determine if wetland geomorphic class percentages are good predictors of instream habitat or water chemistry PCs. NMDS was used to ordinate macroinvertebrate taxonomic composition using Bray-Curtis dissimilarity following methods described in McCune and Grace (2002). NMDS axis scores represented a gradient in taxonomic composition. This gradient was related to wetland geomorphic class percentages using multiple regression or RTA, as in the previous examples.

Regression models developed using data from this study were used to predict ecological support functions for all 1-km stream reaches in the study area. Landscape and reach scale variables found to be useful predictors for each ecological support function were plugged into specific regression equations to predict ecological support functions based on each variable considered. The predicted associations were subsequently mapped over the entire study area across all 1st order stream reaches.

Results

Our pre-field efforts to screen potential sites for access proved successful for thirty of the forty sites. We were unable to sample ten sites for a range of reasons, including the decommissioning of logging bridges and roads that appeared to be present on aerial imagery; the absence of a stream at the designated site; and the presence of a brown bear cache in the vicinity of one site that prevented sampling as a safety precaution. Many of the sites that were successfully sampled required extraordinary efforts to access, resulting in very long field days; including bushwhacking long distances through dense willow and alder and downed dead spruce deposited by the recent spruce bark beetle outbreak (Figure 8). The 30 streams that were sampled were stratified across the five major drainages and the four dominant wetland settings. The sites sampled ranged from higher elevation headwater streams in the eastern portions of the watersheds to lower elevation headwater streams in the western portion of the Kenai Lowlands. Headwater streams in the higher elevation eastern portion of the study area were typically flowing through a narrow riparian fringe, set into an incised valley; whereas headwater streams in the lower elevation western portion of the study area flowed through broad low gradient fen wetlands (Figure 9).



Figure 8. Access to headwater sites often required hiking long distances off trail through downed spruce and willow.

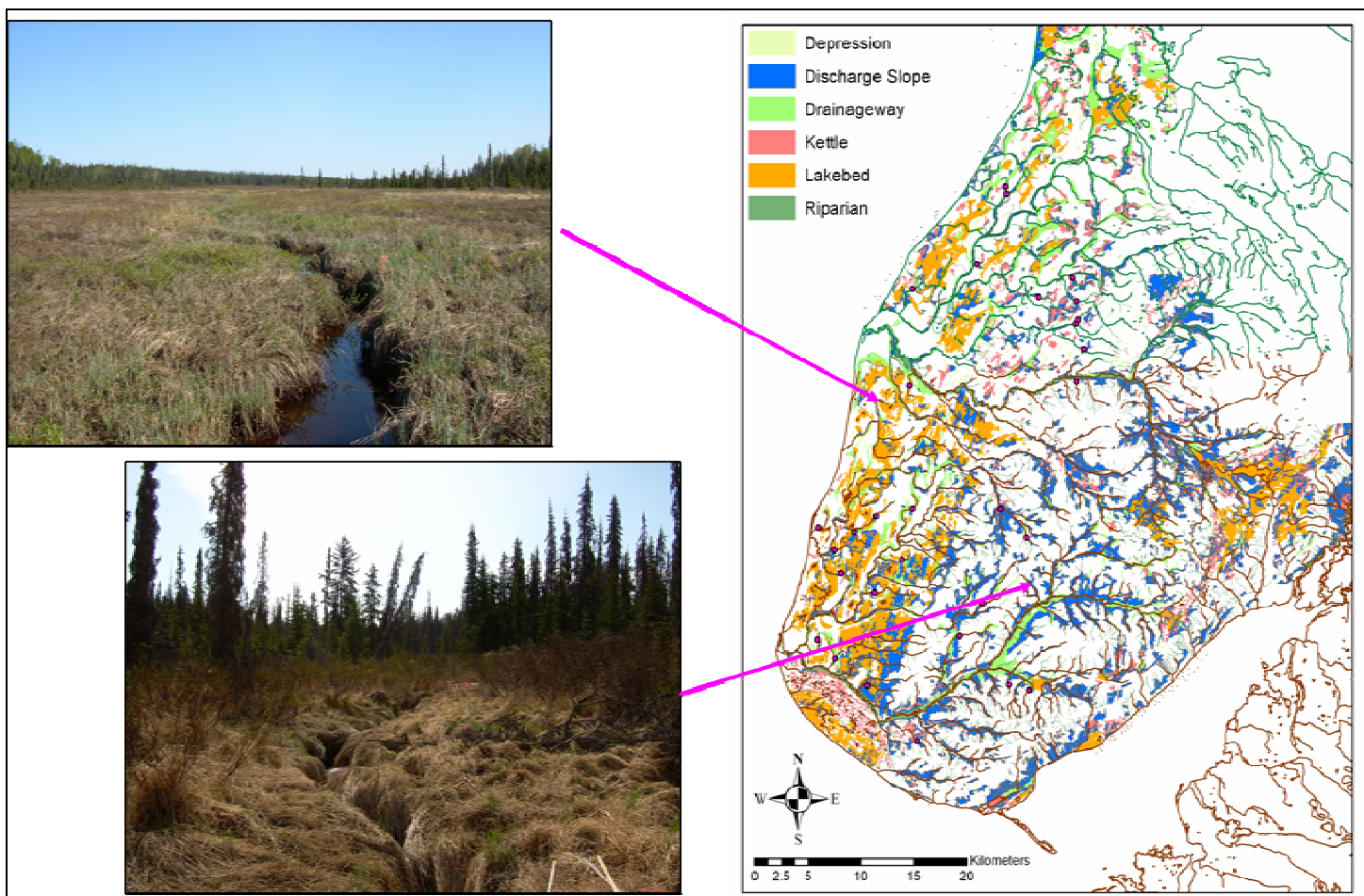


Figure 9. Thirty headwater stream sites in the Kenai Lowlands study area were sampled. The sampled streams spanned an elevation gradient, with the highest sites located in the east and lower elevation sites in the western portion of the study area. Headwater streams in the eastern portion of the study area were associated with narrow riparian fringe wetlands, whereas headwater streams in the lower elevation western portion of the study area were associated with broad fens.

Fish, primarily juvenile Dolly Varden and coho, were captured in nearly every stream sampled, although densities of fish captured varied greatly. Table 1 shows total fish by species captured in headwater streams of each of the five main watersheds (note that sculpins and sticklebacks are only accounted for in the “Total Fish” column in the top table. On the far right of the top table is the fish per site (150 meters fished at each site) for coho, dollies, and total salmonids.

Table 1. Total fish by species sampled in each of the major drainage basins. Coho = *Oncorynchus kisutch*; DV = *Salvelinus malma*; Steelhead = *Oncorhynchus mykiss*; kings = *Oncorynchus tshawytscha*.

watershed	# sites	total fish	total coho	total DV	total steel head	total kings	total salmonids	coho/ 150m	DV/ 150m	salmonids /150m
Deep	5	109	22	84	3	0	109	4.4	16.8	21.8
Ninilchik	7	267	59	128	1	0	188	8.4	18.3	26.9
Stariski	7	386	146	240	0	0	386	20.9	34.3	55.1
N. Fork Anchor	5	386	58	231	1	0	290	11.6	46.2	58
S. Fork Anchor	6	653	95	390	164	4	653	15.8	65	108.8
Total	30	1801	380	1073	169	4	1626	12.7	35.8	54.2

Eighty-five different stream macroinvertebrate taxon were identified in the samples (Appendix A-1). This is the first thorough compilation of stream macroinvertebrates in headwaters of the Kenai Lowlands.

Landscape metrics for the drainage area upstream of each sample site were calculated for each of the thirty sites sampled using GIS, resulting in 37 landscape scale variables (Table 2 and Appendix 2), and Principal Component Analyses (PCAs) were employed to explore the relationship between bundles of variables that are strongly correlated (share the same variance among sample units). In one PCA, we analyzed correlations between the 37 landscape metrics and wetland class; in another PCA, we looked at correlations between landscape scale metrics and 67 reach scale abiotic and habitat metrics.

The landscape-wetland class PCA detected large differences in watershed and wetland characteristics among sites across the 5 basins. The great variability in watershed and wetland characteristics in sites across all drainage basins, suggests that drainage basins do not have unique characteristics that drive landscape patterns, but rather that similar patterns occur across all drainage basins on the Kenai Lowlands (Figures 10 and 11). The first axis (PC1) is best described as a topographic gradient that also influenced the distribution of certain wetland classes. Positive scores on PC1 corresponded to sites with high watershed slopes, high elevations, far from salt water, eastern, and moderately high percentages of riparian wetlands whereas negative scores were sites with lower elevations, high percent wetlands, and high percentage of mixed lakebed/kettle/drainageway (LB/K/DW) wetlands (Figures 12 and 13). The second PC axis (PC2) represented a gradient in wetland classes. Positive scores on PC2 corresponded to sites with high percentage of drainageway wetland, whereas negative scores indicated sites with high percentage of discharge slope and, to a lesser degree, riparian wetlands. The heavy loading of discharge slope wetlands on PC2 rather than PC1 suggests discharge slope wetlands were not strongly associated with the topographic/elevation gradient that ran from the interior of the peninsula towards the coast, but rather occurred sporadically all along the topographic gradient. See Table 3 for an explanation of landscape metrics that were important in explaining variance among sites.

Table 2. A partial listing of landscape metrics included in the landscape scale-wetland class PCA. 37 landscape metrics were calculated for each site. The metrics included in this table were the ones found to be most significant in explaining variance among sites. For a complete list of metrics calculated, see Appendix 2.

Landscape metric	Metric Calculation
TopoInd (topographic index)	An index of potential wetness for a cell (point) based on upslope area and slope, also referred to as “wetness index”. It’s a measure of how wet a point should be based just on upstream drainage area and slope. Calculated automatically from a DEM using an ArcView extension.
IDW (inverse distance weighted to wetland classes)	A calculation of the amount of each wetland type (and uplands) in a site’s watershed weighted by inverse distance to the site. In other words, discharge slopes (for instance) close to the site have a stronger influence on the site than those farther away.
ID2 (inverse distance weighted squared)	Same as IDW, but weighting by distance-squared, rather than just distance. This gives even higher weight to proximity of the habitat type to the site.
Salt_m (distance to salt water)	
RelRat (relief ratio)	The elevation range of the basin for each site divided by the length of the basin.
Wslope (watershed slope)	Mean percent slope of the watershed.
East_DS North_DS	Downstream GPS coordinates for sample reach.
Elev_DEM (elevation)	Elevation determined by digital elevation model.

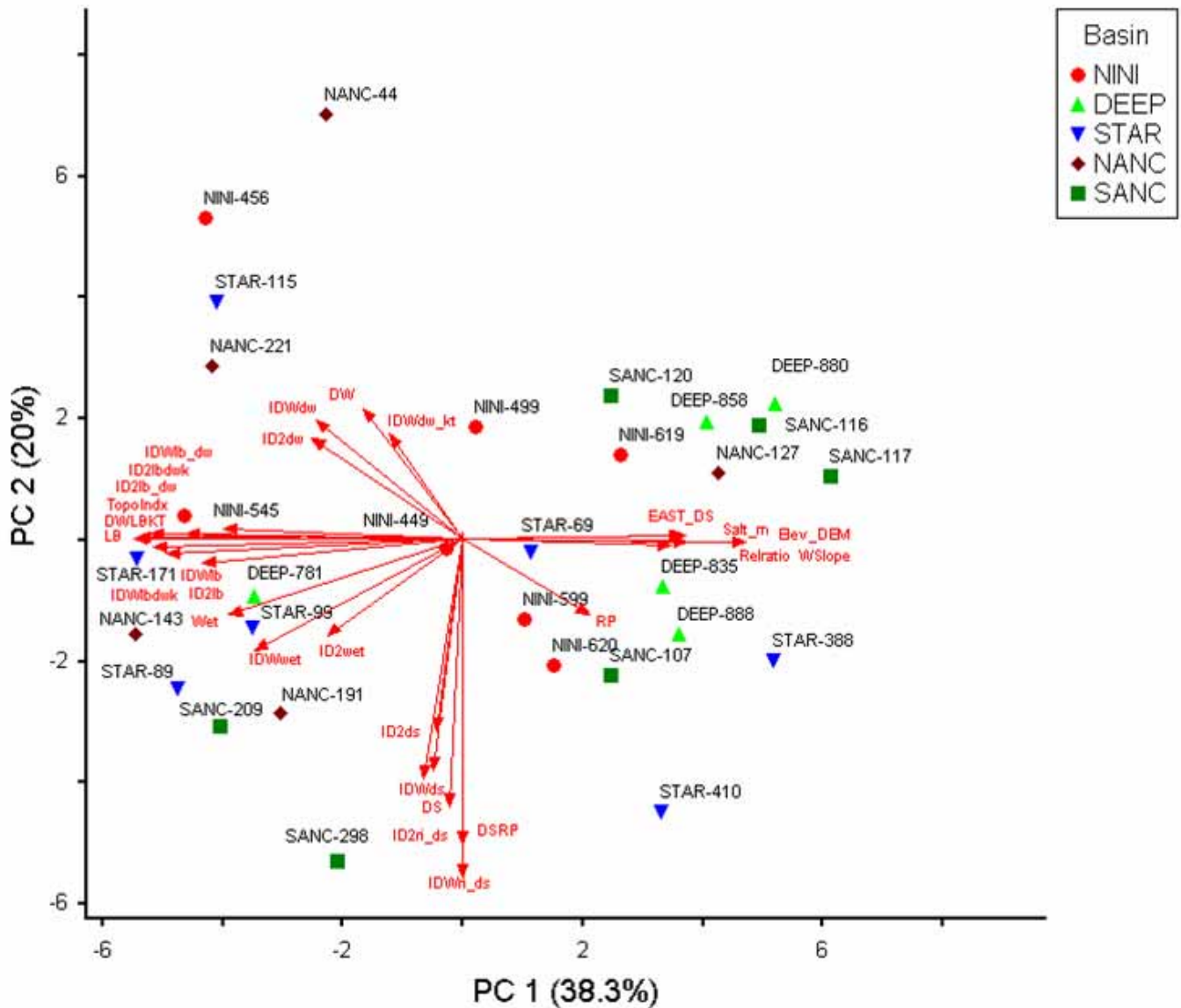


Figure 10. PCA ordination (cumulative variance=58.8% for first 2 PCs) of HWS sites based on 37 landscape variables associated with each contributing watershed. Symbols correspond to one of the five major drainage basins: NINI=Ninilchik; DEEP=Deep; STAR=Starisky; NANC=North Anchor; SANC=South Anchor. The first axis was strongly related to topography, with high gradient watersheds dominated by riparian forested wetlands and located in the interior of the peninsula located to the right, and low-gradient watersheds dominated by drainageway, lakebed, and kettle wetlands to the left. The second axis explained less variance and separated sites based on the amount of discharge slope and combined discharge slope-riparian wetland conditions from those with primarily drainageway wetlands in their watersheds. The even distribution of basin codes throughout the ordination shows that basins were not governing the distribution of wetland classes, nor did the range of topographic relief vary among the major basins.

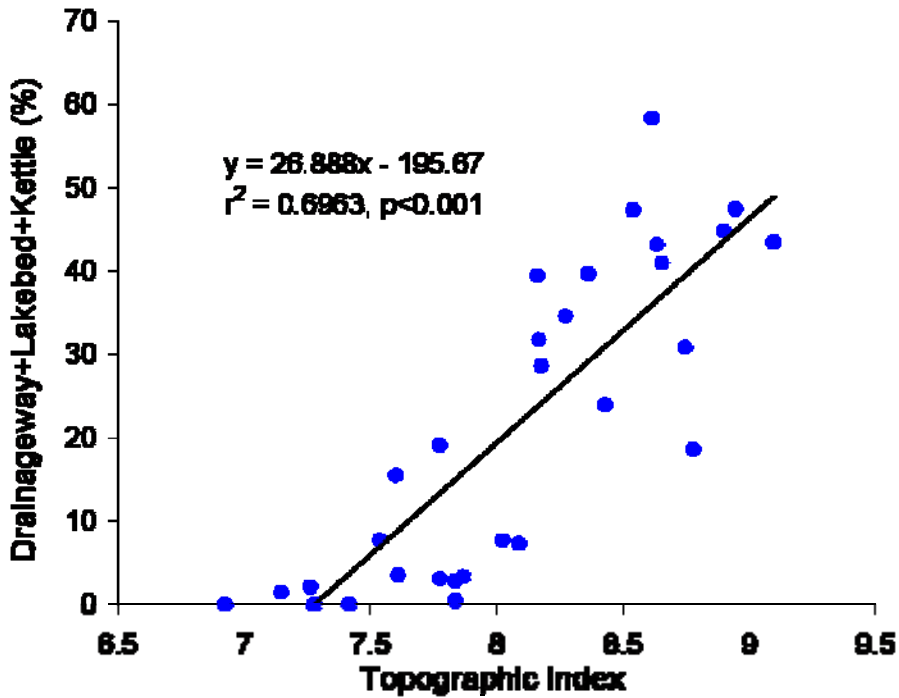


Figure 12. Regression model showing the strong relationship between drainageway, lakebed and kettle (DWLBKT) wetland classes and topographic index; lower topographic index sites (higher gradient) had low percentages of DWLBKT; whereas higher topographic index (lower gradient sites) exhibited high amounts of DWLBKT wetlands.

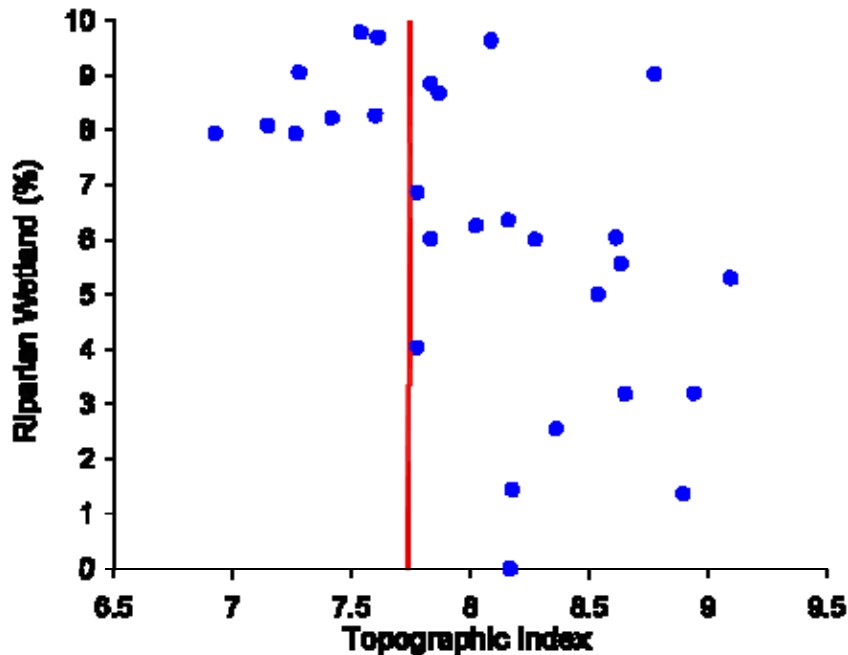


Figure 13. Riparian wetlands were present at all sites. At low topographic index values, all the sites had similarly high amounts of riparian wetland; whereas sites with higher topographic index values had more variable amounts of riparian wetlands. The vertical red line indicates the threshold.

The landscape-wetland class PCA provided the framework for a second PCA, exploring correlations between landscape metrics, wetland setting, and reach-scale metrics. Sixty-seven reach-scale instream abiotic and habitat characteristics were analyzed in the second PCA (see Appendix 2 for a complete list). This PCA showed strong physical habitat gradients corresponding to the topographic gradient; and strong water-chemistry patterns, most notably dissolved oxygen, temperature and dissolved nitrogen (Figure 14). The first three PC axes of this PCA explained more variance than expected by chance ($p < 0.05$). The first PC explained 27% of the variance among the 67 variables. Numerous water chemistry and habitat variables corresponded to this axis. Sites with positive scores on PC1 had high dissolved oxygen, slightly alkaline pH, elevated inorganic nitrogen, especially nitrate-N, strongly undercut banks, high average velocity, heavy canopy cover, high amounts of instream woody debris, and mineral substrates dominated by sand, gravel, cobble, and boulders. Sites with negative scores on PC1 had higher water temperatures, deep channels with low water velocity, substantial overhanging herbaceous vegetation, and soft, peat substrates. The second PC axis explained 10% of the variance and primarily represented a gradient in channel dimensions spanning narrow to wide channels (stream size). Positive scores on PC2 corresponded to sites with narrower wetted channels and bank widths, whereas negative scores indicated sites with wider channels and bank widths, somewhat lower pH, and higher levels of filamentous algae and leafpacks. Negative scores were associated with more variable bank heights, woody debris, and non-peat fine sediment substrate.

Table 3. A partial listing of metrics included in the reach scale-wetland class PCA. 67 landscape metrics were calculated for each site. The metrics included in this table were the ones found to be most significant in explaining variance among sites. For a complete list of metrics calculated, see Appendix 2.

Landscape metric	Metric Calculation
DO-PCT	Percent dissolved oxygen.
LWD-CT	Large woody debris count.
CB	Cobble substrate.
GC	Coarse gravel substrate.
SUcut/ Ucut	Undercut banks. S indicates standard deviation.
SA	Sandy sustrate.
Den	Densiometer reading of tree canopy cover
Thal-m	Thalweg depth in meters.
Pct_fn	Fine particles – indicates peat substrate,
Bhabove	Overhanging herbaceous vegetation

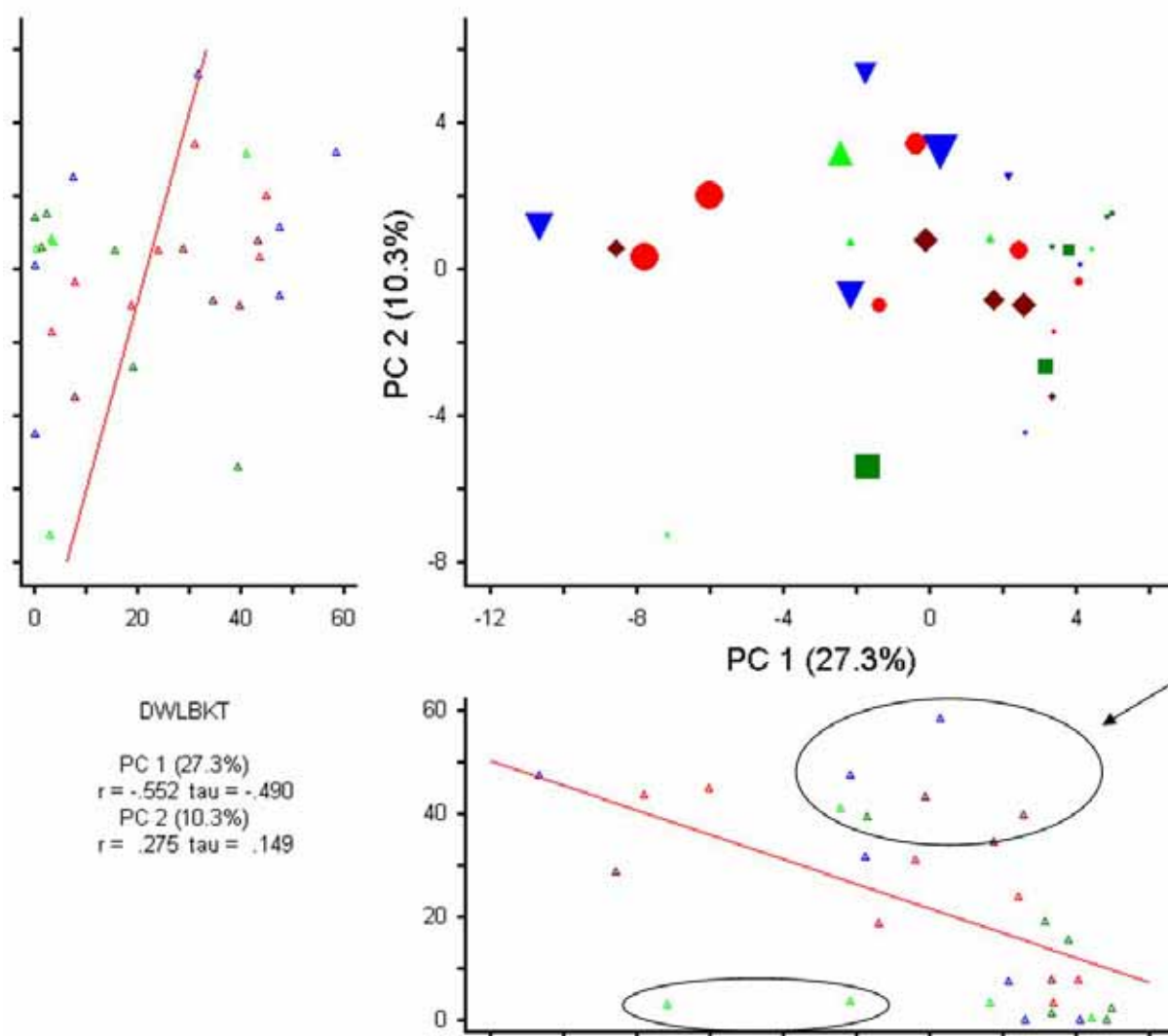


Figure 16. Quantitative overlay of the combined percent cover of drainageway, lakebed, and kettle wetlands (DWLBKT) onto the reach-scale water chemistry and habitat PCA diagram shown in Figure 13. These were the most common wetland classes at lower topographic relief and thus showed a similar correlation pattern to reach-scale water chemistry and instream habitat as topographic gradient. However, some sites at low topographic relief had low amounts of these wetlands, yet exhibited similar water chemistry and habitat as other low gradient streams. Similarly, a few sites with intermediate topographic relief had high amounts of these wetlands yet looked different in terms of habitat than lower-gradient sites with similar high-levels of DWLBKT. Thus, although these wetland classes showed similar correlations to habitat as topography, it appears that topography is a better predictor of instream conditions than these wetland classes.

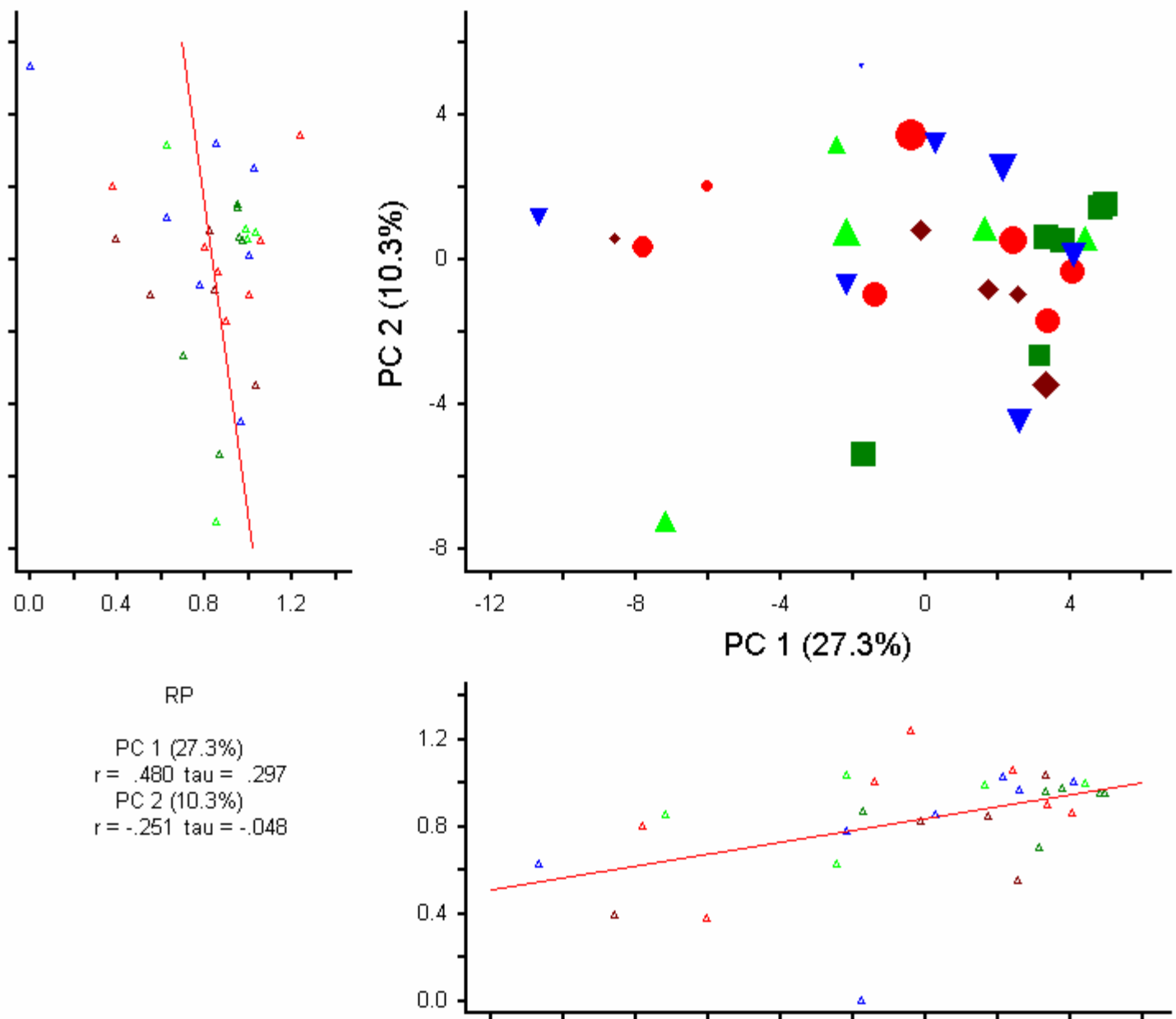


Figure 17. Quantitative overlay of log-transformed percent cover of riparian wetlands (RP) onto the reach-scale water chemistry and habitat PCA diagram shown in Figure 12. These were the most common wetland classes at higher topographic relief and thus showed a similar correlation pattern to reach-scale water chemistry and instream habitat as topographic gradient, but were not as predictive as topography.

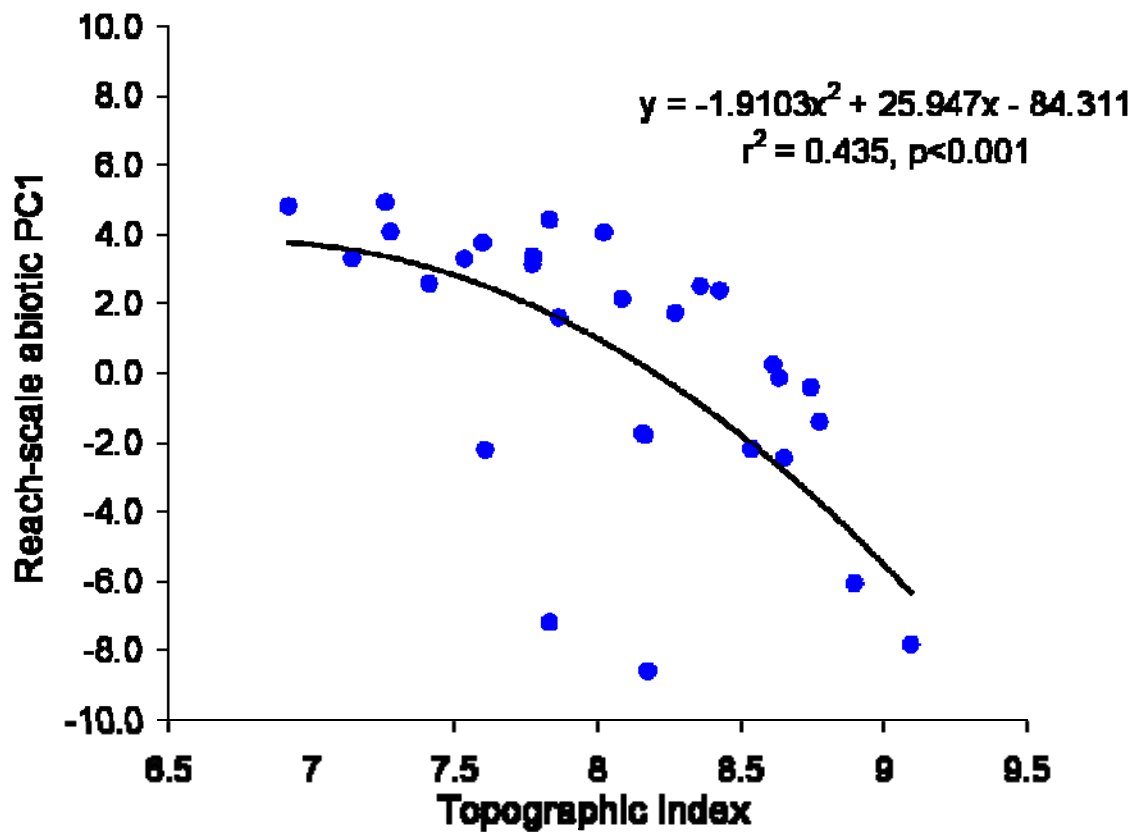


Figure 19. Reach-scale variables were strongly related to the topographic index and showed the steepest degree of change between intermediate and low gradient watersheds (moderate to high topographic index scores).

Thus, some or all reach-scale variables described by PC1 changed nonlinearly with increasing topographic index (Figures 20 through 30).

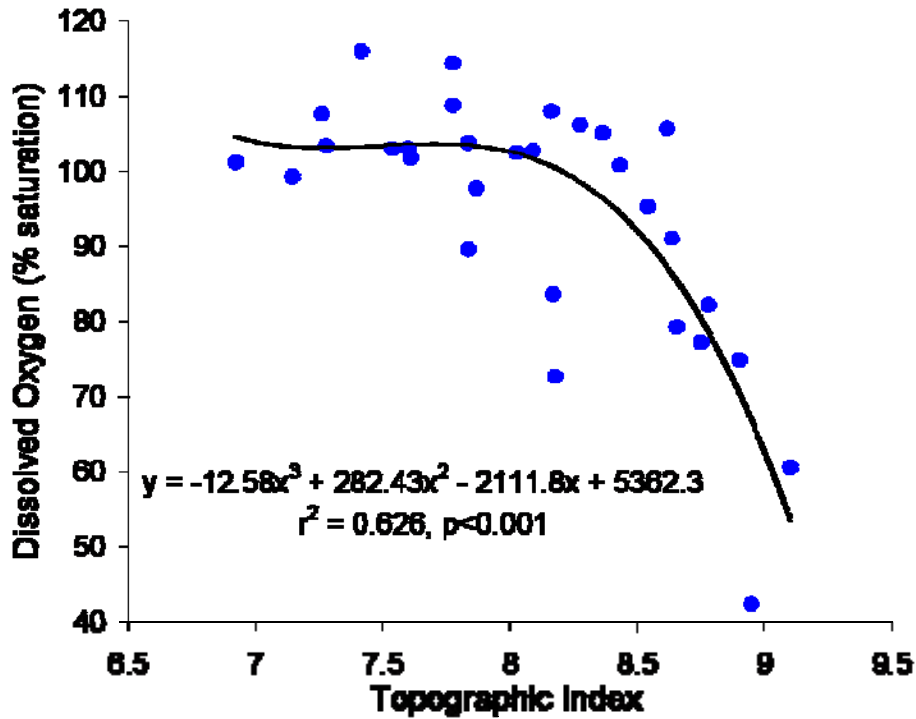


Figure 20. Polynomial regression fit between dissolved oxygen (% saturation) and topographic index. Dissolved oxygen showed precipitous declines in watersheds with topographic index scores > 8.

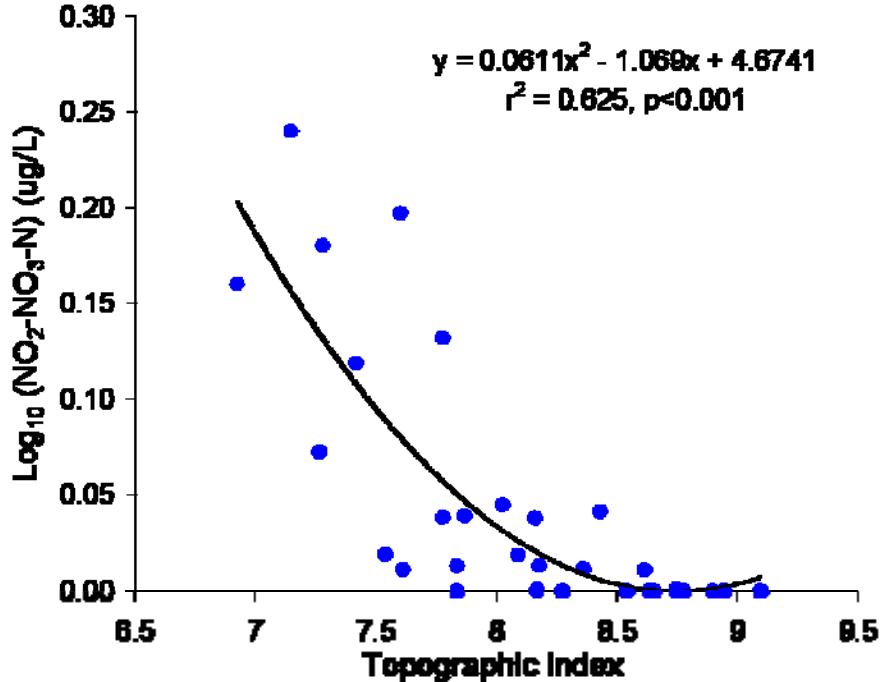


Figure 21. Relationship between nitrate-nitrite-N and topographic index. Nitrate-nitrite-N was highest at high watershed slopes and declined sharply between topographic index values of 7 and 8. Above 8.5, nitrate-nitrite-N trended toward zero.

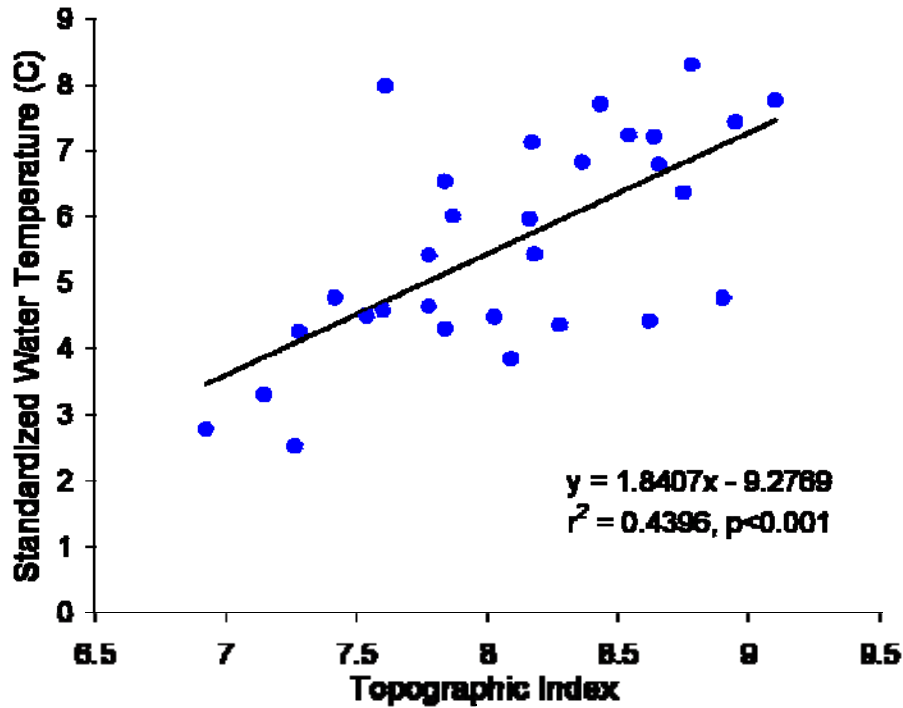


Figure 22. Relationship between topography and water temperature, normalized to the first day of the study period (mid-May). Water temperature linearly increased with increasing topographic index scores, being lowest in high-gradient (and higher elevation) watersheds.

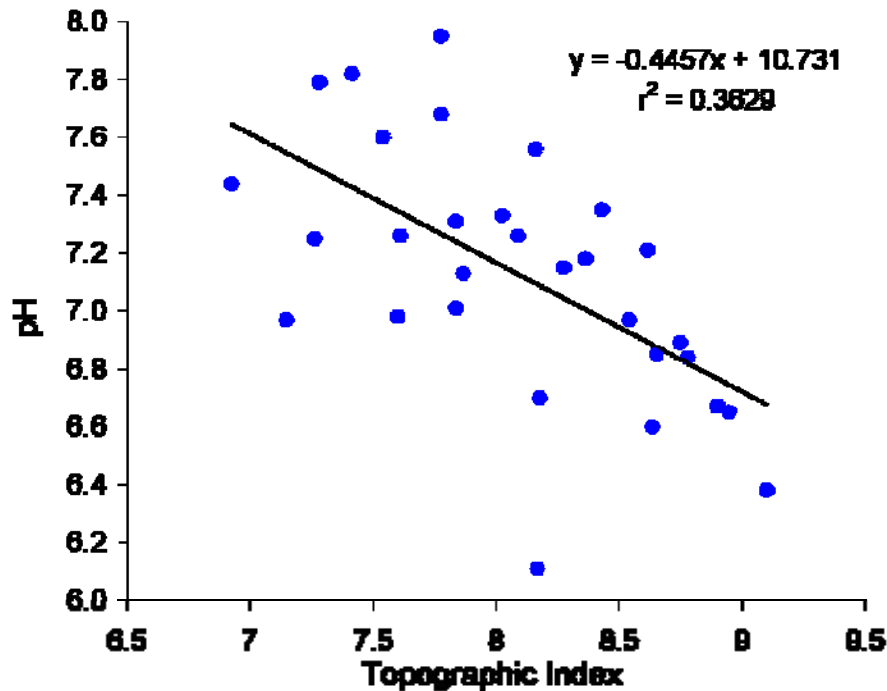


Figure 23. Relationship between pH and topography. Streams became more slightly acid in wetter, low gradient watersheds, suggesting a greater influence of organic acids on stream water chemistry.

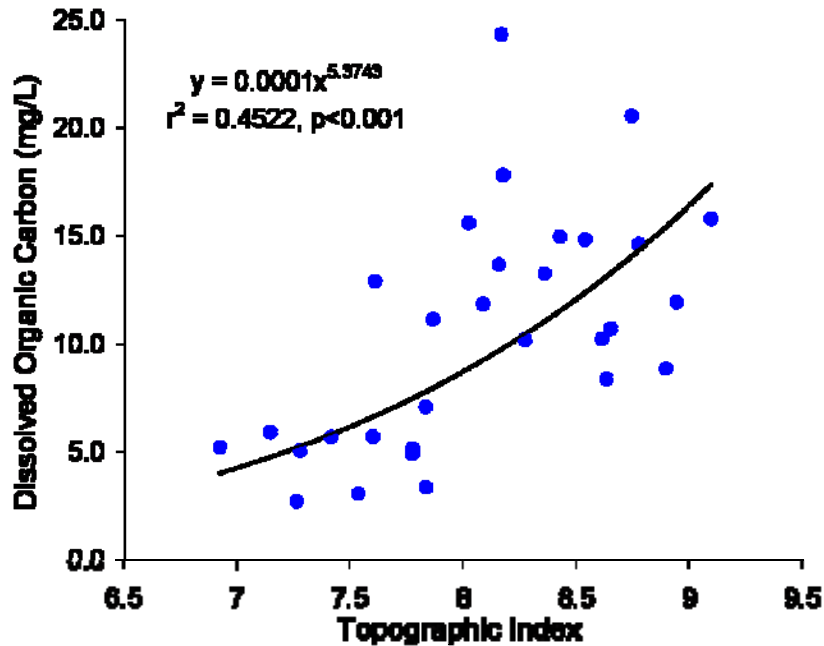


Figure 24. Regression of dissolved organic carbon (DOC) on topographic index scores. DOC ranged from 2-6 mg/L at topographic index scores between 7 and 8, but sharply increased above scores of 8. This is consistent with the patterns of DO, temperature, and pH, suggesting more interaction between wetland organic matter and shallow groundwater in watersheds with lower topographic relief.

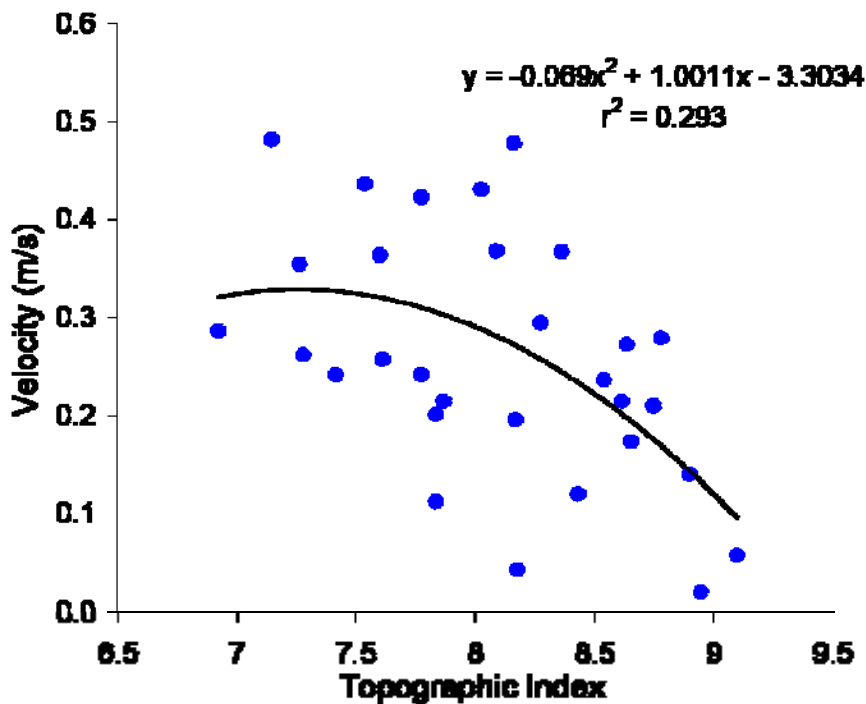


Figure 25. Mean stream velocity as a function of watershed topographic index. Not surprisingly, sites with higher topographic index scores tended to have slower-moving stream flows, but this relationship also showed considerable local variability that was not explained by watershed topography.

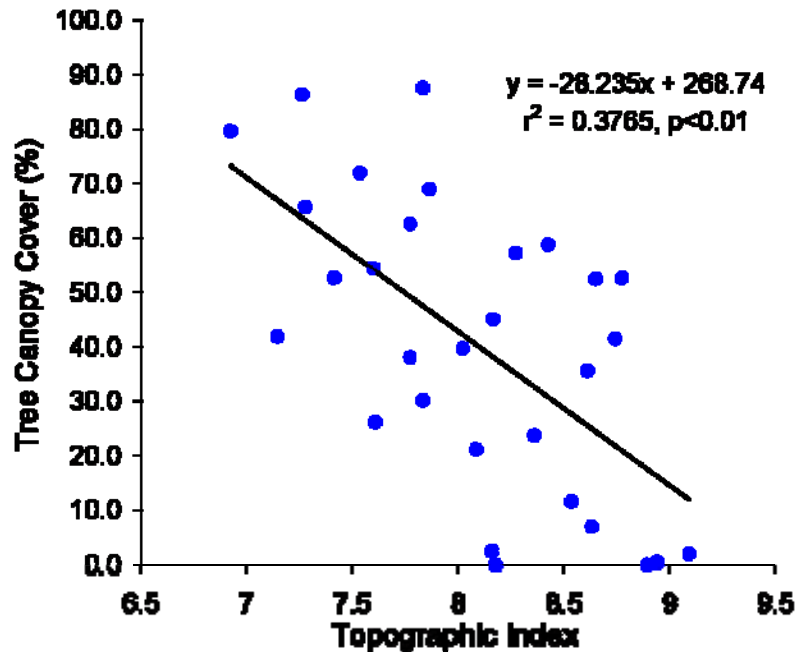


Figure 26. Aggregate tree canopy cover along each stream reach as a function of topography. Tree cover was much higher at higher watershed slopes, reflecting the narrowing bands of wetland vegetation around the streams. Tree cover was also strongly related to channel characteristics (undercut banks, mineral substrate, woody debris).

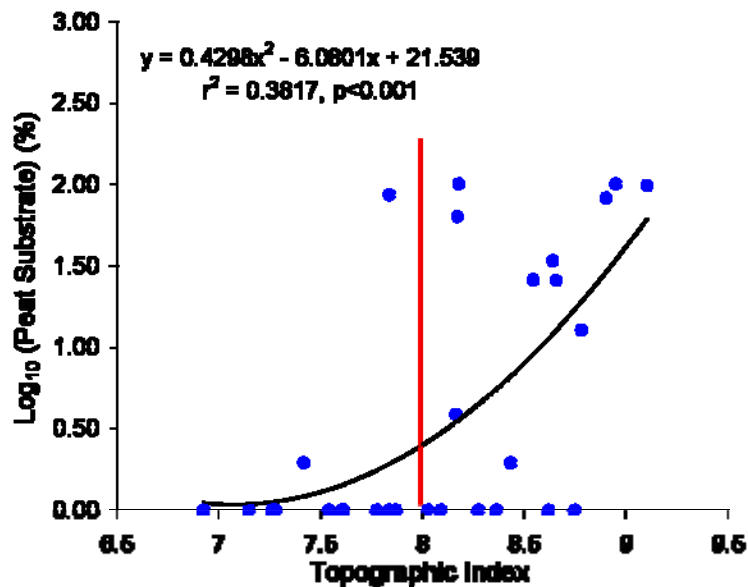


Figure 27. Proportion of stream substrate as fine organic matter (peat) along the topographic gradient. Peat dramatically increased in streams with topographic index values above 8, and was essentially absent below this score. This finding is highly consistent with the patterns of water chemistry (DO, pH, DOC) and shows that streams in most low gradient watersheds began to resemble depositional, wetland habitats compared to the erosional, mineral substrate streams at higher gradients. The red line indicates a threshold showing that at high topographic index values, all the sites had similarly high amounts of peat substrate. Below the line it is highly variable.

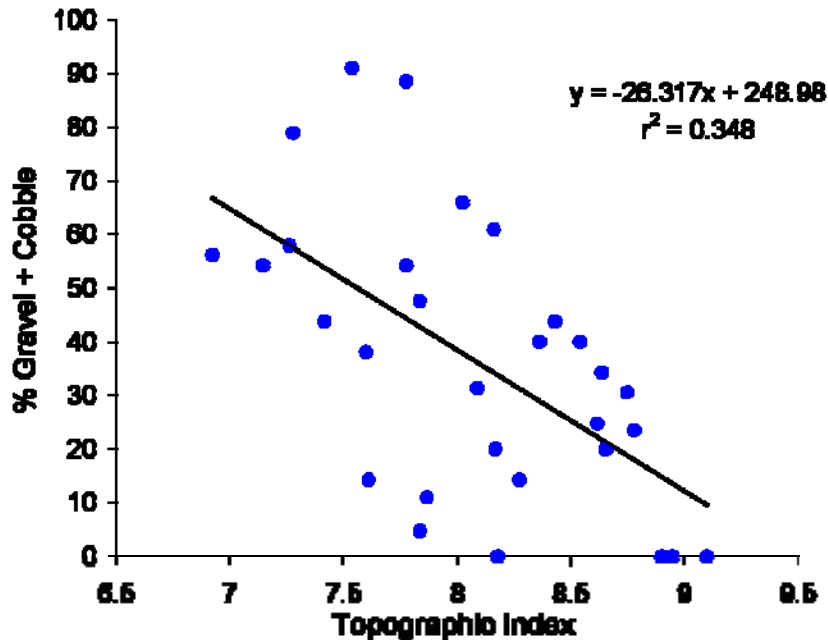


Figure 28. Distribution of gravel and cobble (small rock) substrate along the topographic gradient. Consistent with the previous graph showing an increase of organic sediments with increasing topographic index scores, % gravel and cobble were much higher at low index scores (at higher watershed slopes). This turns out to be a critical finding because of the requirement of fine gravel, erosional habitats for spawning salmonids and successful rearing of eggs.

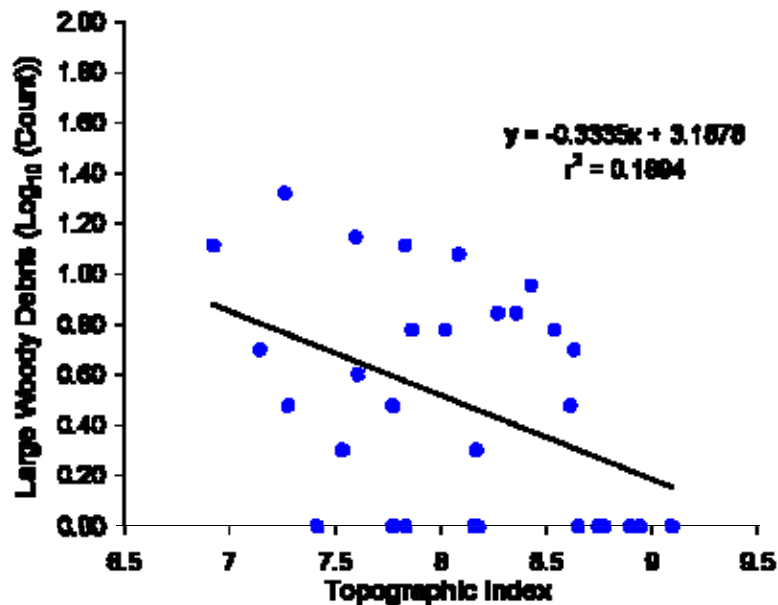


Figure 29. Distribution of large woody debris along the topographic gradient. Woody debris tended to be more abundant in higher gradient streams supporting more near-stream tree canopy, although the relationship was relatively weak. Note that streams with very high topographic index (>8.5) scores had little to no large woody debris.

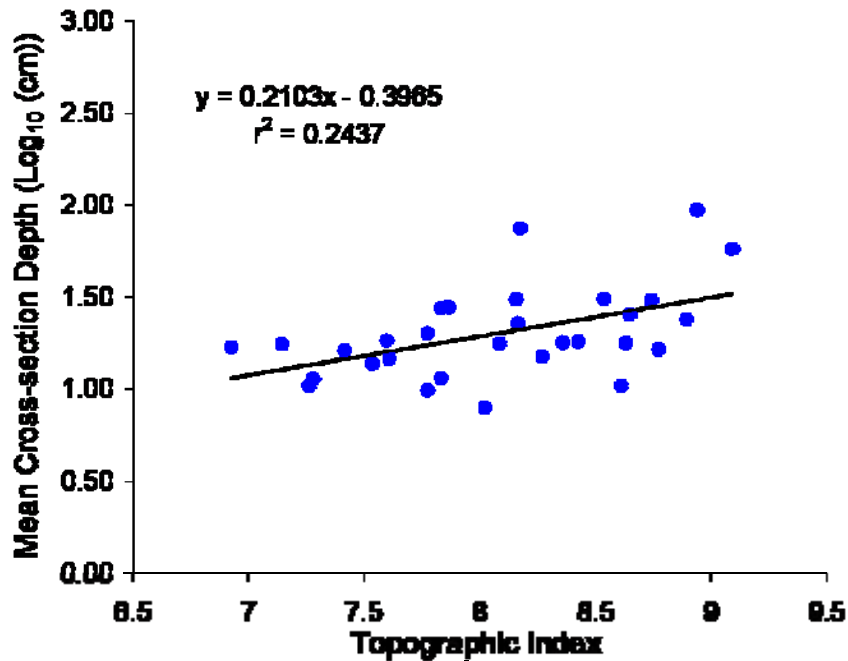
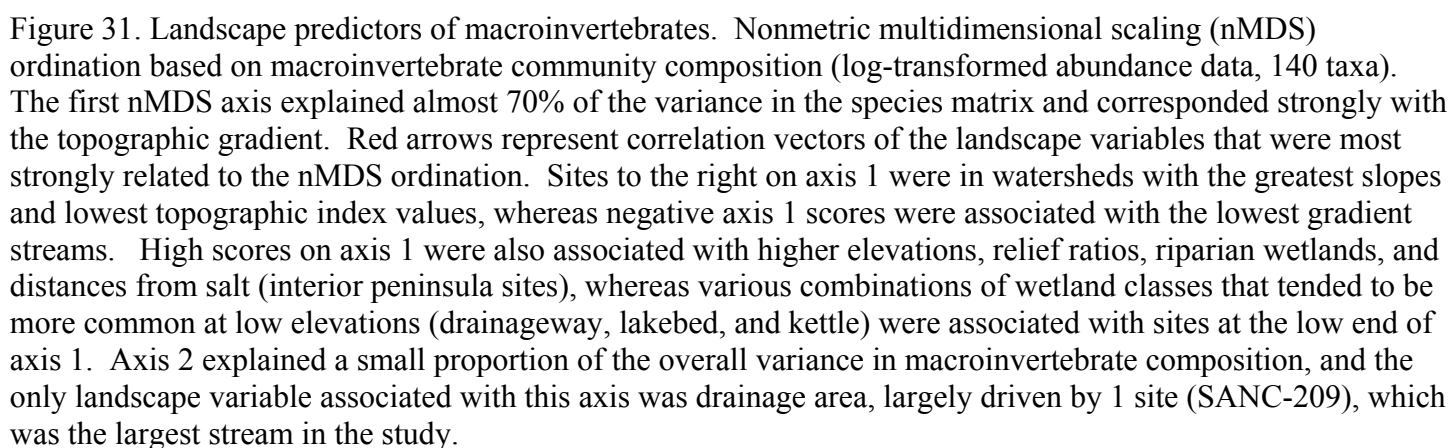


Figure 30. Mean stream depth along the topographic gradient. Streams were shallower (and faster) at low topographic index scores.

Nonmetric multidimensional scaling (nMDS) ordination was used to assess the relationship of macroinvertebrate assemblages to landscape and reach-scale variables associated with the 30 sampled stream sites (Figures 31 through 41). Regression tree and nonparametric changepoint analysis were used to assess fish distributions. The results show nonlinear distributions for small cohos (<10 cm), large cohos (>10 cm), small Dolly Varden (<12 cm) and large Dolly Varden (>12 cm), with small Dolly Varden predominant in the highest gradient streams, small coho predominate in intermediate gradient streams, and large coho predominate in the low gradient streams, while large Dolly Varden were widely scattered in streams of all gradients (Figures 42-51).



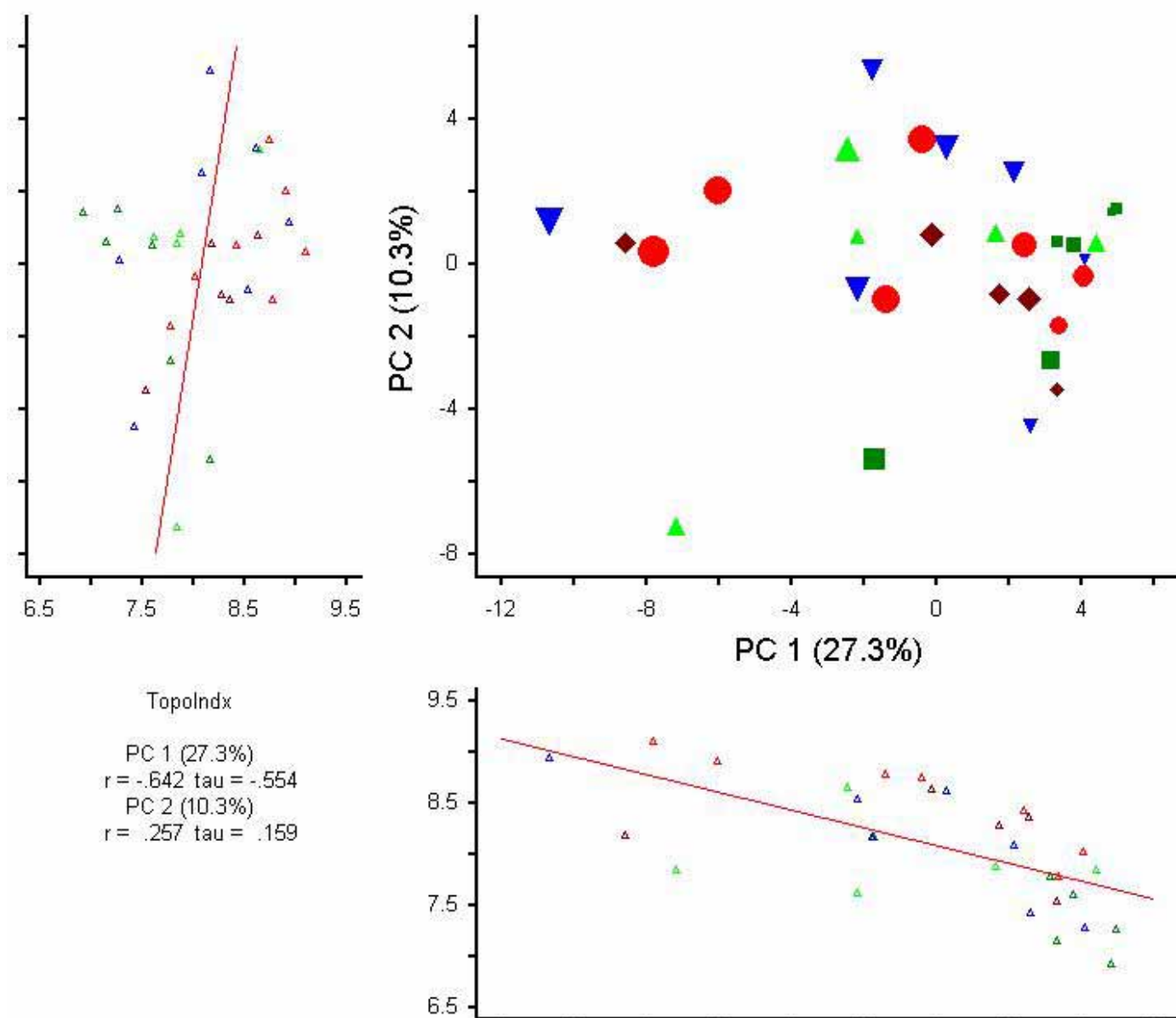


Figure 32. Quantitative overlay of topographic index scores onto the macroinvertebrate nMDS ordination, showing that sites with high topographic index scores were clearly separated to the left on axis 1. In other words, sites with similar topographic index scores tended to support similar macroinvertebrate communities, and these communities were consistently dissimilar from those at higher topographic index scores.

Figure 33. Reach-scale abiotic predictors of macroinvertebrates. Nonmetric multidimensional scaling (nMDS) ordination of the 30 stream sites based on macroinvertebrate community composition (see previous figures for details). Red arrows represent correlation vectors of the local, instream abiotic variables that were most strongly related to the nMDS ordination. Text codes refer to species centroids in the ordination space (see Appendix 2). Species locations correspond to their central tendency among sites, thus species at each end of the axes are most strongly related to environmental variables describing those axes, and species near the middle are either widespread or occur primarily at sites in the middle of the ordination diagram. Site IDs were excluded from this ordination. Macroinvertebrates from sites on the right side of axis 1 were associated with high dissolved oxygen, nitrate-nitrite-N, pH, velocity, sand, gravel, cobble, tree canopy, undercut banks, and large woody debris, whereas sites to the left were typified by high % peat, deeper channels, high embeddeness of substrates, overhanging herbaceous vegetation (mostly *Calamagrostis*), water temperature, and dissolved organic carbon. Sites on axis 2 were somewhat correlated with small woody debris (top half of ordination), undercut banks, and large woody debris. Sites on the bottom half of the ordination tended to have more filamentous algae, and were typically more open-canopied sites.

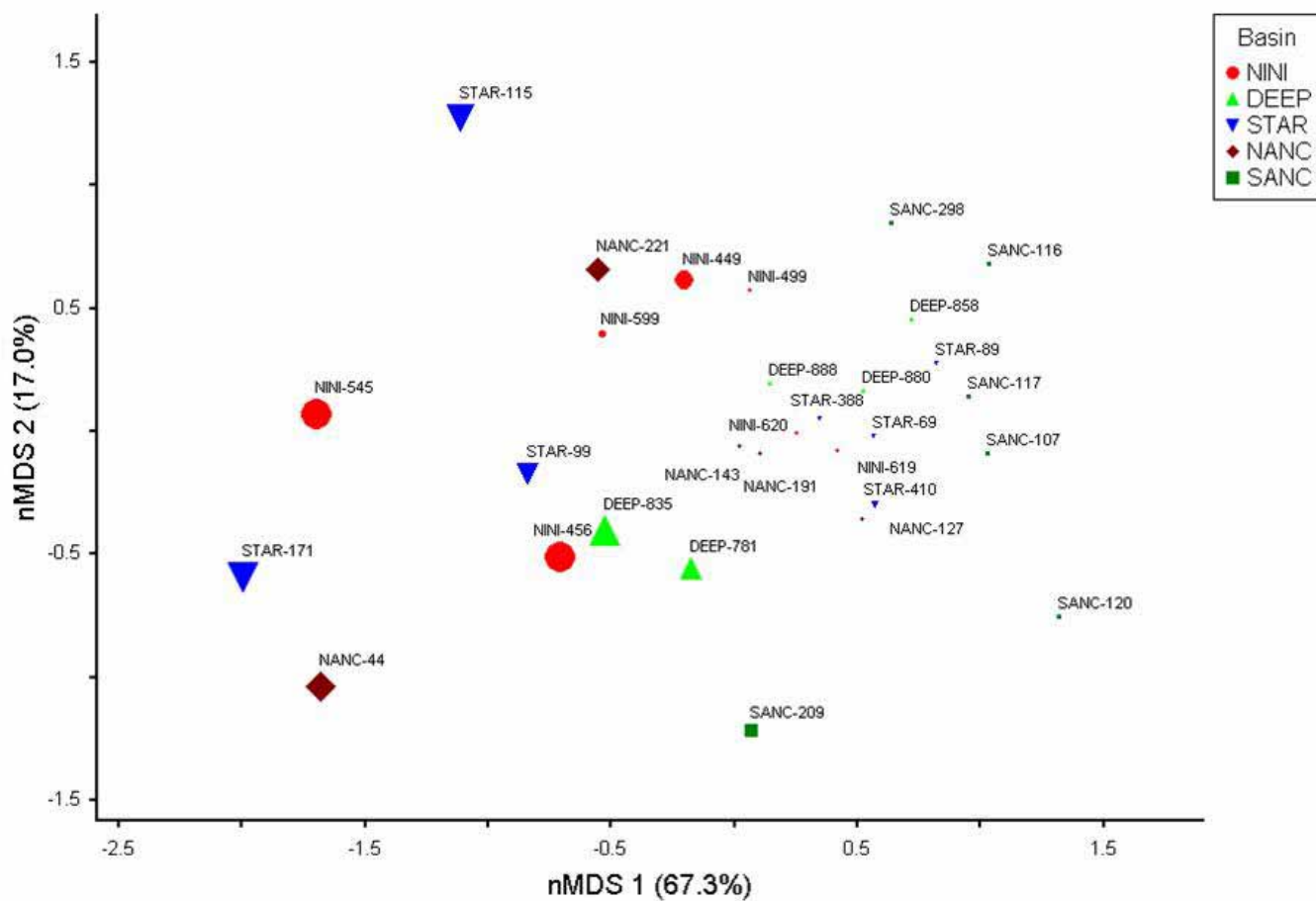


Figure 34. Quantitative overlay of % peat substrate onto the macroinvertebrate nMDS ordination, showing that sites with high amounts of peat were clearly separated to the left on axis 1, also corresponding to the high topographic index scores.

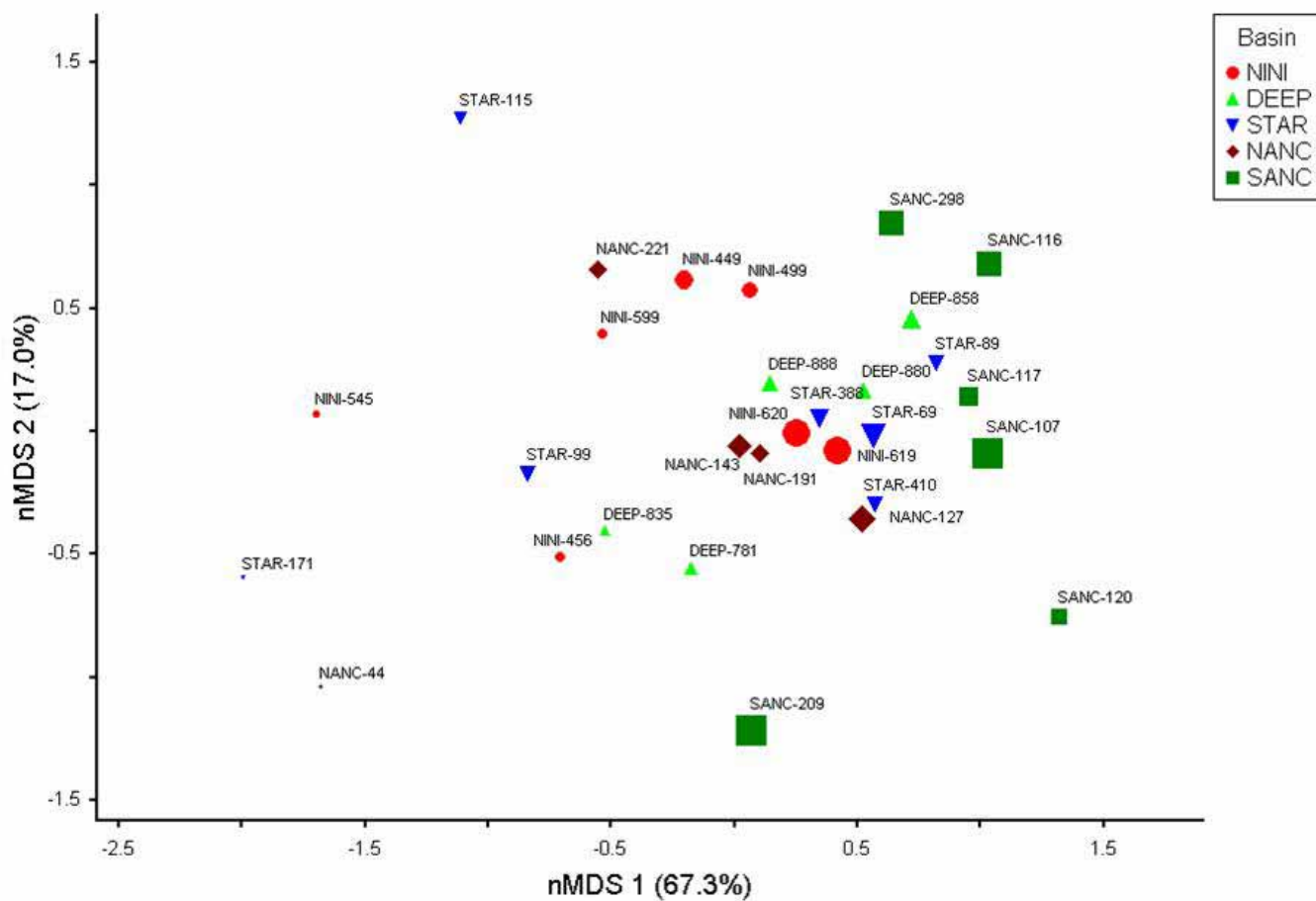


Figure 35. Quantitative overlay of stream velocity (m/s) onto the macroinvertebrate nMDS ordination.

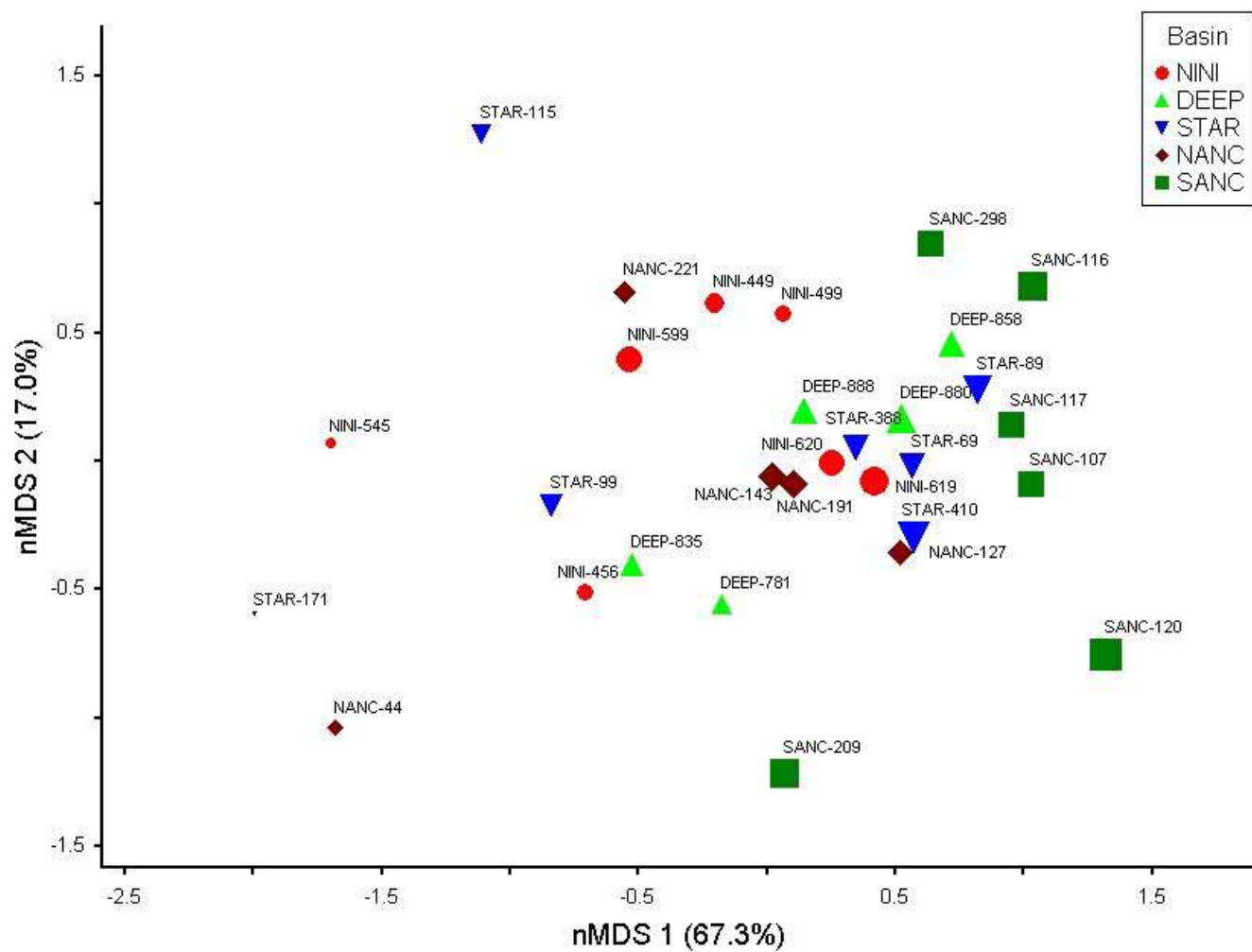


Figure 36. Quantitative overlay of dissolved oxygen onto the macroinvertebrate nMDS ordination.

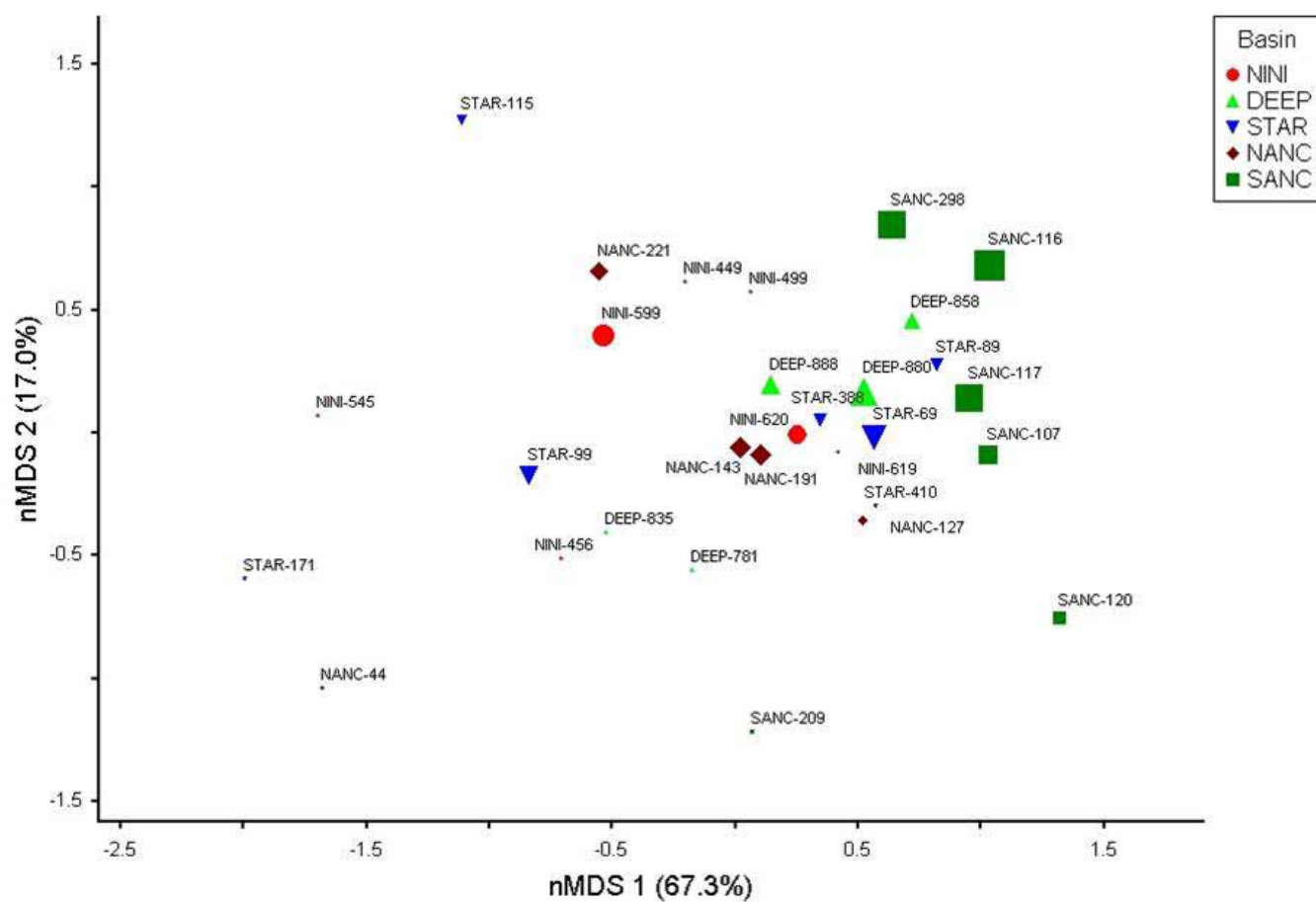


Figure 37. Quantitative overlay of large woody debris onto the macroinvertebrate nMDS ordination.

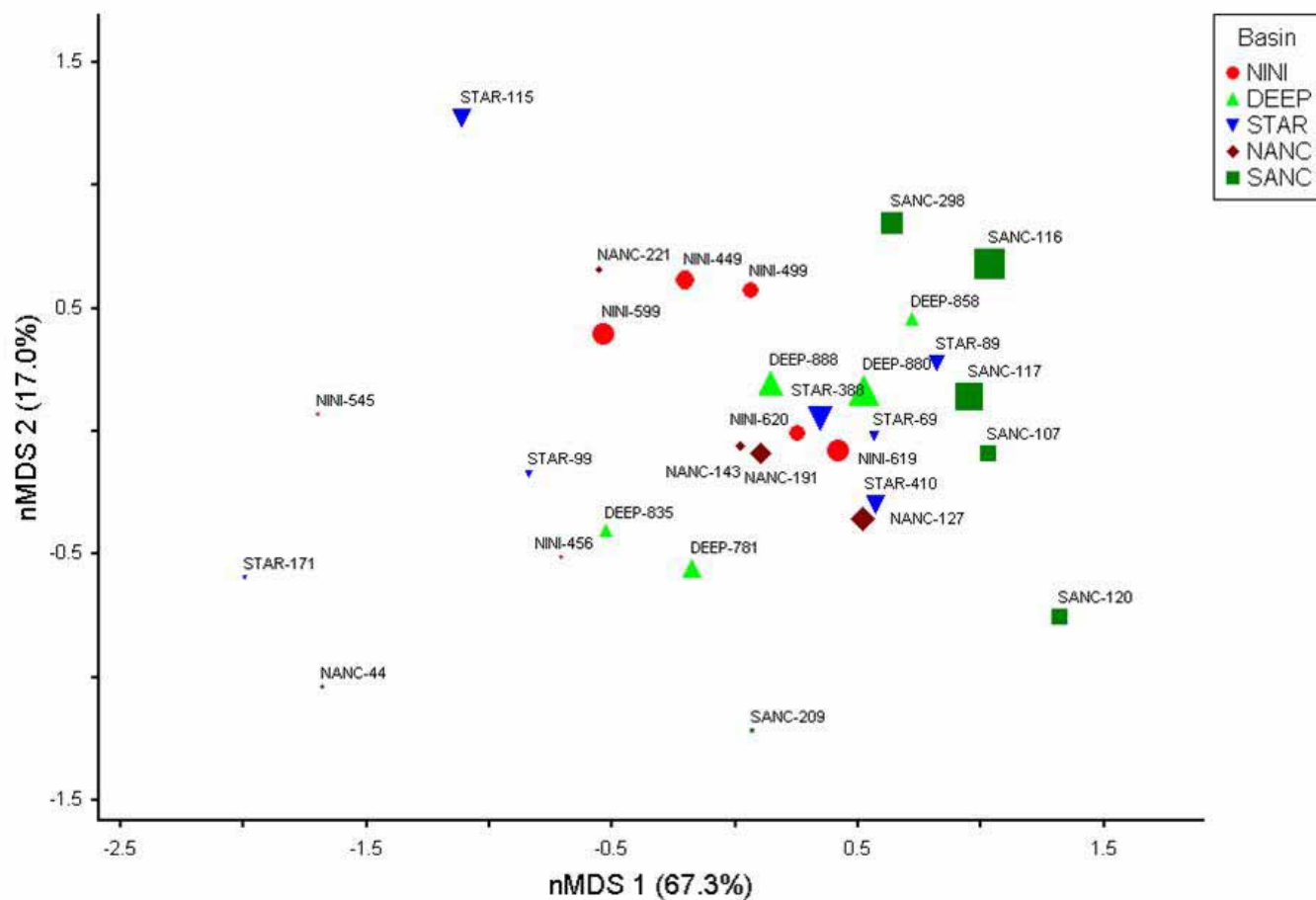


Figure 38. Quantitative overlay of % tree canopy cover onto the macroinvertebrate nMDS ordination.

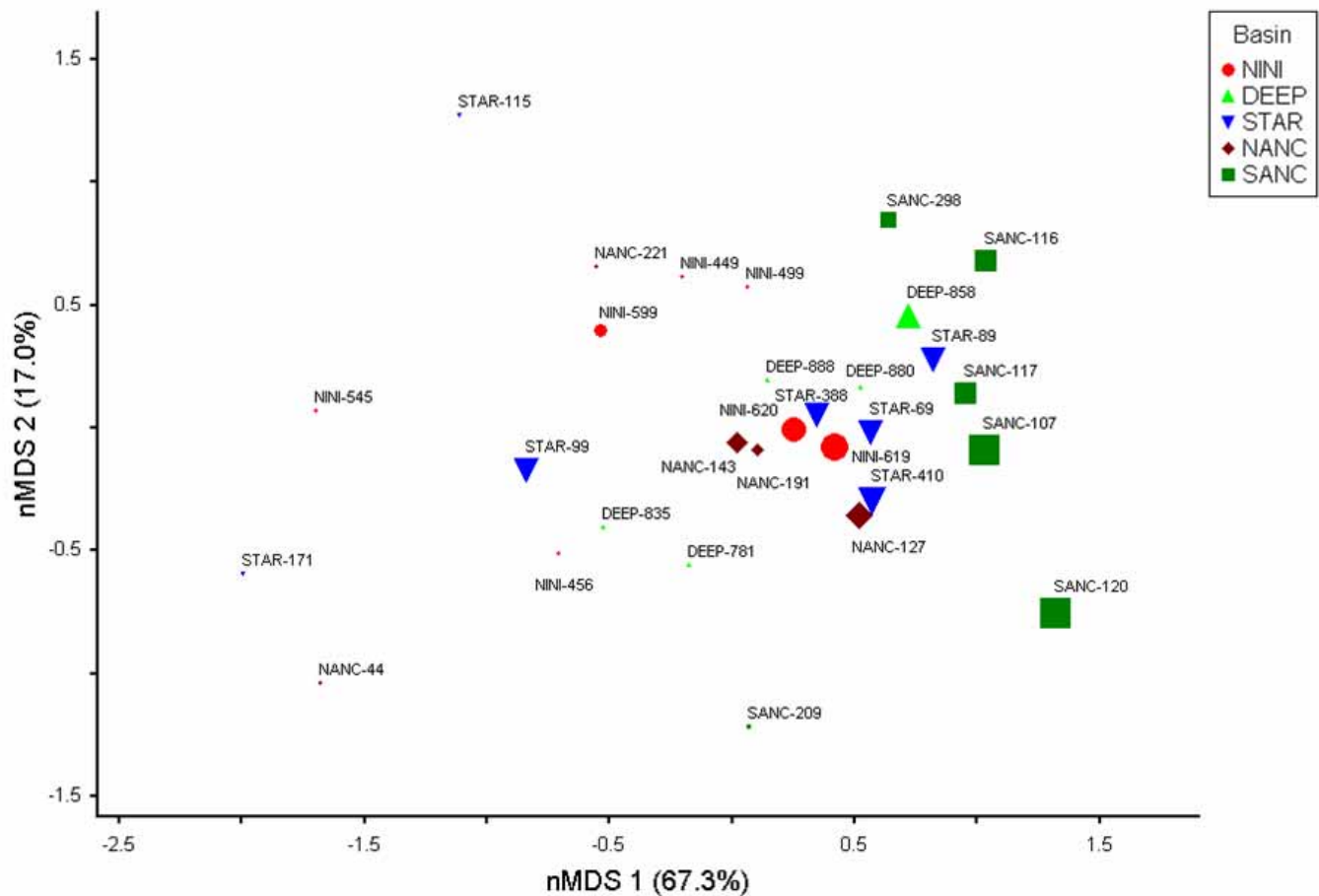


Figure 39. Quantitative overlay of small (<10 cm) Dolly Varden densities onto the macroinvertebrate nMDS ordination, illustrating these fish were associated with high gradient systems and corresponded well to macroinvertebrate community composition.

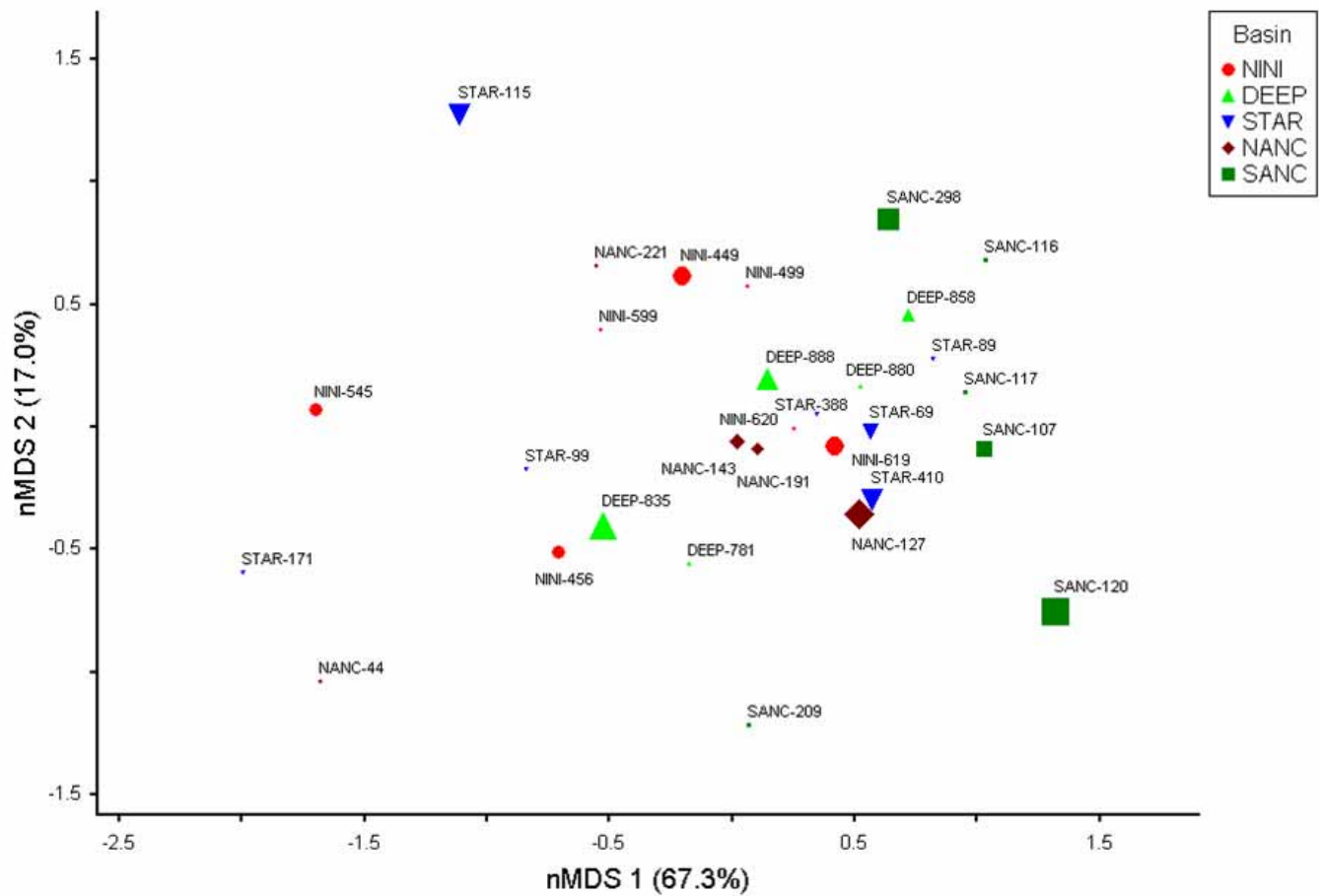


Figure 40. Quantitative overlay of large (>10 cm) Dolly Varden densities onto the macroinvertebrate nMDS ordination. Large Dolly Varden were widely distributed along the topographic gradient and did not appear to be associated with a particular type of macroinvertebrate community.

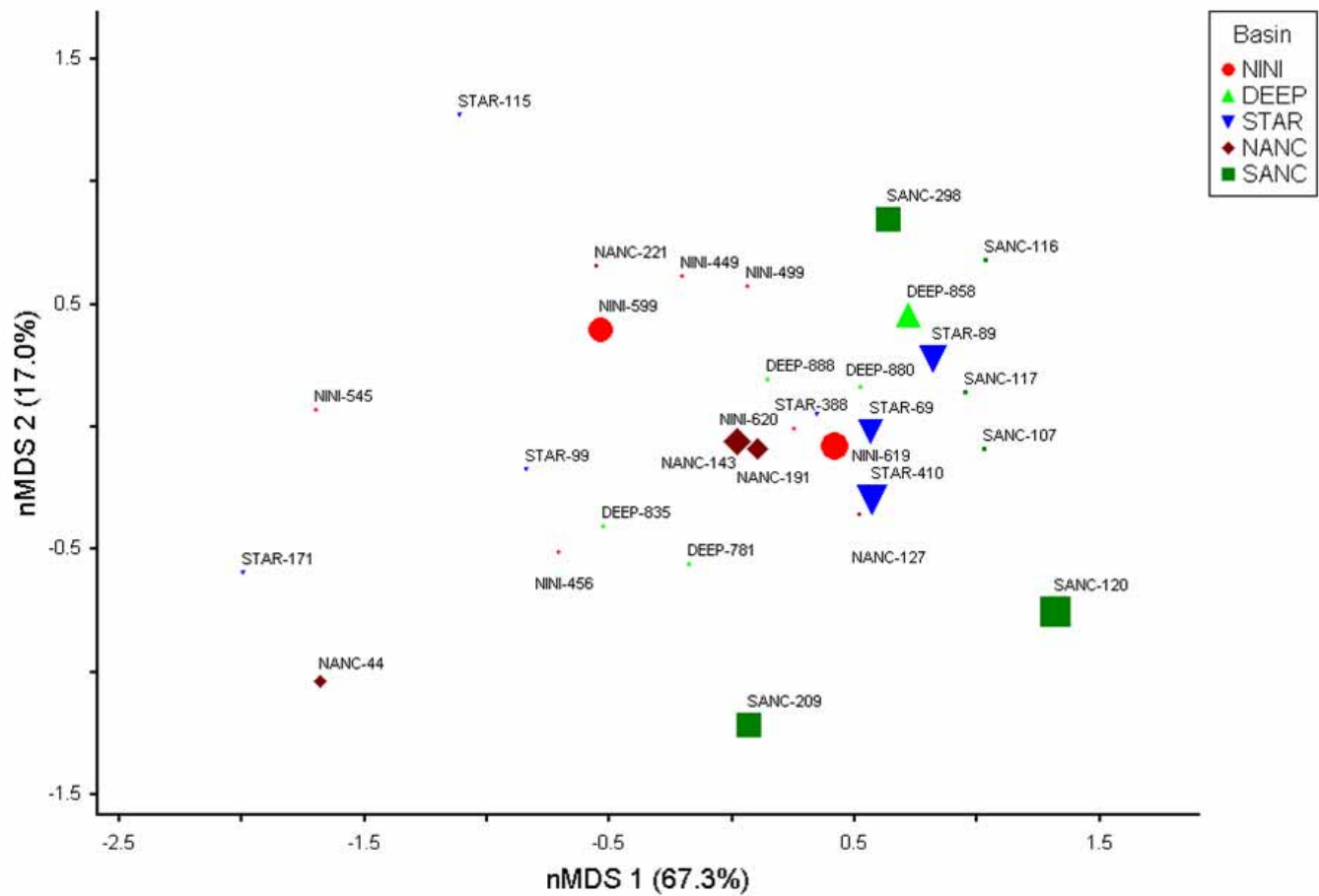


Figure 41. Quantitative overlay of small (<10 cm) juvenile coho densities onto the macroinvertebrate nMDS ordination. Small cohos were distributed similarly as small Dolly Varden, but did not occur at the highest-gradient sites (far right on axis 1).

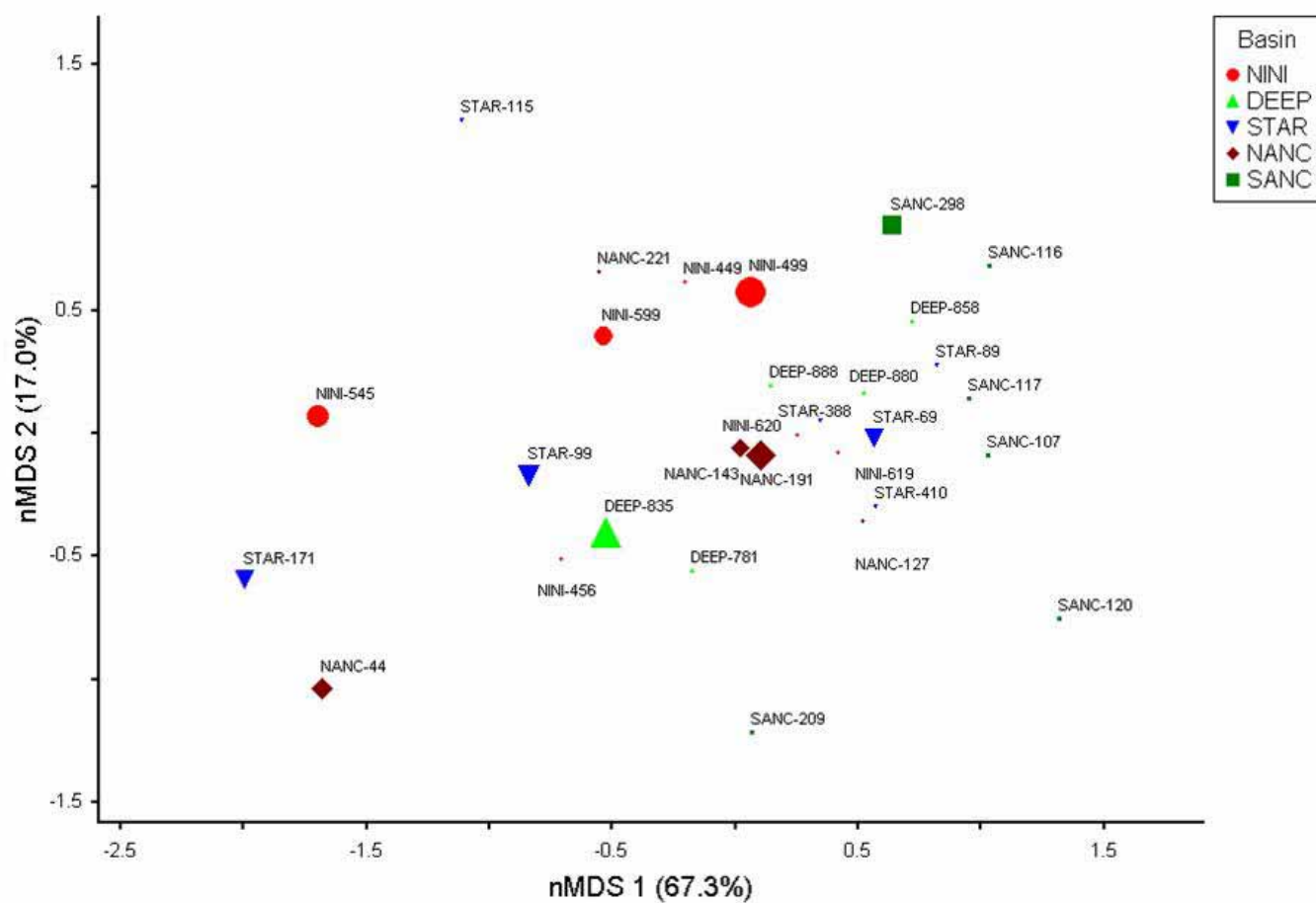


Figure 42. Quantitative overlay of large (>10 cm) juvenile coho densities onto the macroinvertebrate nMDS ordination. Large cohos did not occur at high gradient sites, and corresponded well to macroinvertebrates that were found at intermediate/low gradient locations.

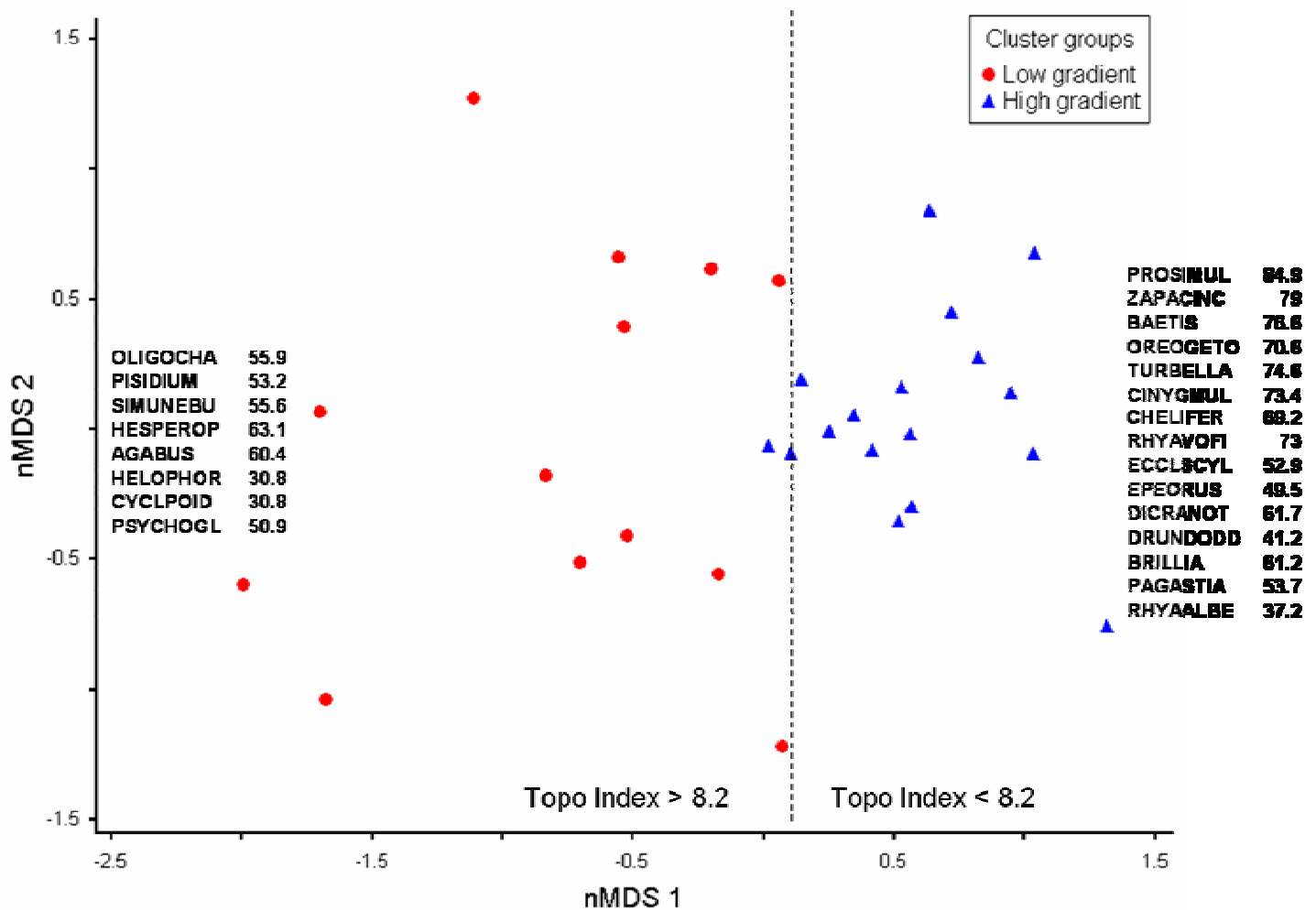


Figure 43. Classification of macroinvertebrate communities determined by hierarchical agglomerative cluster analysis (flexible beta=-0.30). Two clear groups of taxa emerged from this analysis: high gradient, low topographic index communities represented primarily by clinger-type, riffle dwelling taxa such as mayflies, stoneflies, blackflies, and free-living caddisflies, and a second group that corresponded to low gradient, high topographic wetness streams characterized by taxa more commonly found in pond or wetland ecosystems (e.g., aquatic segmented worms, fingernail clams, large-bodied cased caddisflies, dytiscid beetles, and microcrustaceans). For illustration purposes, groups are shown in the nonmetric multidimensional scaling (nMDS) ordination of the 30 stream sites based on macroinvertebrate community composition (see previous graphs for details). Key indicator species of each group are shown adjacent to the sample units in the ordination (see Appendix 1). Values adjacent to each taxon correspond to their indicator scores (% of perfect indication, scale 0-100%) determined by Indicator Species Analysis (Dufrene and Legendre 1997).

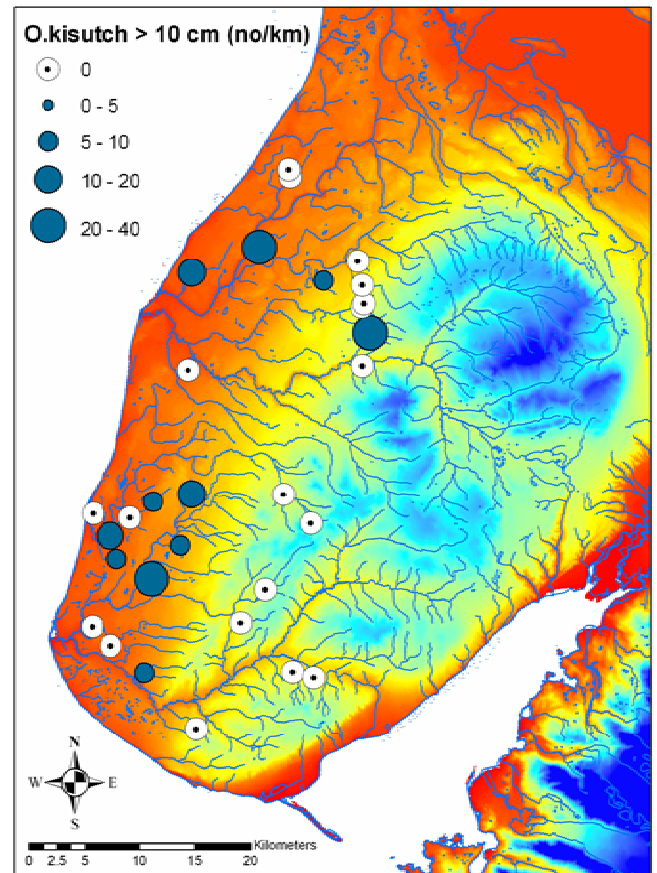
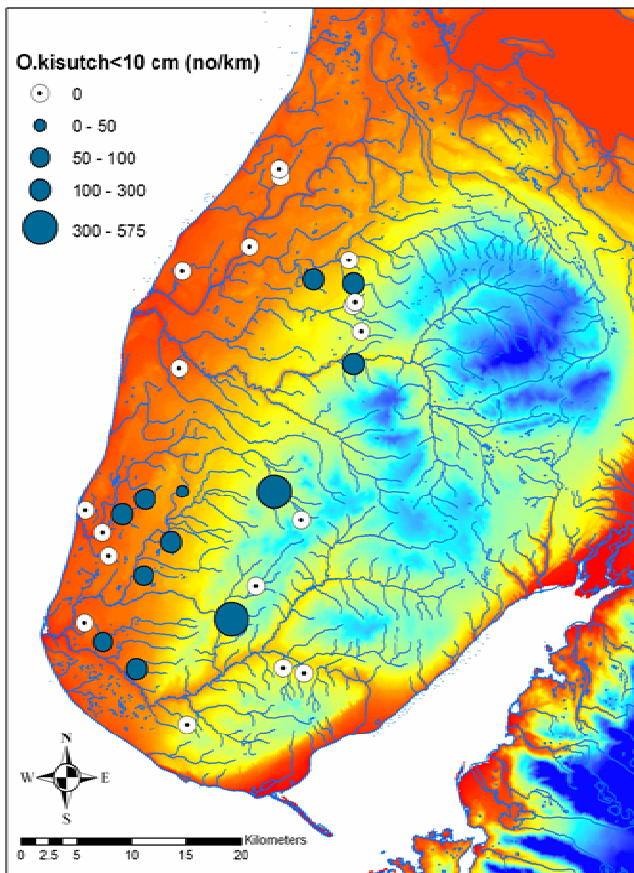
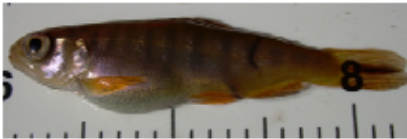


Figure 44. Distribution of small (<10 cm) and large (>10 cm) juvenile coho salmon among study sites. The maps clearly show the propensity to find large cohos in low slope/elevation areas, whereas small cohos achieved greatest densities in slightly higher slope streams at intermediate elevations.

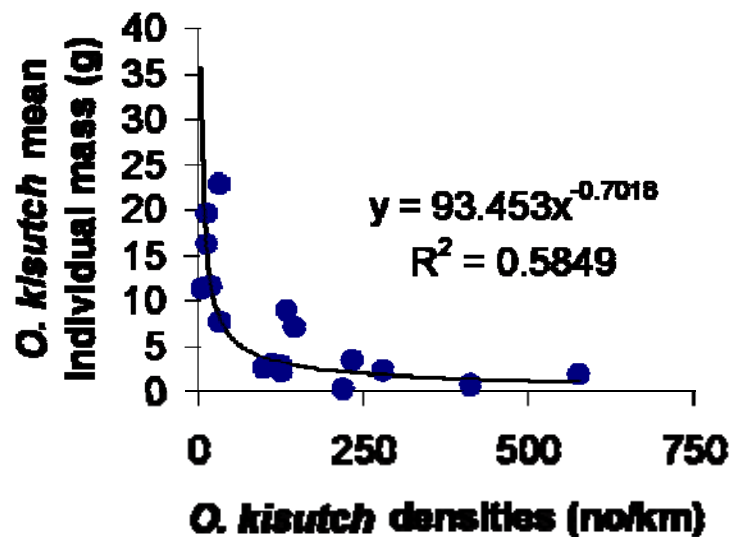


Figure 45. Relationship between juvenile coho mean individual mass and densities among streams where cohos occurred. The strong negative exponential relationship indicated that streams with high densities of cohos tended to support small, first-year fish, whereas streams dominated by larger, second or third year fish had low overall densities. This relationship suggests the biomass per kilometer of stream reach was density dependent and that different streams supported different size classes of fish, yet similar levels of population biomass. No overall relationship was evident for biomass/km and landscape or environmental variables.

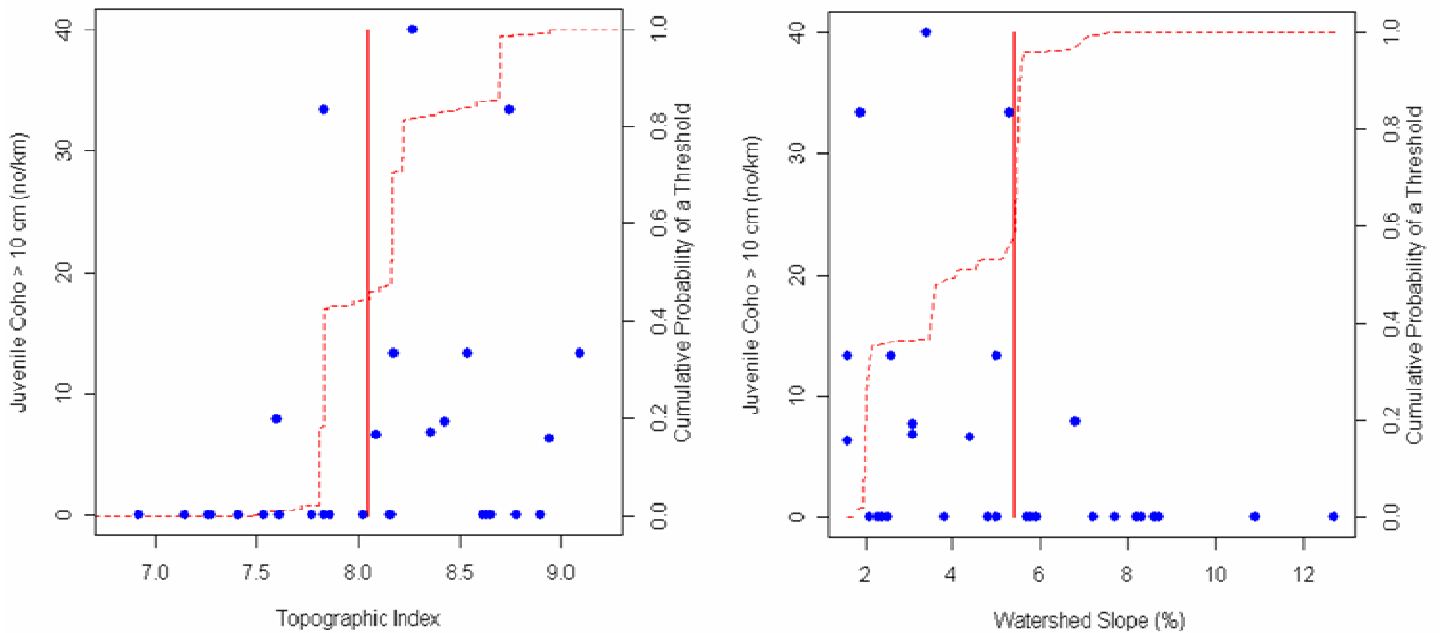


Figure 46. Landscape predictors of large (>10 cm) juvenile coho salmon. Regression tree and nonparametric changepoint analysis illustrated a nonlinear distribution of large juvenile coho (>10 cm) with respect to Topographic Index (left) and Watershed Slope (right). Large cohos were only found in low-to-moderate gradient streams. High topographic index scores were an indicator of lower contributing slopes in the watershed and increased wetness, hence show a reverse pattern to watershed slope. The vertical line indicates the value of x that best explained variance in large juvenile salmon densities, whereas the dashed line represents the cumulative probability that increasing values of x will result in a threshold response in juvenile salmon densities.

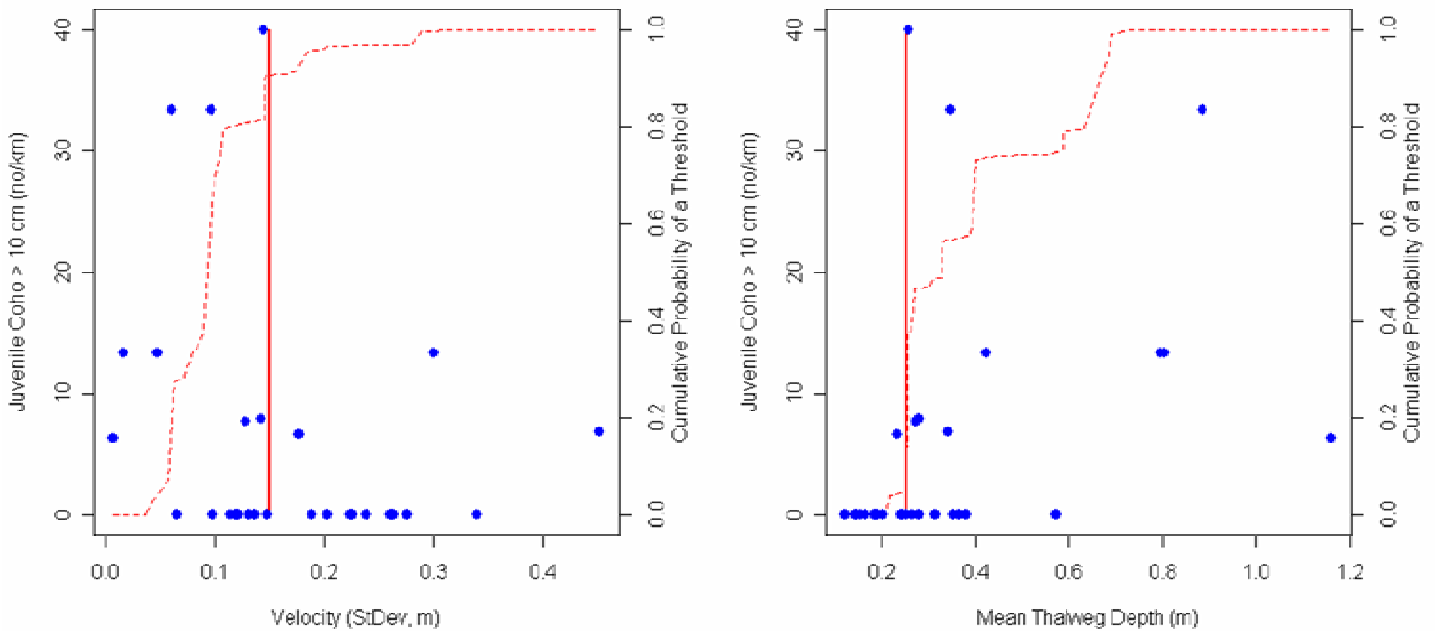


Figure 47. Reach-scale abiotic predictors of large (>10 cm) juvenile coho salmon. Regression tree and nonparametric changepoint analysis illustrated a nonlinear distribution of large juvenile coho (>10 cm) with respect to velocity (left) and mean water depth in the thalweg (right). Large cohos were mostly found in reaches with consistent, low flow velocity and moderate to deep channels, with no large juveniles found in reaches that averaged < 0.25 m depth. The vertical line indicates the value of x that best explained variance in large juvenile salmon densities, whereas the dashed line represents the cumulative probability that increasing values of x will result in a threshold response in juvenile salmon densities.

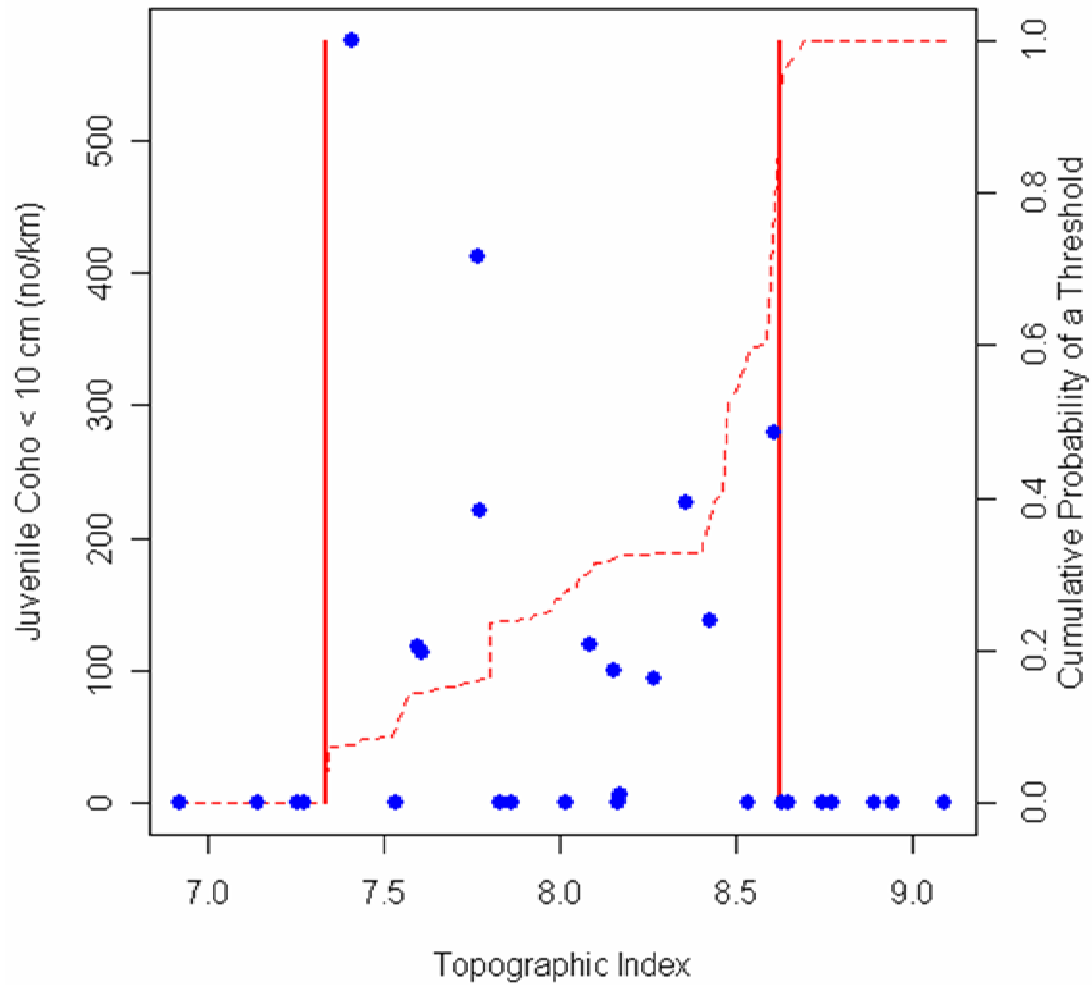


Figure 48. Landscape predictors of small (<10 cm) juvenile coho salmon. Regression tree and nonparametric changepoint analysis illustrated a nonlinear distribution of small juvenile coho (<10 cm) with respect to Topographic Index.

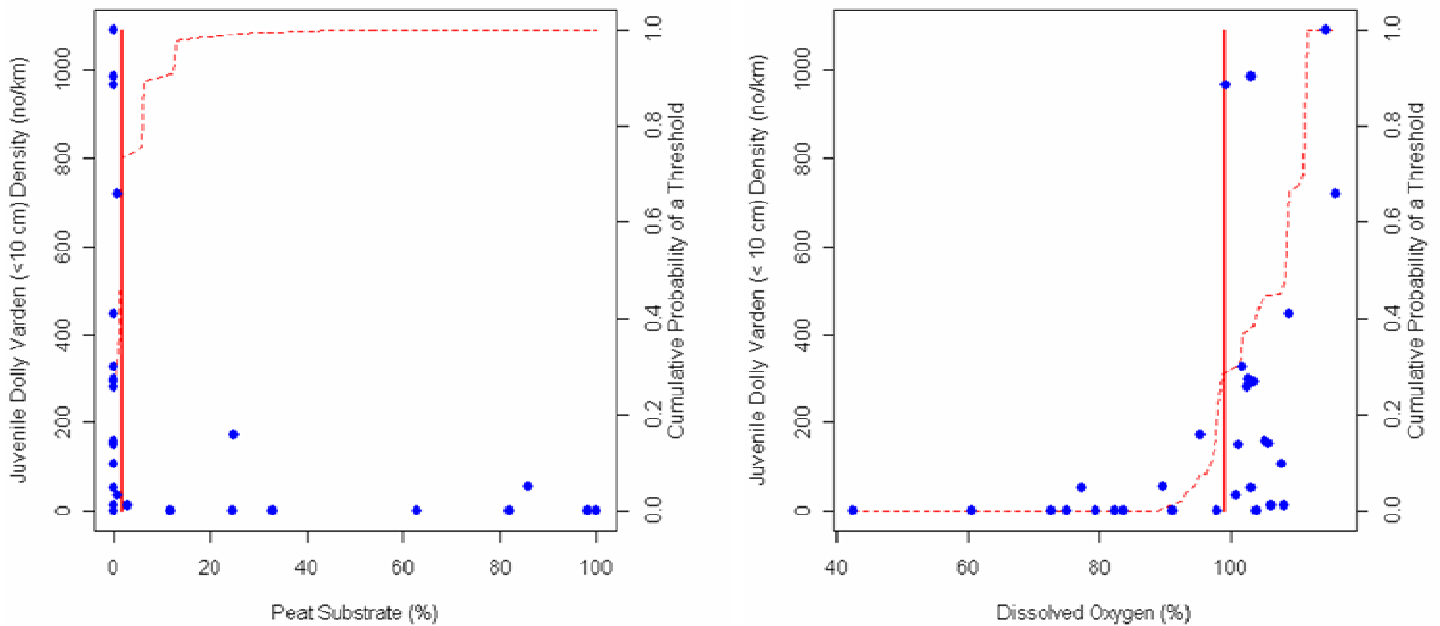


Figure 49. Reach-scale abiotic predictors of small (<10 cm) juvenile coho salmon. Regression tree and nonparametric changepoint analysis identified sharp thresholds in response to the percentage of channel substrate composed of organic sediment (peat) (left), and dissolved oxygen as percent of saturation (right). First-year cohos were essentially absent in peaty streams with soft bottoms as well as streams with sufficient biological oxygen demand to reduce DO below 100% saturation during daylight hours. Both of these responses make ecological and physiological sense because streams with mostly peat bottoms did not support adequate mixing of water near the sediment interface nor fine, clean mineral substrates for spawning adults. Depressed dissolved oxygen during daylight probably corresponded to even lower levels during night, which would stress juveniles, particularly during the earliest stages of development.

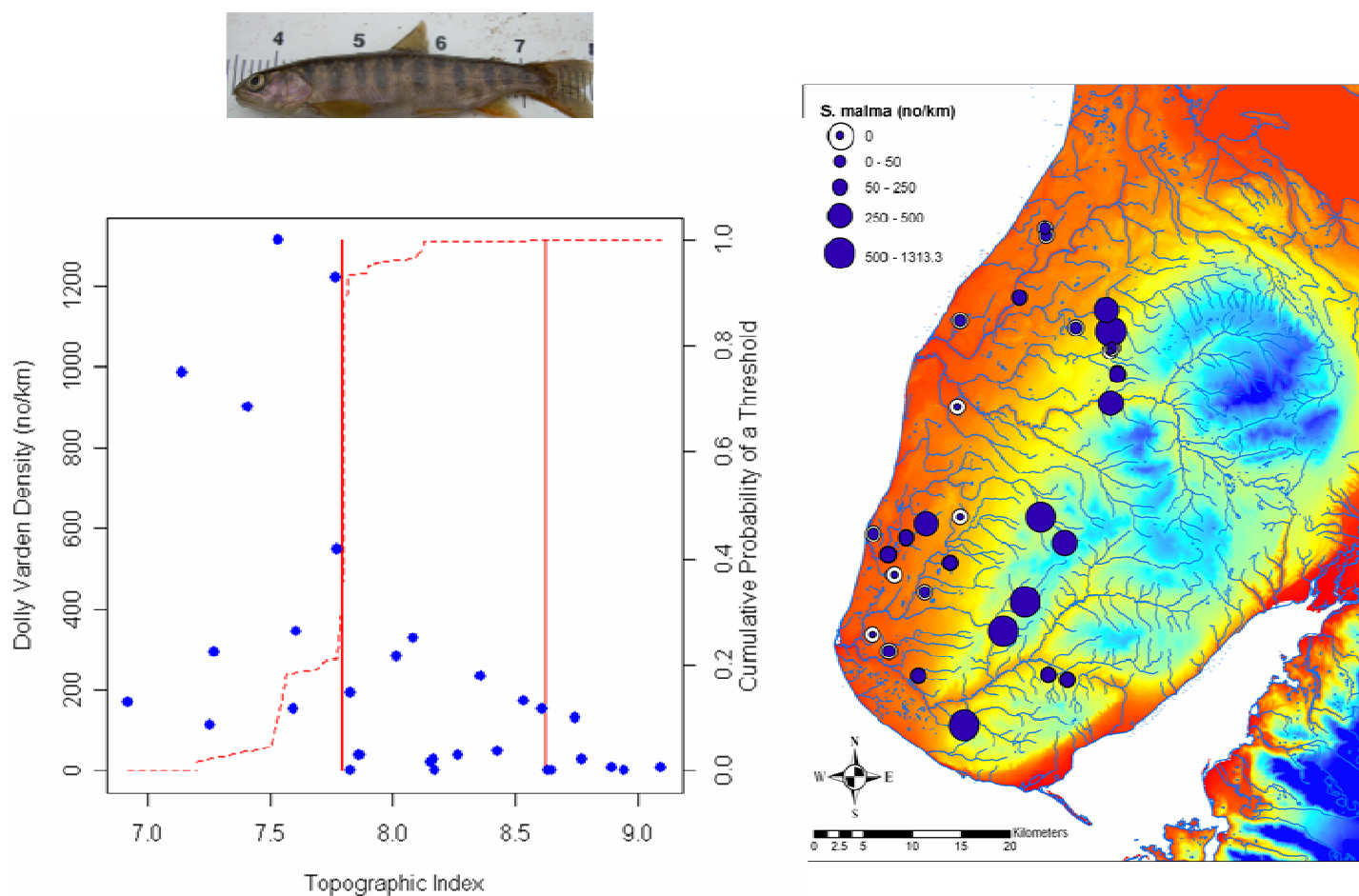


Figure 50. Landscape-scale predictors of Dolly Varden densities. Regression tree and nonparametric changepoint analysis identified three tiers of abundance of Dolly Varden along the topographic gradient. Dolly Varden were in the greatest densities and present in every stream in high gradient watersheds. Dolly Varden were present in most watersheds of intermediate slopes, but at reduced densities when compared to high slope watersheds. Dolly Varden were least abundant in low gradient streams, but these fish tended to be large (>12 cm), much like the pattern exhibited by juvenile cohos. However, in this case, larger Dolly Varden were scattered throughout most streams along the topographic gradient rather than being isolated to low-gradient areas.

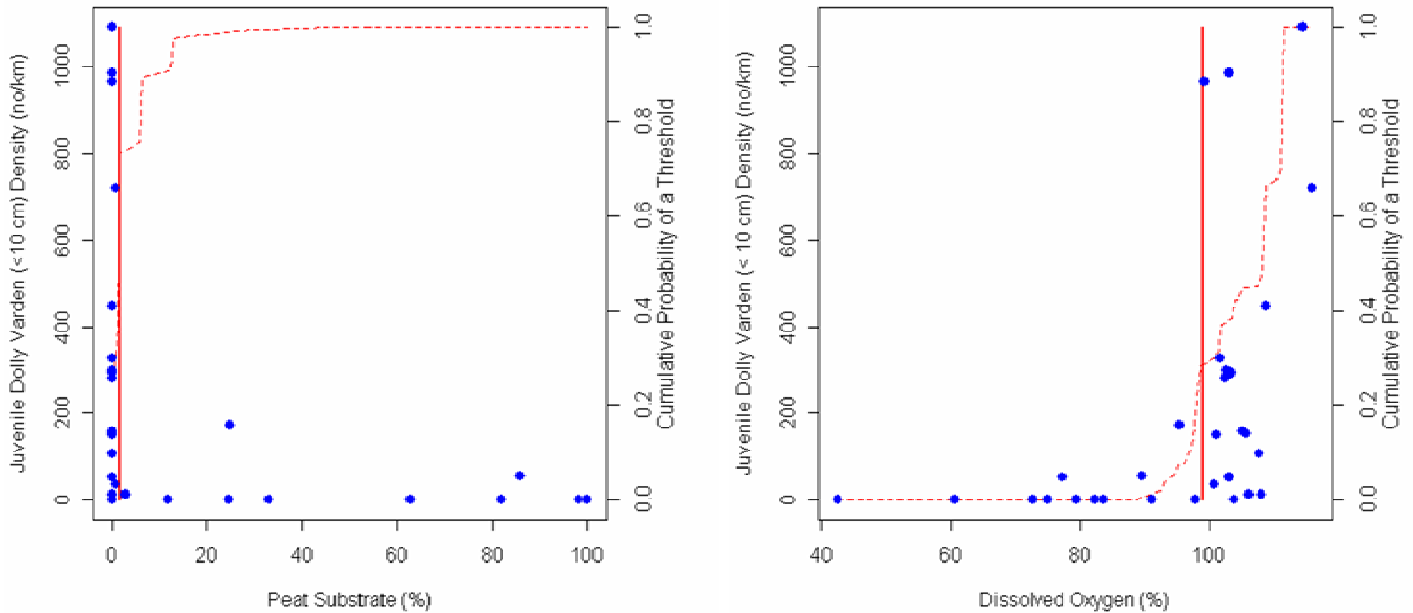


Figure 51. Reach-scale abiotic predictors of Dolly Varden densities. Regression tree and nonparametric changepoint analysis found sharp, threshold declines in small (<10 cm) Dolly Varden densities when stream channels had any peat substrate (left) and when dissolved oxygen dropped below 98% saturation (right). Dolly Varden >10 cm did not respond in a consistent manner to these or any other variable, but these fish represented a very small proportion of total Dolly Varden densities. Juvenile Dolly Varden were responding to these abiotic factors for similar reasons as small, juvenile coho (lack of spawning substrate, excessive biological oxygen demand in the sediments), except they were also found in faster, shallower streams at higher watershed slopes than small juvenile cohos.

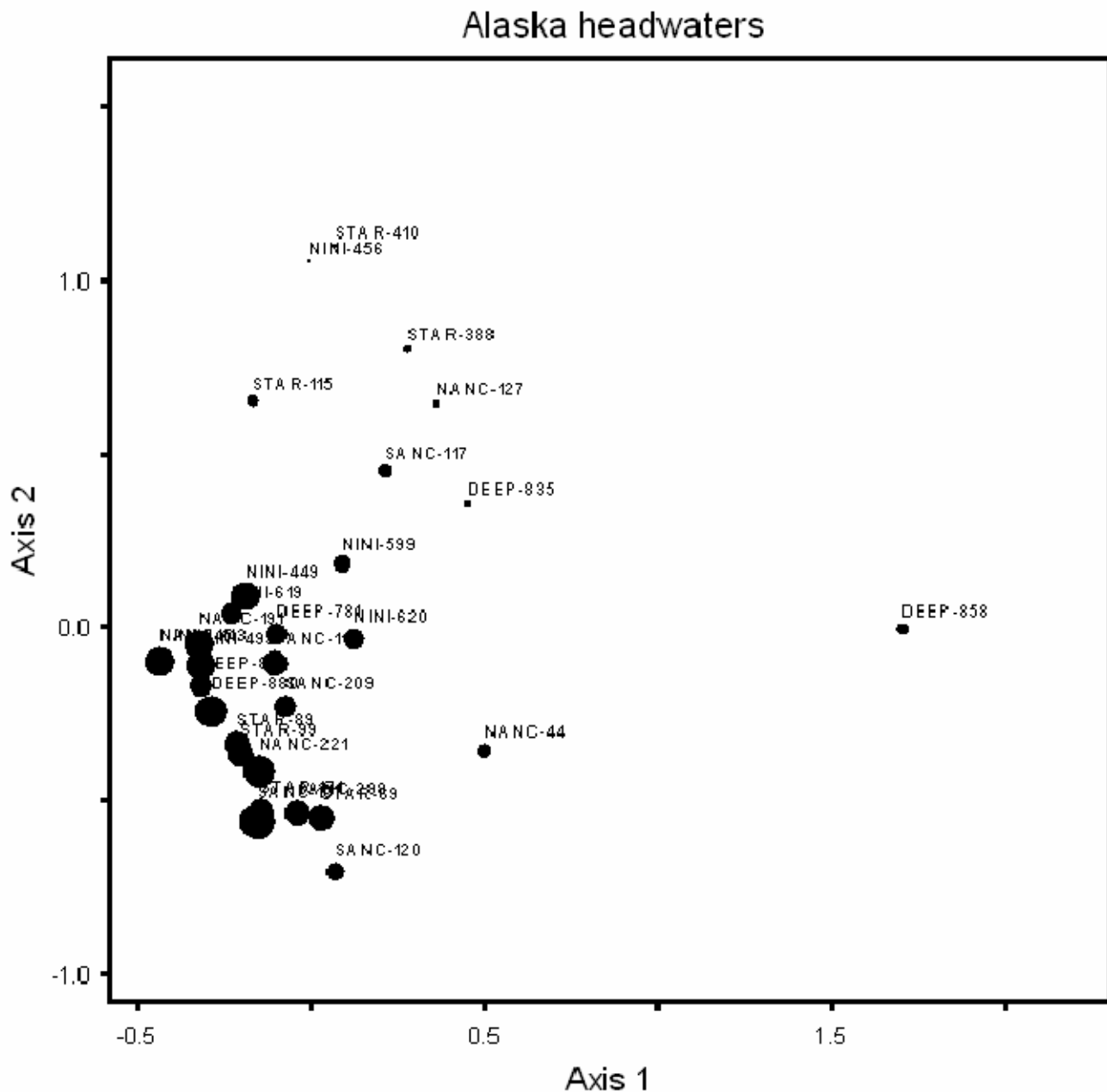


Figure 52. Ordination of the riparian vegetation data clearly shows that all sites have a strong connection to the *Calamagrostis* zone. Larger circles indicate greater cover. Adding other riparian vegetation variables, for example density of trees and shrubs, to the ordination did not add predictive power. Overall, the data show a large amount of similarity between sites. In all sites sampled a band of *Calamagrostis* predominated immediately adjacent to the stream, regardless of great differences in wetland types across geomorphic settings.

Differences in nutrient content were detected from green shoot plant material collected at several sites in each of the major drainages, with material from the South Fork of the Anchor having the highest means. The data set is very small but the observed differences are interesting, especially differences in nitrogen (Table 4).

Table 4. Differences in macronutrient levels of herb material from the four major drainage basins (North and South Forks of the Anchor River combined).

Watershed	N	P	K	Ca	Mg
Deep	2.85 ± .34 ^{ab}	0.44 ± .13 ^a	2.65 ± .57 ^a	0.72 ± .31 ^a	0.32 ± .10 ^a
Ninilchik	2.76 ± .18 ^b	0.25 ± .02 ^a	1.97 ± .41 ^a	0.61 ± .13 ^a	0.25 ± .05 ^a
Anchor	3.73 ± .30 ^a	0.42 ± .08 ^a	1.92 ± .16 ^a	1.20 ± .68 ^a	0.35 ± .12 ^a
Starisky	2.98 ± .13 ^{ab}	0.41 ± .05 ^a	2.50 ± .49 ^a	0.50 ± .12 ^a	0.27 ± .06 ^a
P	0.046	0.0771	0.6271	0.5613	0.8626

Discussion and Attribution of the Wetlands Management Tool

Salmonids have complex life cycles, and use a variety of stream habitats during different stages of development. It is important that we develop research that addresses the dynamic nature of stream fish life histories, and that the results of the research are presented in a spatially explicit, georeferenced format (Fausch et al. 2002).

Though this project is necessarily limited in scope, it addresses these needs by providing models for correlations between landscape metrics, and reach scale instream habitat variables, including water chemistry, hydrology, substrate and cover characteristics for headwater stream systems. The models are based on abiotic and biota data gathered from streams representing dominant geomorphic settings on the Kenai Lowlands. The great variability in watershed and wetland characteristics detected in sites across all drainage basins, suggests that drainage basins do not have unique characteristics that drive landscape patterns, but rather that similar patterns occur across all drainage basins on the Kenai Lowlands, thus improving our ability to apply the models to attribute the Kenai Lowlands Management Tool.

Although we originally set out to attribute wetland polygons, our detailed analysis of the data set shows that topography is the primary driver of abiotic characteristics, habitat features and stream biota. Many instream factors that are indicators of wetland influence are strongly related to the topographic variables, but less so to the wetland classes. The wetlands themselves were formed by topographic variation, and so it makes sense that topographic gradient would be as good, or better, of an indicator of stream condition as the wetland classes. There are complex correlative relationships between large scale processes (topography), wetland settings, and more local scale conditions (Figure 52). For example, topography drives the setting for watershed slope, which in turn drives dominant vegetation communities. At high gradient sites, trees predominate, providing stream canopy cover and more woody debris and root wads, as well as maintaining stream banks through roots, contributing to undercut banks and more erosional, mineral substrates. These local scale habitat characteristics provide conditions suitable for spawning, and developing eggs and fry. The ability of topographic features, in particular topographic index to predict instream characteristics made it a key landscape metric for attributing the Wetland Management Tool.

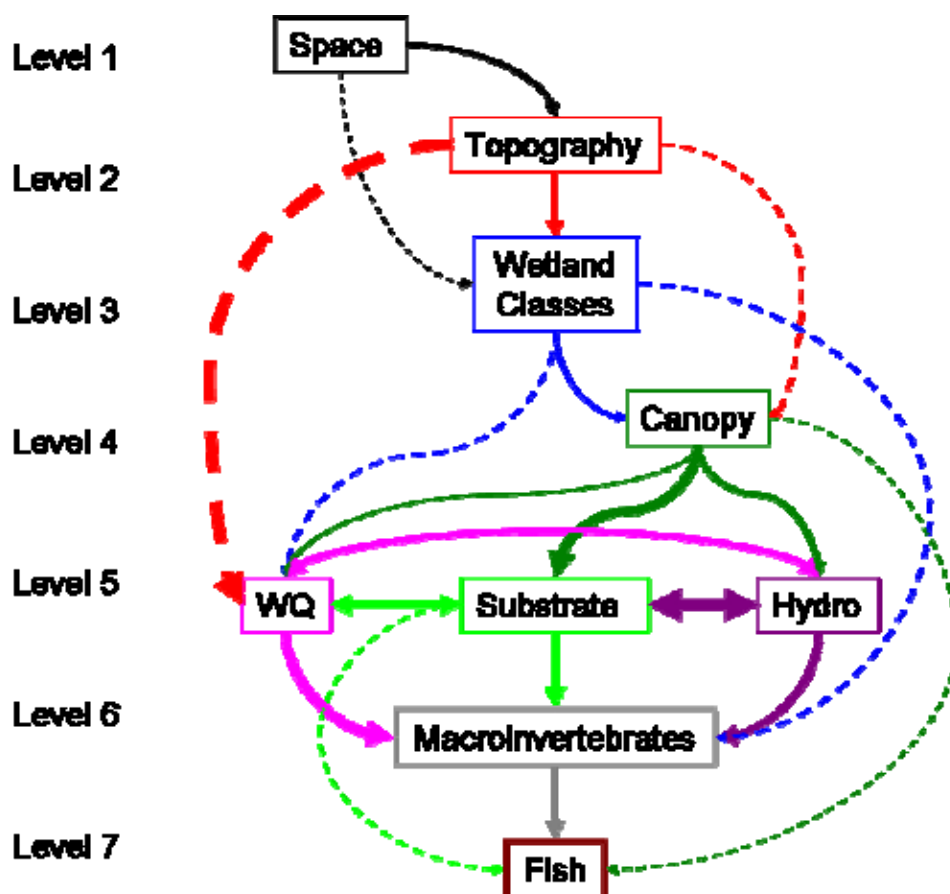


Figure 52. Partial Mantel path model illustrating the independent effect of space (spatial proximity among sites), topography (Topographic Index), wetland classes (DW, LB, KT, RP, DS), canopy (reach-scale tree canopy cover), water quality (DO, temp, DOC, pH, nitrate-nitrate-N), substrate (10 variables), hydrology (mean depth, mean velocity, variance of depth), macroinvertebrate community structure (140 taxa), and fish community structure. Line widths correspond to strength of the relationship. Solid lines are direct effects between a variable and a level immediately below it in the variable hierarchy. Dashed lines represent an indirect effect between a predictor and a response variable separated by one or more levels in the hierarchy. Indirect effects describe significant correlations after the effect of all variables between the predictor and response have been removed (residual, partial correlation). For all correlations, the effect of spatial autocorrelation has been removed, as is the effect of any competing variable on the same level of the path diagram (e.g., the effect of substrate and hydrology on macroinvertebrates is accounted for when computing the direct effect of water quality on macroinvertebrates, thus water quality influenced macroinvertebrates independently of substrate and hydrology, and the same was true for substrate and hydrology). For additional details, see King et al. (2005).

We applied the models across the broad landscape, using the headwater stream segments as carriers of information. All 1 km headwater stream segments in the five drainage basins were attributed. To attribute the segments, we calculated the topographic index, and then used the resulting values to predict water chemistry, hydrology, and habitat features, and associated macroinvertebrate and fish assemblages. Topographic index is a summary statistic that incorporates the degree of topographic convergence to the site. High convergence would strongly suggest groundwater inputs laterally. It is also referred to as “wetness index”; a measure of how wet a

point should be based just on upstream drainage area and slope. Stream segments were ‘binned’ according to topographic index: high gradient, low wetness streams had topographic indices between 6.5-7.5; intermediate gradient and wetness streams had topographic indices of 7.6-8.5; low gradient, high wetness streams had topographic indices of 8.6-9.5. Water chemistry, hydrology, substrate, and habitat variables, as well as macroinvertebrate and fish assemblages were assigned according to binned topographic indices (Table 5).

Table 5. Attribute criteria, associated data and supporting information applied to the Wetland Management Tool.

Abiotic and Habitat Variables	Attribution		
Water Chemistry	1 (topoindex values of 6.5-7.5)	2 (topoindex values of 7.6-8.5)	3 (topoindex values of 8.6-9.5)
	DO 100-120%	DO 80-100%	DO 40-80%
	DOC 0.15-0.25 µg/L	DOC 0.05-0.15 µg/L	DOC 0.0-0.05 µg/L
	Temp. 2-5°C	Temp. 5.1-7°C	Temp. 7.1-9°C
	DOC 0-10 mg/L	DOC 10-15 mg/L	DOC 15-25 mg/L
	Nitrite-nitrate .05-.25 µg/L	Nitrite-nitrate .02-.05 µg/L	Nitrite-nitrate 0-.02 µg/L
	pH 7-8	pH 6.5-6.9	pH 6.0-6.5
Habitat	Gravel 50-90%	Gravel 20-50%	Gravel 0-20%
	Canopy cover 40-90%	Canopy cover 20-40%	Canopy cover 0-20%
	LWD more abundant	LWD less abundant	LWD absent
Hydrology	Velocity .2-.5 m/s	Velocity .1-.2 m/s	Velocity 0-.1 m/s
	Mean depth log ₁₀ 0.5-1.25 cm	Mean depth log ₁₀ 1.25-1.25 cm	Mean depth log ₁₀ 1.50-2.0 cm
Peat	0%		.25-2.5%
Macroinvertebrate assemblages	Topoindex <8.2: assemblages dominated by clinger-type, riffle dwelling taxa such as mayflies, stoneflies, blackflies, and free-living caddisflies. Topoindex >8.2: assemblages dominated by taxa more common in pond or wetland ecosystems (e.g., aquatic segmented worms, fingernail clams, large-bodied cased caddisflies, dytiscid beetles, and microcrustaceans.		
Large coho salmon (>10 cm)	Topoindex 0-8, number of large coho = 0 Topo index > 8.0, number of large coho = 10-40/km		
Small coho salmon (<10 cm)	Topoindex 6.5-7.4, number of small coho = 0 Topoindex 7.5-8.6, number of small coho = 100-600/km Topoindex 8.7-9.5, number of small coho = 0		
Dolly Varden	Topoindex 6.5-7.7, number of Dolly Varden = 200-1200/km Topoindex 7.8-8.6, number of Dolly Varden = 0-400/km Topoindex 8.7-9.5, number of Dolly Varden = 0-170		
Wetland Drainage Basins	% wetlands in each geomorphic class (lakebed, drainageway, kettle, riparian) in the upstream watershed for each headwater stream segment		

‘Pop-up’ dialogue boxes were associated with the attributes to describe the relevance of the variables:

Water Chemistry dialog: DOC, DO, nitrite-nitrate N, temperature, and pH, show consistent patterns suggesting more interaction between wetland organic matter and shallow groundwater in watersheds with lower topographic relief. High DO levels in higher gradient streams and lower temperatures provide suitable conditions for developing eggs and fry.

Habitat dialog: % gravel and cobble are much higher at higher watershed slopes. These habitats are important because they satisfy the requirement of fine gravel, erosional habitats for spawning salmonids and successful rearing of eggs. Higher watershed slopes also tend to have a higher percentage of tree canopy cover, which provides large woody debris and root wads inputs to streams, both of which are important cover and forage habitat for juvenile salmon. Riparian trees also provide bank stabilization, which helps maintain gravel substrates free of fine particles, contribute to formation of undercut banks, which provide fish habitat.

Peat dialog: peat dramatically increased in streams with topographic index values above 8, and was essentially absent below this score. This finding is highly consistent with the patterns of water chemistry (DO, pH, DOC) and shows that streams in most low gradient watersheds began to resemble depositional, marsh habitats compared to the erosional, mineral substrate streams at higher gradients.

Hydrology dialog: streams with higher watershed slopes are faster and shallower than headwater streams at lower watershed slopes.

Macroinvertebrate dialog: High gradient headwater streams support macroinvertebrate communities represented primarily by clinger-type, riffle dwelling taxa such as mayflies, stoneflies, blackflies, and free-living caddisflies. Low gradient headwater streams support macroinvertebrate communities characterized by taxa more commonly found in pond or wetland ecosystems (e.g., aquatic segmented worms, fingernail clams, large-bodied cased caddisflies, dytiscid beetles, and microcrustaceans). All these organisms provide valuable food resources for salmonids.

Fish dialog: Large (older) cohos are found only in low gradients, high wetness streams, young cohos are abundant in middle gradient streams, while Dolly Varden (particularly small dollies) are most abundant in the highest gradient, high elevation streams, but are of intermediate abundance in intermediate gradient streams and large Dolly Varden tend to be widely scattered throughout all gradients.

Conclusions

Results from this project clearly show that there are differences among headwater streams in water chemistry, hydrology, and habitat features that are driven by topography, and strongly connected to surrounding wetlands. These differences are manifested in different macroinvertebrate assemblages, and in densities of salmonid year-classes. The important message for managing headwater streams is that these strong differences don’t make one stream more or less valuable. We found salmonids in nearly every headwater stream setting. Extrapolated estimates of total numbers of fish in headwater streams for each major drainage basin indicates that close to ¼ million salmonids are sustained in the headwater streams of the Kenai Lowlands (Table 6).

Table 6. Extrapolated estimates of total numbers of fish in headwater streams for each major drainage basin of the study area. Note that this assumes that what we sampled was representative of the whole watershed, however we have no idea how fish densities vary longitudinally or temporally along an individual 1st-order stream. It also includes the streams in which we found no fish (or no cohos). Therefore, these extrapolated estimates of total fish numbers only represent one stretch of stream at a single point in time.

watershed	1 st order stream meters	extrapolated fish numbers		total salmonids
		coho	Dolly Varden	
Deep	222,828	6,536	24,957	32,384
Ninilchik	130,905	7,356	15,958	23,438
Stariski	56,420	7,845	12,896	20,741
N. Fork Anchor	76,948	5,951	23,700	29,753
S. Fork Anchor	204,348	21,570	88,551	148,266
Total	691,449	164,872	249,844	249,844

Obviously there is sufficient food to support fish in all the headwater streams where we found fish, but perhaps some stream settings, and associated macroinvertebrate communities favor certain species and or age classes. With salmonid species that spend a year or more in freshwater, such as coho salmon, and Dolly Varden, stream conditions must be within tolerable limits throughout the year, and important habitat configurations must be available. It is therefore important to maintain a diversity of habitats for their benefit.

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~Coowe Walker, Ryan King and Dennis Whigham

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Appendix A-1

Macroinvertebrate Species List

GROUP	FAMILY	TAXON	TAXON_ID
AMPHIPODA	GAMMARIDAE	GAMMARUS	GAMMARUS
ARYNCHOBDELLIDA	ERPOBDELLIDAE	ERPOBDELLIDAE	ERPOBDELL
ARYNCHOBDELLIDA	HIRUDINIDAE	HAEMOPSIS MARMORATA	HAEMMARM
BIVALVIA	PISIDIIDAE	PISIDIUM	PISIDIUM
CLADOCERA	CLADOCERA	CLADOCERA	CLADOCER
COLEOPTERA	DYTISCIDAE	AGABUS	AGABUS
COLEOPTERA	DYTISCIDAE	AGABUS SERIATUS GR	AGABSERI
COLEOPTERA	DYTISCIDAE	COLYMBETES DAHURICUS	COLYDAHU
COLEOPTERA	DYTISCIDAE	DYTISCIDAE	DYTISCID
COLEOPTERA	DYTISCIDAE	HYDROPORUS	HYDROPOR
COLEOPTERA	DYTISCIDAE	NEOSCUTOPTERUS HORNII	NEOSHORN
COLEOPTERA	DYTISCIDAE	SANFILLIPODYTES	SANFILLI
COLEOPTERA	HELOPHORIDAE	HELOPHORUS	HELOPHOR
COLEOPTERA	STAPHYLINIDAE	STAPHYLINIDAE	STAPHYLI
COLLEMBOLA	COLLEMBOLA	COLLEMBOLA	COLLEMBOL
CYCLOPOIDA	CYCLOPOIDA	CYCLOPOIDA	CYCLPOID
DIPTERA	CERATOPOGONIDAE	BEZZIA	BEZZIA
DIPTERA	CERATOPOGONIDAE	CERATOPOGONIDAE	CERATOPO
DIPTERA	CERATOPOGONIDAE	PROBEZZIA	PROBEZZI
DIPTERA	CHAOBORIDAE	EUCORETHRA UNDERWOODI	EUCOUNDE
DIPTERA	CHIRONOMIDAE	ABLABESMYIA	ABLABESM
DIPTERA	CHIRONOMIDAE	BRILLIA	BRILLIA
DIPTERA	CHIRONOMIDAE	CHIRONOMUS	CHIROMUS
DIPTERA	CHIRONOMIDAE	CORYNONEURA	CORYNONE
DIPTERA	CHIRONOMIDAE	CRICOTOPUS	CRICOTOP
DIPTERA	CHIRONOMIDAE	DIAMESA	DIAMESA
DIPTERA	CHIRONOMIDAE	DIPLOCLADIUS	DIPLOCLA
DIPTERA	CHIRONOMIDAE	EUKIEFFERIELLA	EUKIEFFE
DIPTERA	CHIRONOMIDAE	GLYPTOTENDIPES	GLYPTOTE
DIPTERA	CHIRONOMIDAE	GYMNOMETRIOCNEMUS	GYMNOMET
DIPTERA	CHIRONOMIDAE	HELENIELLA	HELENIEL
DIPTERA	CHIRONOMIDAE	HETEROTRISSECLADIUS	HETEROTR
DIPTERA	CHIRONOMIDAE	KRENOPELOPIA	KRENOPEL
DIPTERA	CHIRONOMIDAE	KRENOSMITTIA	KRENOSMI
DIPTERA	CHIRONOMIDAE	LASIODIAMESA	LASIODIA
DIPTERA	CHIRONOMIDAE	LIMNOPHYES	LIMNOPHY
DIPTERA	CHIRONOMIDAE	MACROPELOPIA	MACROPEL
DIPTERA	CHIRONOMIDAE	METRIOCNEMUS	METRIOCN
DIPTERA	CHIRONOMIDAE	METRIOCNEMUS FUSCIPES	METRFUSC
DIPTERA	CHIRONOMIDAE	MICROPSECTRA	MICROPSE
DIPTERA	CHIRONOMIDAE	MONOPELOPIA	MONOPELO
DIPTERA	CHIRONOMIDAE	NANOCLADIUS	NANOCLAD
DIPTERA	CHIRONOMIDAE	NATARSIA	NATARSIA
DIPTERA	CHIRONOMIDAE	ODONTOMESA	ODONTOME
DIPTERA	CHIRONOMIDAE	ORTHOCLADIINAE	ORTHINAE
DIPTERA	CHIRONOMIDAE	ORTHOCLADIUS	ORTHOCLA
DIPTERA	CHIRONOMIDAE	ORTHOCLADIUS	ORTHOCLA
DIPTERA	CHIRONOMIDAE	(SYMPOSIOTRISSECLADIUS)	ORTHSYMP

DIPTERA	CHIRONOMIDAE	PAGASTIA	PAGASTIA
DIPTERA	CHIRONOMIDAE	PARACHIRONOMUS	PARACHIR
DIPTERA	CHIRONOMIDAE	PARACLADOPELMA	PARACLAD
DIPTERA	CHIRONOMIDAE	PARAKIEFFERIELLA	PARAKIEF
DIPTERA	CHIRONOMIDAE	PARAMERINA	PARAMERI
DIPTERA	CHIRONOMIDAE	PARAMETRIOCNEMUS	PARAMETR
DIPTERA	CHIRONOMIDAE	PARAPHAENOCLADIUS	PARAPHAE
DIPTERA	CHIRONOMIDAE	PARATANYTARSUS	PARATANY
DIPTERA	CHIRONOMIDAE	PARORTHOCLADIUS	PARORTHO
DIPTERA	CHIRONOMIDAE	POLYPEDILUM	POLYPEDI
DIPTERA	CHIRONOMIDAE	POTTHASTIA	POTTHAST
DIPTERA	CHIRONOMIDAE	PROCLADIUS	PROCLADI
DIPTERA	CHIRONOMIDAE	PRODIAMESA	PRODIAME
DIPTERA	CHIRONOMIDAE	PROTANYPUS	PROTANYP
DIPTERA	CHIRONOMIDAE	PSECTROCLADIUS	PSECTROC
DIPTERA	CHIRONOMIDAE	PSEUDODIAMESA	PSEUDODI
DIPTERA	CHIRONOMIDAE	PSILOMETRIOCNEMUS	PSILOMET
DIPTERA	CHIRONOMIDAE	RHEOCRICOTOPUS	RHEOCRIC
DIPTERA	CHIRONOMIDAE	RHEOTANYTARSUS	RHEOTANY
DIPTERA	CHIRONOMIDAE	SMITTIA	SMITTIA
DIPTERA	CHIRONOMIDAE	STEMPELLINELLA	STEMPELL
DIPTERA	CHIRONOMIDAE	STILOCLADIUS	STILOCLA
DIPTERA	CHIRONOMIDAE	TANYTARSUS	TANYTARS
DIPTERA	CHIRONOMIDAE	THIENEMANNIELLA	THIENEMA
DIPTERA	CHIRONOMIDAE	TVETENIA	TVETENIA
DIPTERA	CULICIDAE	AEDES	AEDES
DIPTERA	CULICIDAE	CULICIDAE	CULICIDA
DIPTERA	DIXIDAE	DIXELLA	DIXELLA
DIPTERA	DIXIDAE	DIXIDAE	DIXIDAE
DIPTERA	EMPIDIDAE	CHELIFERA	CHELIFER
DIPTERA	EMPIDIDAE	EMPIDIDAE	EMPIDIDA
DIPTERA	EMPIDIDAE	OREOGETON	OREOGETO
DIPTERA	EMPIDIDAE	TRICOCLINOCERA	TRICOCLI
DIPTERA	EPHYDRIDAE	EPHYDRIDAE	EPHYDRID
DIPTERA	PSYCHODIDAE	PERICOMA	PERICOMA
DIPTERA	SCIOMYZIDAE	SCIOMYZIDAE	SCIOMYZI
DIPTERA	SIMULIIDAE	PROSIMULIUM	PROSIMUL
DIPTERA	SIMULIIDAE	PROSIMULIUM ESSELBAUGHI	PROSESSE
DIPTERA	SIMULIIDAE	SIMULIUM	SIMULIUM
DIPTERA	SIMULIIDAE	SIMULIUM NEBULOSUM	SIMUNEBU
DIPTERA	SIMULIIDAE	SIMULIUM TRANSIENS	SIMUTRAN
DIPTERA	STRATIOMYIIDAE	STRATIOMYIIDAE	STRATIOM
DIPTERA	TIPULIDAE	ANTOCHA	ANTOCHA
DIPTERA	TIPULIDAE	DICRANOTA	DICRANOT
DIPTERA	TIPULIDAE	HESPEROCONOPA	HESPEROC
DIPTERA	TIPULIDAE	HEXATOMA	HEXATOMA
DIPTERA	TIPULIDAE	MOLOPHILUS	MOLOPHIL
DIPTERA	TIPULIDAE	PEDICIA	PEDICIA
DIPTERA	TIPULIDAE	TIPULA	TIPULA
DIPTERA	TIPULIDAE	TIPULIDAE	TIPULIDA

EPHEMEROPTERA	AMELETIDAE	AMELETUS	AMELETUS
EPHEMEROPTERA	BAETIDAE	BAETIDAE	BAETIDAE
EPHEMEROPTERA	BAETIDAE	BAETIS	BAETIS
EPHEMEROPTERA	BAETIDAE	CALLIBAETIS FLUCTUANS	CALLFLUC
EPHEMEROPTERA	EPHEMERELLIDAE	DRUNELLA DODDSI	DRUNDODD
EPHEMEROPTERA	EPHEMERELLIDAE	EPHEMERELLA AURIVILLII	EPHEAURI
EPHEMEROPTERA	HEPTAGENIIDAE	CINYGMA	CINYGMA
EPHEMEROPTERA	HEPTAGENIIDAE	CINYGMULA	CINYGMUL
EPHEMEROPTERA	HEPTAGENIIDAE	EPEORUS	EPEORUS
EPHEMEROPTERA	HEPTAGENIIDAE	HEPTAGENIIDAE	HEPTAGEN
GASTROPODA	LYMNAEIDAE	LYMNAEIDAE	LYMNAEID
GASTROPODA	PLANORBIDAE	GYRAULUS DEFLECTUS	GYRADEFL
HETEROPTERA	CORIXIDAE	CALLICORIXA	CALLICOR
HETEROPTERA	CORIXIDAE	SIGARA	SIGARA
HETEROPTERA	GERRIDAE	GERRIS	GERRIS
NEMATA	NEMATA	NEMATA	NEMATA
ODONATA	AESCHNIDAE	AESCHNIDAE	AESCHNID
OLIGOCHAETA	OLIGOCHAETA	OLIGOCHAETA	OLIGOCHA
OSTRACODA	OSTRACODA	OSTRACODA	OSTRACOD
PLECOPTERA	CAPNIIDAE	CAPNIIDAE	CAPNIIDA
PLECOPTERA	CHLORPERLIDAE	ALASKAPERLA ORIBARIS	ALASORIB
PLECOPTERA	CHLORPERLIDAE	ALLOPERLA	ALLOPERL
PLECOPTERA	CHLORPERLIDAE	CHLOROPERLIDAE	CHLOROPE
PLECOPTERA	CHLORPERLIDAE	SUWALLIA	SUWALLIA
PLECOPTERA	CHLORPERLIDAE	SWELSTA	SWELTSA
PLECOPTERA	LEUCTRIDAE	LEUCTRIDAE	LEUCTRID
PLECOPTERA	NEMOURIDAE	ZAPADA CINCTIPES	ZAPACINC
PLECOPTERA	PERLODIDAE	ISOPERLA	ISOPERLA
TRICHOPTERA	BRACHYCENTRIDAE	BRACHYCENTRUS AMERICANUS	BRACAMER
TRICHOPTERA	GLOSSOMATIDAE	GLOSSOSOMA	GLOSSOSO
TRICHOPTERA	LEPIDOSTOMATIDAE	LEPIDOSTOMA	LEPIDOST
TRICHOPTERA	LIMNEPHILIDAE	CHYRANDA CENTRALIS	CHYRCENT
TRICHOPTERA	LIMNEPHILIDAE	ECCLISOCOSMOECUS SCYLLA	ECCLSCYL
TRICHOPTERA	LIMNEPHILIDAE	HESPEROPHYLAX	HESPEROP
TRICHOPTERA	LIMNEPHILIDAE	LIMNEPHILIDAE	LMNPHDAE
TRICHOPTERA	LIMNEPHILIDAE	LIMNEPHILUS	LIMNEPHI
TRICHOPTERA	LIMNEPHILIDAE	ONOCOSMOECUS UNICOLOR	ONOCUNIC
TRICHOPTERA	LIMNEPHILIDAE	PSYCHOGLYPHA	PSYCHOGL
TRICHOPTERA	PHILOPOTAMIDAE	DOLOPHILODES	DOLOPHIL
TRICHOPTERA	RHYACOPHILIDAE	RHYACOPHILA ALBERTA GR	RHYAALBE
TRICHOPTERA	RHYACOPHILIDAE	RHYACOPHILA VOFIXA GR	RHYAVOFI
TROMBIDIFORMES	TROMBIDIFORMES	TROMBIDIFORMES	TROMBIDI
TURBELLARIA	TURBELLARIA	TURBELLARIA	TURBELLA

Appendix A-2

Principal Component Analysis Data

Strong positive correlations=**blue**; Strong negative correlations=**red**; Most influential variables in **bold**.

Eigenvectors (Correlations with axes)			Eigenvalues (percent of variance explained by each axis)			
Variable	PC1	PC2	PC Axis	Eigenvalue	% of Variance	Cum.% of Var.
TOTAREA	-0.2206	-0.215	1	14.168	38.293	38.293
NORTH_DS	0.0079	0.2205	2	7.411	20.031	58.324
EAST_DS	0.7694	0.1017	3	3.589	9.699	68.022
Salt_m (distance to salt water)	0.7643	0.1137	4	2.96	7.999	76.022
Main_m (distance to main stem)	-0.1619	0.3564	5	1.857	5.02	81.041
Near_m (distance to nearest tributary)	-0.189	0.0268	6	1.804	4.877	85.918
Elev_GPS (elevation)	0.8667	-0.1073	7	1.152	3.112	89.03
WSlope (mean percent watershed slope)	0.7727	-0.1063	8	0.871	2.355	91.385
Relratio (relief ratio- the elevation range of the basin for each site divided by the length of the basin.)	0.744	-0.1448	9	0.767	2.072	93.458
d-Link** (stream link magnitude)	-0.1529	0.1015	10	0.513	1.386	94.844
d-Link ²	-0.3921	-0.1494				
Topographic index (see Table 3 in text for explanation)	-0.7978	0.1692	**Downstream link (D-link) describes the upstream and downstream influences on a stream section within a drainage network and is calculated as the link magnitude of the next downstream confluence. *IDW = inverse distance weighted (see Table 3 in text for explanation). Wet=any wetland class lb = lakebed dw = drainageway ds = discharge slope rip = riparian kt = kettle ri_ds = riparian and discharge slope lb_dw = lakebed and drainageway lbdwk = lakebed, drainageway and kettle dw_kt = drainageway and kettle			
Wet (total wetlands)	-0.7906	-0.4469				
DS (% discharge slope wetlands)	-0.325	-0.7952				
DW (% drainage way wetlands)	-0.5145	0.5849				
LB (% lakebed wetlands)	-0.9075	-0.1512				
DWLBKT (% drainageway, lakebed & kettle)	-0.93	0.0843				
RP (% riparian wetlands)	0.5803	-0.4509				
DSRP (% discharge slope & riparian)	0.011	-0.8949				
IDWwet* (to any wetlands)	-0.7456	-0.5429				
IDWlb	-0.887	-0.2009				
IDWdw	-0.6226	0.5563				
IDWds	-0.2791	-0.7798				
IDWrip	0.3273	-0.4356				
IDWri_ds	-0.0072	-0.9428				
IDWlb_dw	-0.9339	0.042				
IDWlbdwk	-0.9333	0.0726				
IDWdw_kt	-0.4406	0.5268				
ID2wet (ID2 =inverse distance weighted by distance squared)	-0.6018	-0.5048				
ID2lb	-0.8296	-0.258				
ID2dw	-0.6341	0.5134				
ID2ds	-0.2648	-0.7109				
ID2rip	-0.0553	-0.4493				
ID2ri_ds	-0.1874	-0.8395				
ID2lb_dw	-0.9117	0.1317				
ID2lbdwk	-0.8563	0.1303				
ID2dw_kt	-0.4092	0.453				

Principal Component Analysis of 37 watershed and landscape variables among the 30 headwater streams sampled.

Eigenvectors (Correlations with axes)				Eigenvalues (percent of variance explained by each axis)				
Variable	PC1	PC2	PC3	PC Axis	Eigenvalue	% of Variance	Cum.% of Var.	
Q_cms	0.0601	-0.3637	-0.2385	1	18.277	27.28	27.28	
TEMP_ADJ (temperature)	-0.5838	-0.0126	-0.27	2	6.924	10.334	37.614	
SPCOND (specific conductivity)	-0.3633	-0.137	0.4332	3	5.546	8.278	45.892	
DO_PCT (dissolved oxygen)	0.8333	-0.335	-0.0585	4	4.47	6.672	52.564	
PH	0.6403	-0.5135	0.3442	5	4.06	6.06	58.624	
PO4_P (phosphate)	-0.2265	-0.2321	0.6557	6	3.317	4.95	63.575	
NH3_N (ammonia)	-0.1693	0.442	-0.0225	7	3.136	4.681	68.256	
NO2NO3_N (nitrite and nitrate)	0.5328	-0.0942	0.1678	8	2.854	4.26	72.515	
DIN (dissolved inorganic nitrogen)	0.6204	-0.0337	0.2549	9	2.509	3.745	76.26	
TP (total phosphorous)	-0.4678	0.227	0.2216	10	1.902	2.84	79.099	
TN (total nitrogen)	0.1912	0.3903	-0.1954					
TOC (total organic carbon)	-0.4073	0.4432	-0.1817					
XSECDPTH (cross section depth)	-0.8255	0.0427	-0.0653					
XSEC_SDP	-0.6964	-0.0448	-0.3601					
EMBED (embeddedness)	-0.8272	0.1517	-0.1733					
SBA	0.3253	-0.4563	-0.0519					
SUCut (standard deviation undercut bank)	0.7372	0.118	-0.2873					
BA	0.711	0.4993	-0.097					
Ucut (undercut bank)	0.6967	0.3535	-0.3887					
SWet_wth (standard deviation wetted width)	-0.3741	-0.5664	-0.3792					
SBk_wth (standard deviation bank full width)	-0.3188	-0.6886	-0.4386					
SBk_ht (standard deviation bank height)	-0.1963	-0.3653	-0.5816					
SIncs_ht (standard deviation incised bank height)	-0.1341	-0.3896	-0.5942					
SDepth (depth) (standard deviation depth)	-0.6506	-0.3185	-0.3594					
Svelo (velocity) (standard deviation velocity)	0.5637	-0.2634	-0.1627					
Wet_wth (wetted width)	-0.2774	-0.7853	-0.2839					
Bk_wth (bankfull width)	-0.2604	-0.7644	-0.3294					
Bk_ht (bank height)	-0.4506	-0.0794	-0.3787					
Incsd_ht (incised height)	-0.1766	-0.1929	-0.4111					
Thal_m (thalweg depth)	-0.8714	-0.2232	-0.1284					
Velocity (velocity)	0.706	-0.2253	-0.1201					
Fila_alg (filamentous algae)	-0.515	-0.4343	0.2718					
Micr_alg (micro algae)	0.3371	-0.3685	0.0827					
Macrophyt (macrophytes)	-0.0994	-0.2525	0.6743					
LWD (large woody debris)	0.4763	0.301	-0.2766					
SWD (small woody debris)	0.6002	-0.1139	-0.1412					
Roots	0.3339	0.1221	-0.0365					
Overhang	0.6111	-0.0043	-0.0747					

Ucut ²	0.8506	0.3367	-0.1945					
Leafpack	0.224	-0.5457	0.3066					
Boulders	0.586	-0.2288	0.0185					
Artifical (artificial structures)	0.0804	0.1612	0.3731					
Den (canopy cover measured by densitometer)	0.7468	0.0445	0.014					
LWD_CT (large woody debris count)	0.6582	0.2435	-0.2524					
LWD_M3 (large woody debris per cubic meter)	0.5067	0.1128	-0.3015					
BHABBOUL (BHABboulders)	0.5679	-0.1012	-0.0604	BHAB = the proportion of total habitat that was sampled for benthos				
BHABCOBB (BHABcobbles)	0.5082	-0.4081	0.2465					
BHABFILA (BHABfilamentous algae)	-0.4104	0.2156	0.1432					
BHABGRAV (BHABgravel)	0.4529	-0.2974	0.1239					
BHABLWD (BHABlarge woodydebris)	0.0929	0.1302	-0.4399					
BHABMACR (BHABmacro algae)	-0.076	0.0533	-0.0698					
BHABOVER (BHABoverhanging vegetation)	-0.6834	0.1961	0.046					
BHABROOT (BHABrootwads)	-0.225	0.198	0.0829					
BHABFINE (BHABfine particles)	-0.1552	-0.3249	-0.3532					
BHABSWD (BHABsmall woody debris)	0.3477	0.3922	-0.4753					
BHABUCUT (BHABundercut bank)	0.2159	-0.1938	-0.23					
PT (peat substrate)	-0.8751	0.0381	-0.0303					
FN (fine substrate)	0.3561	0.2807	-0.5112					
SA (sand substrate)	0.673	0.1784	-0.3774					
GF (fine gravel)	0.5737	0.2362	-0.0787					
GC (course gravel)	0.6372	-0.1414	0.1208					
CB (cobble)	0.7182	-0.3834	0.0997					
SMB (small boulders)	0.5524	-0.3657	-0.1417					
LGB (large boulders)	0.4566	-0.2256	0.0847					
WD (woody debris)	-0.1707	0.2883	0.1627					
PCT_FN (fine particles)	-0.8382	0.3564	-0.2689					
ST_LINE (straightline distance from top to bottom of reach)	0.0905	0.11	-0.0231					

PCA of 67 reach-scale instream habitat and water chemistry variables among the 30 headwater streams sampled.

